

Chapter 5.

Machine Learning 1

- Supervised Learning

AI Course

Chapter Description

Chapter objectives

- ✓ Be able to introduce machine learning-based data analysis according to the business objective, strategy, and policy and manage the overall process.
- ✓ Be able to select and apply a machine learning algorithm that is the most suitable to the given problem and perform hyperparameter tuning.
- ✓ Be able to design, maintain, and optimize a machine learning workflow for AI modeling using structured and unstructured data.

Chapter contents

- ✓ Unit 1. Machine Learning Based Data Analysis
- ✓ Unit 2. Application of Supervised Learning Model for Numerical Prediction
- ✓ Unit 3. Application of Supervised Learning Model for Classification
- ✓ Unit 4. Decision Tree
- ✓ Unit 5. Naïve Bayes Algorithm
- ✓ Unit 6. KNN Algorithm
- ✓ Unit 7. SVM Algorithm
- ✓ Unit 8. Ensemble Algorithm

Unit 1.

Machine Learning Based Data Analysis

- 1.1. What is Machine Learning?
- 1.2. Python scikit-learn Library for Machine Learning
- 1.3. Preparation and Division of Dataset
- 1.4. Data Pre-processing for Making a Good Training Dataset
- 1.5. Practicing to Find an Optimal Method to Solve Problems with scikit-learn

What is Machine Learning?

- What is machine learning?



- A statistical model that learns from **data**.
- A rather **simple** model can make **complex** predictions.

- Samuel's definition in the early phase of artificial intelligence

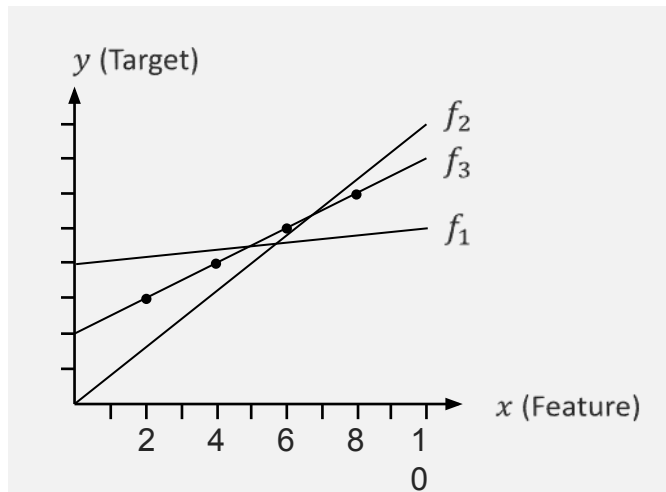
- “Programming Computers to learn from experience should eventually eliminate the need for much of this detailed programming effort.” – Samuel, 1959

- Modern definition

- “A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .” – Mitchell, 1997 (p.2)
- “Programming computers to optimize a performance criterion using example data or past experience.” – Alpaydin, 2010
- “Computational methods using experience to improve performance or to make accurate predictions.” – Mohri, 2012

- Mathematical definition

- Suppose that the x-axis is invested advertising expenses while the y-axis is sales.



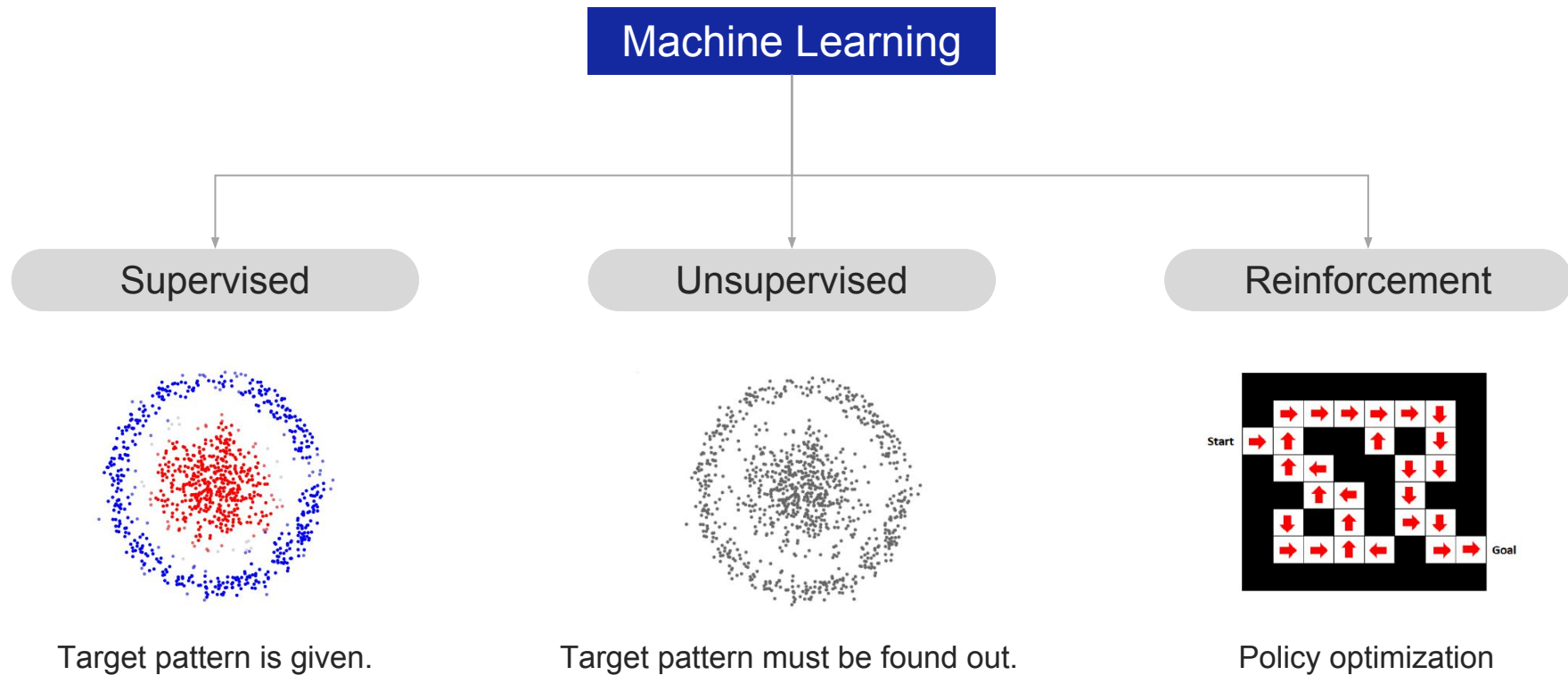
- Question about prediction – What are the sales when random advertising expenses are given?
- Linear regression
 - w and b as parameters

$$y = \underline{w}x + \underline{b}$$

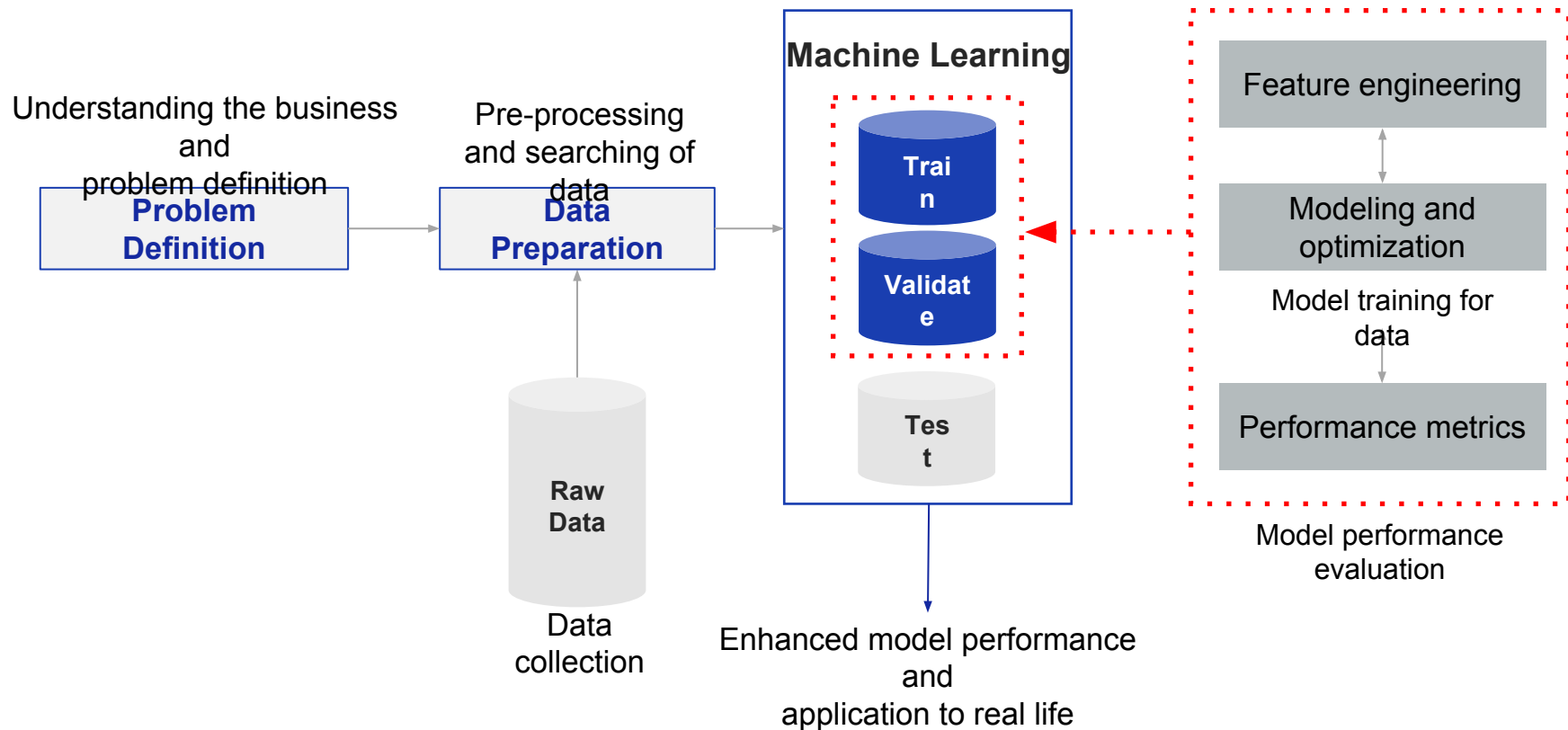
- 'w' is commonly used as an abbreviation of 'weigh.'

- Since the optimal value is unknown in the beginning, start with an arbitrary value and then reach the optimal value by gradually enhancing the performance.
 - From the graph, it starts from f_1 to continue as $f_1 \rightarrow f_2 \rightarrow f_3$.
 - The optimal value is f_3 where $w=0.5$ and $b=2.0$.

- Types of machine learning according to methods of supervision



- Machine learning workflow



- Machine learning types:

Type	Algorithm/Method
Unsupervised learning	Clustering
	MDS, t-SNE
	PCA, NMF
	Association analysis
Supervised learning	Linear regression
	Logistic regression
	Tree, Random Forest, Ada Boost, XGBoost
	Naïve Bayes
	KNN
	Support vector machine (SVM)
	Neural Network

Parameters vs. Hyperparameters

- Parameters

- Learned from data by training and not manually set by the practitioner.
- Contain the data pattern.

Ex Coefficients of linear regression

Ex Weights of neural network

- Hyperparameters

- **Can be set** manually by the practitioner.
- Can be tuned to optimize the machine learning performance.

Ex k in KNN algorithm

Ex Learning rate in neural network

Ex Maximum depth in Tree algorithm

Unit 1.

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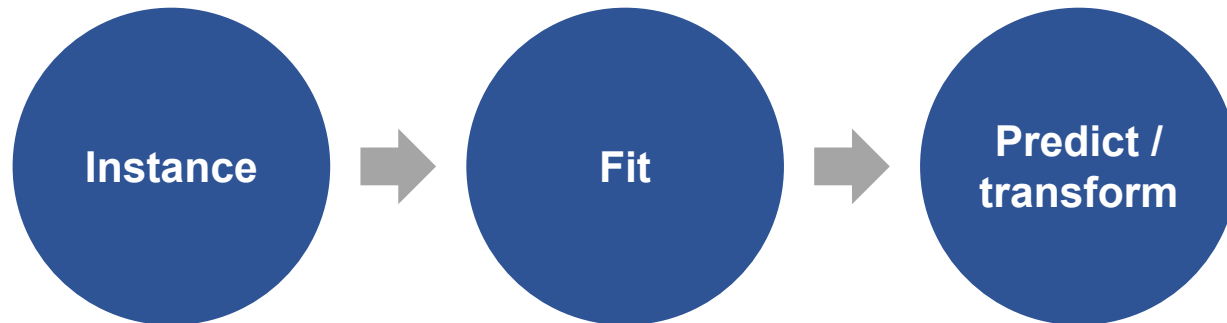
Features of the scikit-learn library

- Features

- Integrated library interface by applying the façade design pattern
- Installed with various kinds of machine learning algorithms, model selection, and data pre-processing functions
- Simple and efficient tool to analyze predicted data
- Based on NumPy, SciPy and matplotlib
- Easily accessible and can be reused in many different situations
- Highly compatible with different libraries
- Does not support GPU
- Can be used as an open source and for commercial purposes

Mechanism of scikit-learn

- scikit-learn is characterized by its intuitive and easy interface, complete with high-level API.



scikit-learn Library

- About the scikit-learn library

- It is a representative Python machine learning library.
- To import a machine learning algorithm as a class:

```
from sklearn.<family> import <machine learning algorithm>
```

Ex from sklearn.linear_model import LinearRegression

- Hyperparameters are specified when the machine learning object is instantiated:

Ex myModel = KNeighborsClassifier(n_neighbors=10) # KNN with k = 10

- About the scikit-learn library

- To train a supervised learning model: `myModel.fit(X_train, Y_train)`
- To train an unsupervised learning model: `myModel.fit(X_train)`
- To predict using an already trained model: `myModel.predict(X_test)`
- To import a preprocessor as class: `from sklearn.preprocessing import <a preprocessor>`
- To split the dataset into a training set and a testing set:
`X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.3, random_state=123)`
- To calculate a performance metric (accuracy): `metrics.accuracy_score(Y_test, Y_pred)`
- To cross validate and do hyperparameter tuning at the same time:
`myGridCV = GridSearchCV(estimator, parameter_grid, cv=k)`
`myGridCV.fit(X_train, Y_train)`

•Practicing scikit-learn

- The sklearn.datasets module includes utilities to load datasets, including methods to load and fetch popular reference datasets. It also features some artificial data generators.

```
In [1]: from sklearn.datasets import load_breast_cancer
```

- Import data with the load_breast_cancer().

```
In [2]: type(load_breast_cancer())
```

```
Out[2]: sklearn.utils.Bunch
```

```
In [3]: data=load_breast_cancer()
```

- Container object exposing keys as attributes.
Bunch objects are sometimes used as an output for functions and methods.
They extend dictionaries by enabling values to be accessed by key, bunch["value_key"], or by an attribute, bunch.value_key.

•Practicing scikit-learn

```
In [5]: pd.DataFrame(data['data'])
```

```
Out[5]:
```

	0	1	2	3	4	5	6	7	8	9	...	20	21	22	23
0	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.30010	0.14710	0.2419	0.07871	...	25.380	17.33	184.60	2019.0
1	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.08690	0.07017	0.1812	0.05667	...	24.990	23.41	158.80	1956.0
2	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.19740	0.12790	0.2069	0.05999	...	23.570	25.53	152.50	1709.0
3	11.42	20.38	77.58	386.1	0.14250	0.28390	0.24140	0.10520	0.2597	0.09744	...	14.910	26.50	98.87	567.7
4	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.19800	0.10430	0.1809	0.05883	...	22.540	16.67	152.20	1575.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
564	21.56	22.39	142.00	1479.0	0.11100	0.11590	0.24390	0.13890	0.1726	0.05623	...	25.450	26.40	166.10	2027.0
565	20.13	28.25	131.20	1261.0	0.09780	0.10340	0.14400	0.09791	0.1752	0.05533	...	23.690	38.25	155.00	1731.0
566	16.60	28.08	108.30	858.1	0.08455	0.10230	0.09251	0.05302	0.1590	0.05648	...	18.980	34.12	126.70	1124.0
567	20.60	29.33	140.10	1265.0	0.11780	0.27700	0.35140	0.15200	0.2397	0.07016	...	25.740	39.42	184.60	1821.0
568	7.76	24.54	47.92	181.0	0.05263	0.04362	0.00000	0.00000	0.1587	0.05884	...	9.456	30.37	59.16	268.6

569 rows × 30 columns



Line 5

- This data becomes x (independent variable, data).

•Practicing scikit-learn

```
In [6]: pd.DataFrame(data['target'])
```

```
Out[6]:
```

	0
0	0
1	0
2	0
3	0
4	0
...	...

569 rows × 1 columns



Line 6

- This data becomes y (dependent variable, actual value).

•Practicing scikit-learn

```
In [1]: print(data.DESCR)
```

```
In [2]: from sklearn.model_selection import train_test_split  
        help(train_test_split)
```



Line 1

- Provides details about the data.
- The help shows that the default value of test_size is 0.25.

•Practicing scikit-learn

```
In [7]: len(data.data)
```

```
Out[7]: 569
```



Line 7

- From the total of 569 observed values, divide the data for training and evaluation into 7:3 or 8:2. 7.5:2.5 is the default value.

- Practicing scikit-learn

- Use `train_test_split()` to split the data for making and evaluating the model.

```
In [ ]: train_test_split
```

```
In [8]: 569*0.75
```

```
Out[8]: 426.75
```

```
In [9]: from sklearn.model_selection import train_test_split
```

```
In [10]: X_train, X_test,y_train, y_test=train_test_split(data.data,data.target)
```

•Practicing scikit-learn

```
In [11]: len(X_train)
```

```
Out[11]: 426
```

```
In [12]: 569*0.25
```

```
Out[12]: 142.25
```

```
In [13]: len(X_test)
```

```
Out[13]: 143
```



Line 11

- 426 observed values (75%) out of total 569 observations are found.



Line 13

- 143 observed values (25%) out of total 569 observations are found.

•Practicing scikit-learn

- ▶ For instanting, use the model's hyperparameter as an argument. Hyperparameter is an option that requires a human setting and greatly affects the model performance.

```
In [1]: 1 from sklearn.datasets import load_breast_cancer
        2 from sklearn.model_selection import train_test_split
        3 from sklearn.tree import DecisionTreeClassifier
        4
        5 data=load_breast_cancer()
        6 X_train, X_test, y_train, y_test=train_test_split(data.data, data.target, random_state=42)
        7
        8 model = DecisionTreeClassifier(criterion='entropy')
        9 model

Out[1]: DecisionTreeClassifier(criterion='entropy')
```

Line 1-5

- Loading the test dataset

Line 1-8

- Instanting the estimator and hyperparameter setting
- Model initialization by using the entropy for branching

- fit

- ▶ Use the fit method with an instance estimator for training. Send the training data and label data together as an argument to the supervised learning algorithm.

- predict

- ▶ The instance estimator that has completed training with fitting can be applied with the predict method. 'Predict' converts the estimated results of the model regarding the entered data.

```
In [2]: model.fit(X_train, y_train)
        y_pred=model.predict(X_test)
        y_pred
```

```
Out[2]: array([[1, 0, 0, 1, 1, 0, 0, 0, 0, 1, 1, 0, 1, 1, 1, 0, 1, 1, 1, 0, 1, 1,
                0, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 0, 1, 0, 1, 1, 0, 1, 1,
                1, 1, 1, 1, 1, 1, 0, 0, 1, 1, 1, 1, 1, 1, 0, 0, 1, 1, 0, 0, 1, 1, 1,
                0, 0, 1, 1, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1, 1, 0, 0, 0, 0, 0,
                1, 1, 1, 1, 1, 1, 1, 1, 0, 0, 1, 0, 0, 1, 0, 0, 1, 1, 1, 0, 1, 1,
                0, 1, 1, 0, 1, 0, 1, 1, 1, 0, 1, 1, 1, 0, 1, 0, 0, 1, 1, 0, 0, 0,
                1, 1, 1, 0, 0, 1, 1, 0, 1, 0, 1])
```



Line 2

- It is an estimated value, so the actual value for X_test may vary. Measure the accuracy by comparing the two values.

- Practicing scikit-learn

```
In [18]: pred=model.predict(X_test)
```

```
In [19]: pred_2d=pred.reshape(len(pred),1)
```

```
In [20]: pred_2d
```

```
Out[20]: array([[1],  
[1],  
[1],  
[0],  
[0],  
[0],  
[0],  
[1],  
[1],  
[1],  
[0],  
[1],  
[1],  
[1],  
[1],  
[0],  
[1],  
[0],  
[1],  
[1],  
[1],  
[1]])
```

- Practicing scikit-learn

```
In [21]: import numpy as np
```

```
In [22]: y_test_2d=y_test.reshape(len(pred), 1)
```

```
In [23]: df1=pd.DataFrame(pred_2d, columns=['pred'])
```

```
In [24]: df2=pd.DataFrame(y_test_2d, columns=['real'])
```

•Practicing scikit-learn

```
In [25]: pd.concat([df1, df2], axis=1)
```

Out[25]:

	pred	real			
0	1	1	138	1	1
1	0	0	139	0	0
2	0	0	140	1	1
3	1	1	141	0	0
4	1	1	142	1	1
...	...	...			



Line 25

- Data frame shows a result where predicted value and actual value differ.

- Practicing scikit-learn

```
In [26]: pd_concat = pd.concat([df1, df2], axis=1)
```

```
In [27]: pd_concat.columns
```

```
Out[27]: Index(['pred', 'real'], dtype='object')
```


•Practicing scikit-learn

```
In [65]: df_concat[df_concat['pred'] == df_concat['real']]
```

```
Out[65]:
```

	pred	real			
0	1	1	137	1	1
1	1	1	138	1	1
2	1	1	139	1	1
3	1	1	141	1	1
4	1	1	142	1	1
...	...	...	133 rows × 2 columns		

•Practicing scikit-learn

```
In [66]: len(df_concat[df_concat['pred'] == df_concat['real']])
```

```
Out[66]: 133
```

```
In [68]: 133/143
```

```
Out[68]: 0.9300699300699301
```



Line 66

- 133/143

•Practicing scikit-learn

```
In [66]: len(df_concat[df_concat['pred'] == df_concat['real']]) # 133/143
```

```
Out[66]: 133
```

```
In [68]: 133/143
```

```
Out[68]: 0.9300699300699301
```

- It showed 93% accuracy, which is quite a rare result. In fact, a process of increasing data accuracy is required during data pre-processing, and standardization is one of the options. The following is a brief summary of standardization.
 - Standardization can be done by calculating standard normal distribution. Another term for standardization is z-transformation, and the standardized value is also referred to as z-score. 94% accuracy would be obtained from KNN wine classification through standardization.
 - Standardization is widely used in data pre-processing in general other than KNN, and the following is the
$$x = \frac{x-m}{\sigma} \text{ (} m: \text{average, } \sigma: \text{standard deviation)}$$
 - The standardization is available as StandardScaler class in the scikit-learn.

- Practicing scikit-learn

```
In [32]: from sklearn.preprocessing import StandardScaler
```

```
In [33]: scaler=StandardScaler()
```

```
In [34]: X_train
```

```
Out[34]: array([[1.247e+01, 1.731e+01, 8.045e+01, ..., 1.053e-01, 3.035e-01,
                7.661e-02],
                [1.706e+01, 2.100e+01, 1.118e+02, ..., 1.827e-01, 2.623e-01,
                7.599e-02],
                [1.808e+01, 2.184e+01, 1.174e+02, ..., 9.181e-02, 2.369e-01,
                6.558e-02],
                ...,
                [9.731e+00, 1.534e+01, 6.378e+01, ..., 1.571e-01, 3.108e-01,
                1.259e-01],
                [9.683e+00, 1.934e+01, 6.105e+01, ..., 3.846e-02, 2.552e-01,
                7.920e-02],
                [1.276e+01, 1.884e+01, 8.187e+01, ..., 8.312e-02, 2.744e-01,
                7.238e-02]])
```

•Practicing scikit-learn

```
In [35]: df_scale_previous=pd.DataFrame(data['data'])
```

```
In [36]: scaler.fit(X_train)
```

```
Out[36]: StandardScaler()
```

```
In [37]: X_train=scaler.transform(X_train)
```

```
In [38]: df_scale_after=pd.DataFrame(X_train)
```



Line 35

- Data frame before standardization

•Practicing scikit-learn

```
In [39]: df_scale_previous
```

```
Out[39]:
```

	0	1	2	3	4	5	6	7	8	9	...	20	21	22	23
0	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.30010	0.14710	0.2419	0.07871	...	25.380	17.33	184.60	2019.0
1	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.08690	0.07017	0.1812	0.05667	...	24.990	23.41	158.80	1956.0
2	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.19740	0.12790	0.2069	0.05999	...	23.570	25.53	152.50	1709.0
3	11.42	20.38	77.58	386.1	0.14250	0.28390	0.24140	0.10520	0.2597	0.09744	...	14.910	26.50	98.87	567.7
4	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.19800	0.10430	0.1809	0.05883	...	22.540	16.67	152.20	1575.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
564	21.56	22.39	142.00	1479.0	0.11100	0.11590	0.24390	0.13890	0.1726	0.05623	...	25.450	26.40	166.10	2027.0
565	20.13	28.25	131.20	1261.0	0.09780	0.10340	0.14400	0.09791	0.1752	0.05533	...	23.690	38.25	155.00	1731.0
566	16.60	28.08	108.30	858.1	0.08455	0.10230	0.09251	0.05302	0.1590	0.05648	...	18.980	34.12	126.70	1124.0
567	20.60	29.33	140.10	1265.0	0.11780	0.27700	0.35140	0.15200	0.2397	0.07016	...	25.740	39.42	184.60	1821.0
568	7.76	24.54	47.92	181.0	0.05263	0.04362	0.00000	0.00000	0.1587	0.05884	...	9.456	30.37	59.16	268.6

569 rows × 30 columns



Line 39

- The differences among column values are huge before standardization.

•Practicing scikit-learn

```
In [40]: df_scale_after
```

```
Out[40]:
```

	0	1	2	3	4	5	6	7	8	9	...	
0	-0.473538	-0.470859	-0.475889	-0.498453	-0.491544	-0.531238	-0.666183	-0.653706	-1.019670	-0.327784	...	-0.
1	0.855004	0.382383	0.836277	0.779301	1.116642	0.038826	0.808755	1.306094	-0.275901	-0.291025	...	1.
2	1.150236	0.576617	1.070667	1.086428	-1.598504	-0.334342	0.288007	0.229435	-0.116787	-1.365872	...	0.
3	1.101030	0.387008	1.363655	0.999010	1.905805	2.996152	2.970181	1.836134	1.152431	1.244051	...	0.
4	-0.378022	-0.436174	-0.367065	-0.416281	-1.120030	-0.285118	-0.180023	-0.209674	-0.842054	-0.763016	...	-0.
...	...	...	...	...	...	...	...	...	...	...	...	...
421	1.364423	0.606677	1.334356	1.337025	0.398575	0.686714	0.919334	0.997811	0.523373	-0.201332	...	1.
422	-0.389599	-1.381908	-0.398875	-0.428519	-0.586812	-0.469367	-0.609222	-0.607334	-0.742145	-0.189568	...	-0.
423	-1.266321	-0.926384	-1.173618	-1.022668	0.782492	1.095292	4.151829	0.768024	2.762080	4.450949	...	-1.
424	-1.280214	-0.001460	-1.287883	-1.064919	-0.802232	-1.037096	-0.829736	-1.018335	-0.819852	-0.049882	...	-1.
425	-0.389599	-0.117076	-0.416454	-0.450374	0.040253	-0.468589	-0.784605	-0.806034	-0.157490	-0.126342	...	-0.

426 rows x 30 columns



Line 40

- After standardization, the column values do not significantly deviate from 0.
- Better performance would be possible compared to the performance before standardization.

- transform

- Feature processing is done with 'transform' to return the result.

```
In [3]: from sklearn.preprocessing import StandardScaler
X_train, X_test, y_train, y_test = train_test_split(data.data, data.target)

print(X_train)

scaler = StandardScaler()
scaler.fit(X_train)
X_train = scaler.transform(X_train)

X_train

[[1.276e+01 1.884e+01 8.187e+01 ... 8.312e-02 2.744e-01 7.238e-02]
 [1.146e+01 1.816e+01 7.359e+01 ... 5.509e-02 2.208e-01 7.638e-02]
 [1.130e+01 1.819e+01 7.393e+01 ... 1.218e-01 3.308e-01 1.297e-01]
 ...
 [1.281e+01 1.306e+01 8.129e+01 ... 3.990e-02 1.783e-01 7.319e-02]
 [1.263e+01 2.076e+01 8.215e+01 ... 1.105e-01 2.226e-01 8.486e-02]
 [1.245e+01 1.570e+01 8.257e+01 ... 1.741e-01 3.985e-01 1.244e-01]]
```



Line 3-3

- Output before pre-processing

- transform

- Feature processing is done with 'transform' to return the result.

```
In [3]: from sklearn.preprocessing import StandardScaler
X_train, X_test, y_train, y_test = train_test_split(data.data, data.target)

print(X_train)

scaler = StandardScaler()
scaler.fit(X_train)
X_train = scaler.transform(X_train)

X_train

[[1.276e+01 1.884e+01 8.187e+01 ... 8.312e-02 2.744e-01 7.238e-02]
 [1.146e+01 1.816e+01 7.359e+01 ... 5.509e-02 2.208e-01 7.638e-02]
 [1.130e+01 1.819e+01 7.393e+01 ... 1.218e-01 3.308e-01 1.297e-01]
 ...
 [1.281e+01 1.306e+01 8.129e+01 ... 3.990e-02 1.783e-01 7.319e-02]
 [1.263e+01 2.076e+01 8.215e+01 ... 1.105e-01 2.226e-01 8.486e-02]
 [1.245e+01 1.570e+01 8.257e+01 ... 1.741e-01 3.985e-01 1.244e-01]]
```



Line 3-7

- Pre-processing – Apply scaling

- transform

- Feature processing is done with 'transform' to return the result.

```
In [3]: from sklearn.preprocessing import StandardScaler
X_train, X_test, y_train, y_test = train_test_split(data.data, data.target)

print(X_train)

scaler = StandardScaler()
scaler.fit(X_train)
X_train = scaler.transform(X_train)

X_train

[[1.276e+01 1.884e+01 8.187e+01 ... 8.312e-02 2.744e-01 7.238e-02]
 [1.146e+01 1.816e+01 7.359e+01 ... 5.509e-02 2.208e-01 7.638e-02]
 [1.130e+01 1.819e+01 7.393e+01 ... 1.218e-01 3.308e-01 1.297e-01]
 ...
 [1.281e+01 1.306e+01 8.129e+01 ... 3.990e-02 1.783e-01 7.319e-02]
 [1.263e+01 2.076e+01 8.215e+01 ... 1.105e-01 2.226e-01 8.486e-02]
 [1.245e+01 1.570e+01 8.257e+01 ... 1.741e-01 3.985e-01 1.244e-01]]
```



Line 3-9

- Result check after pre-processing

- transform

- Feature processing is done with 'transform' to return the result.

```
Out[3]: array([[ -0.34913849, -1.43851335, -0.41172595, ..., -0.91671059,
                -0.92508585, -0.80841115],
               [ -0.20468665,  0.31264011, -0.13367256, ...,  1.43655962,
                1.14955889,  1.56911143],
               [ -0.32931176, -0.21507235, -0.31739376, ..., -0.7237126 ,
                0.53496977, -0.61934827],
               ...,
               [  0.04739597, -0.56293662, -0.06529202, ..., -1.23262438,
                -0.68282718, -1.261137  ],
               [ -0.04040808,  0.09966199, -0.03151368, ...,  1.08847951,
                0.48944465,  1.26159953],
               [ -0.5502381 ,  0.31264011, -0.6040977 , ..., -0.59582424,
                -0.29911546, -0.82948141]])
```

- `fit_transform`

- Fit and Transform is combined as `fit_transform`.

```
In [4]: 1 print(X_test)
        2
        3 X_test = scaler.fit_transform(X_test)
        4
        5 X_test

[[1.247e+01 1.860e+01 8.109e+01 ... 1.015e-01 3.014e-01 8.750e-02]
 [1.894e+01 2.131e+01 1.236e+02 ... 1.789e-01 2.551e-01 6.589e-02]
 [1.546e+01 1.948e+01 1.017e+02 ... 1.514e-01 2.837e-01 8.019e-02]
 ...
 [1.104e+01 1.683e+01 7.092e+01 ... 7.431e-02 2.998e-01 7.881e-02]
 [1.981e+01 2.215e+01 1.300e+02 ... 2.388e-01 2.768e-01 7.615e-02]
 [1.026e+01 1.222e+01 6.575e+01 ... 6.696e-02 2.937e-01 7.722e-02]]
```



Line 4-1 & 4-5

- Before & After



Line 4-3

- Combination of fit and transform

- Major scikit modules

Classification	Module	Embedded Functions
Data example	<code>sklearn.datasets</code>	Dataset for practicing
Feature processing	<code>sklearn.preprocessing</code>	Pre-processing techniques (One-hot encoding, normalization, scaling, etc.)
	<code>sklearn.feature_selection</code>	Technique to search and select a feature that provides a significant impact to the model
	<code>sklearn.feature_extraction</code>	Feature extraction from source data The supporting API for feature extraction regarding image is present in the submodule <code>image</code> , while the supporting API for text data feature extraction is present in the submodule <code>test</code> .
Dimension reduction	<code>sklearn.decomposition</code>	Algorithms related to dimension reduction (PCA, NMF, Truncated SVD, etc.)
Validation, hyperparameter tuning, data separation	<code>sklearn.model_selection</code>	Validation, hyperparameter tuning, data separation, etc. (<code>cross_validate</code> , <code>GridSearchCV</code> , <code>train_test_split</code> , <code>learning_curve</code> , etc.)
Model evaluation	<code>sklearn.metrics</code>	Techniques to measure and evaluate model performance (accuracy, precision, recall, ROC curve, etc.)

- Major scikit modules

Classification	Module	Embedded Functions
Machine learning algorithm	sklearn.ensemble	Ensemble algorithms (Random forest, AdaBoost, bagging, etc.)
	sklearn.linear_model	Linear algorithms (Linear regression, logistic regression, SGD, etc.)
	sklearn.naïve_bayes	Naive Bayes algorithms (Bernoulli NB, Gaussian NB, multinomial distribution NB, etc.)
	sklearn.neighbors	Nearest neighbor algorithms (K-NN, etc.)
	sklearn.svm	Support Vector Machine algorithms
	sklearn.tree	Decision tree algorithms
	sklearn.cluster	Unsupervised learning (clustering) algorithms (Kmeans, DBSCAN, etc.)
Utility	sklearn.pipeline	Serial conversion of feature processing and machine learning algorithms, etc.

Unit 1.

Machine Learning Based Data Analysis

- 1.1. What is Machine Learning?
- 1.2. Python scikit-learn Library for Machine Learning
- 1.3. Preparation and Division of Dataset
- 1.4. Data Pre-processing for Making a Good Training Dataset
- 1.5. Practicing to Find an Optimal Method to Solve Problems with scikit-learn

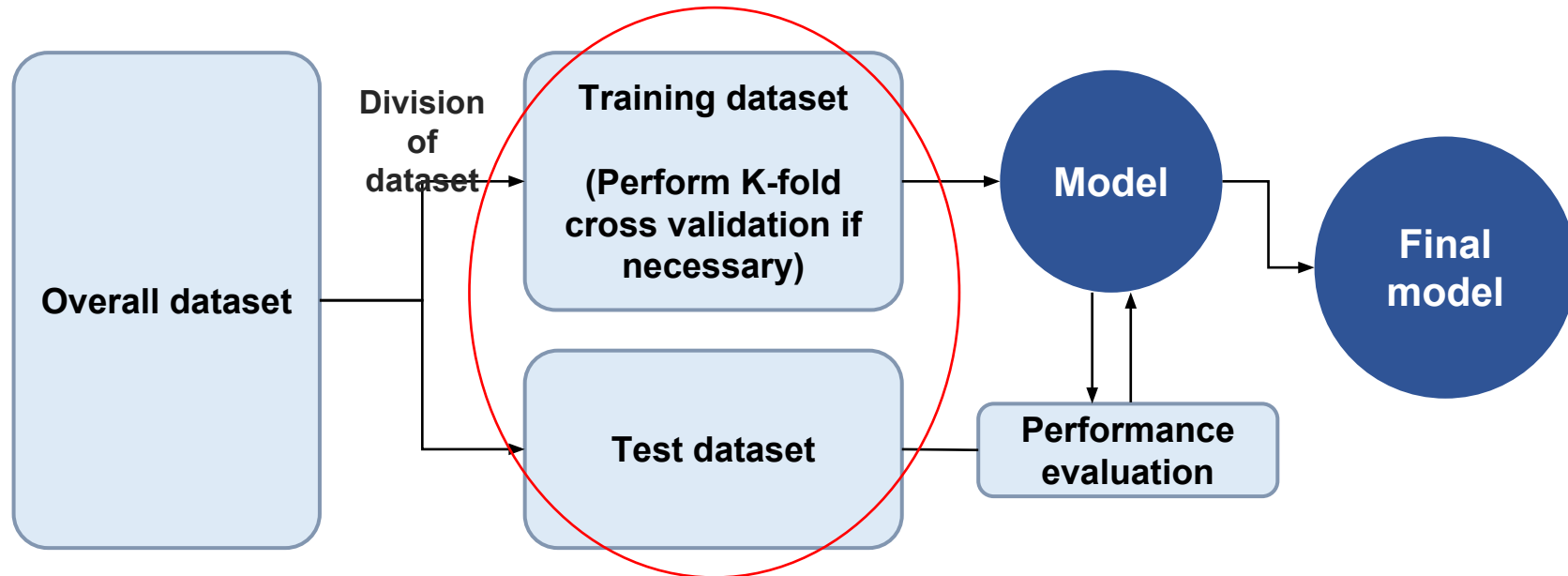
Preparation and Division of Dataset

- Chapter objectives

- Be able to understand the meaning and ripple effect of overfitting and generalization and design dataset division to solve issues.
- Be able to properly divide the training and test datasets for machine learning technique application according to the analysis purpose and dataset features.
- Be able to divide the training and validation datasets and decide the appropriate k-value for cross-validation by deciding the necessity of cross-validation according to the issue and applied technique. Be able to divide the datasets and perform sampling by considering the prediction results based on data features and classified variable distribution.
- Be able to analyze differences among various sampling methods for dataset division and apply appropriate sampling methods.

- Necessity of dataset division

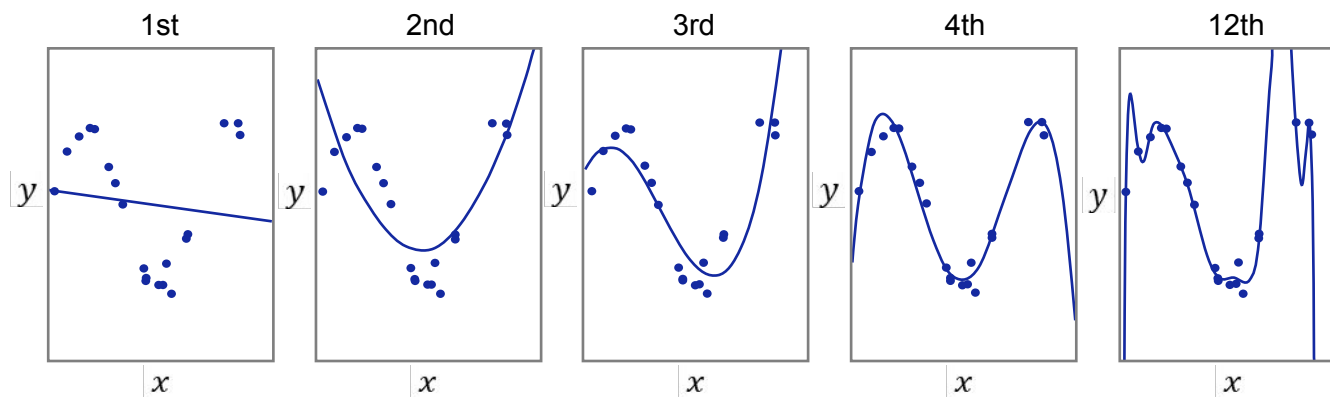
- When analyzing machine learning-based data, especially when applying a supervised learning-based model, do not analyze the overall dataset but analyze by dividing the training and evaluation (test) datasets.



Machine learning modeling process through division of training and test datasets

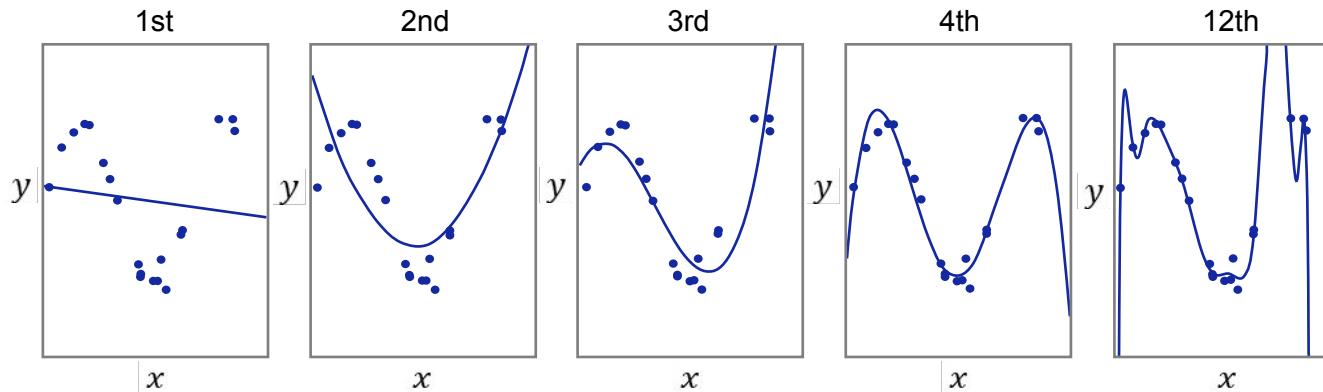
•Overfitting and generalization of modeling

- Strictly speaking, the data included in the provided training dataset can be considered as the values obtained by chance. Hence, a new dataset obtained to predict the values of new objective variables (or response variables) is not the same as the existing training dataset.
- The chance that the patterns from the training data and new data to perfectly accord is extremely low. Thus, when learning the machine learning-based model, overfitting occurs in highlighting the training dataset pattern when reflecting too much of the training dataset. At the same time, generalization for accurate prediction of new data is underperformed.
- To prevent such issues, the datasets are generally divided into training and test datasets. Measure how the machine learning model learned with the training dataset accurately predicts the test dataset's objective variables (or response variables). The resulting standard will become the model performance evaluation standard.



Overfitting and underfitting

- Overfitting and generalization of modeling



Overfitting and underfitting

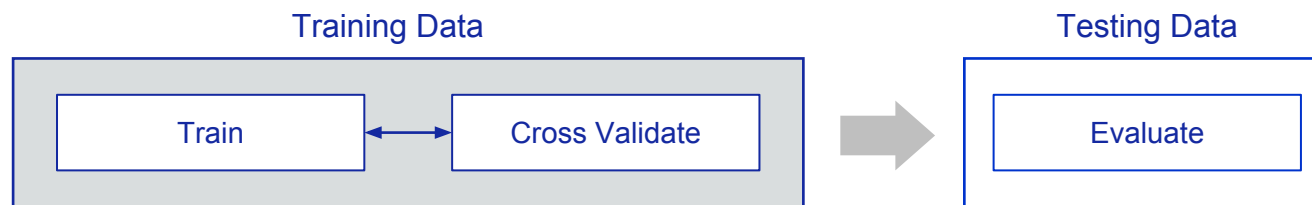
- ▶ Even if machine learning finds the optimal solution in data distribution, a wide margin of error occurs since the model has a small capacity. Such a phenomenon is referred to as underfitting; the linear equation model on the leftmost figure above is an example.
- ▶ An easy alternative is to use higher-degree polynomials, which are non-linear equations.
- ▶ The rightmost figure provided above is applied with the 12th-order polynomial.
- ▶ The model capacity got larger, and there are 13 parameters for estimation.

$$y = w_{12}x^{12} + w_{11}x^{11} + w_{10}x^{10} \cdots w_1x^1 + w_0$$

Method and Process of Dataset Division

- Cross-Validation:

- The data should be split into a training set and a testing set.
- In principle, the testing set should be used only once! It should not be reused!
- If the training set is also used for evaluation, the errors can be unrealistically small.
- We would like to evaluate realistic errors while training by splitting the training data into two.



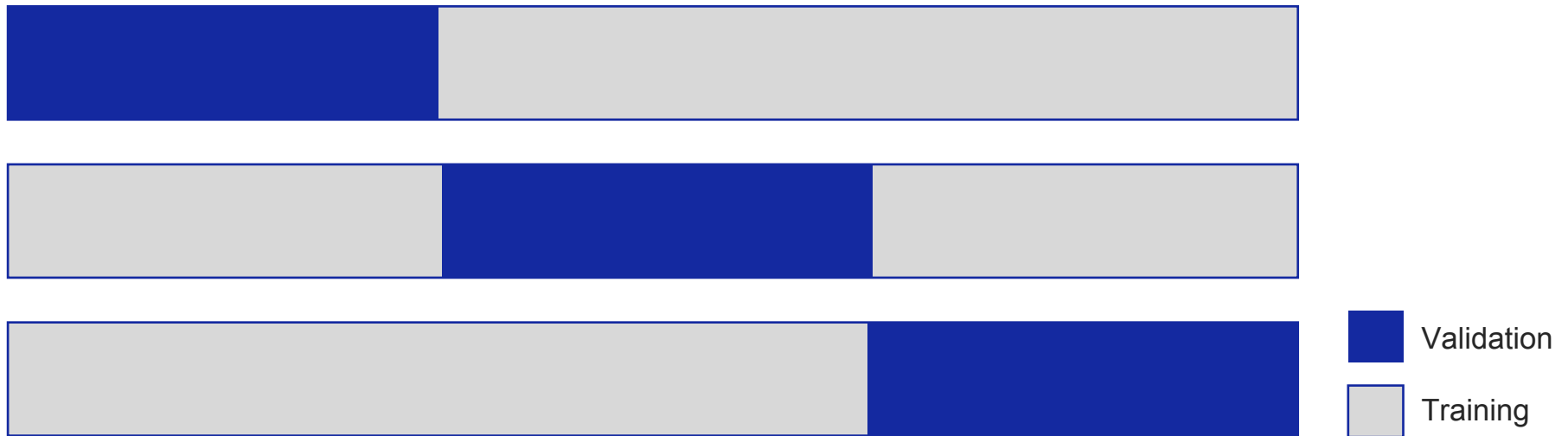
- Cross-Validation and Hyperparameter Optimization:

- As we can repeatedly evaluate errors while training, it is also possible to tune the hyperparameters.

- Cross-Validation:

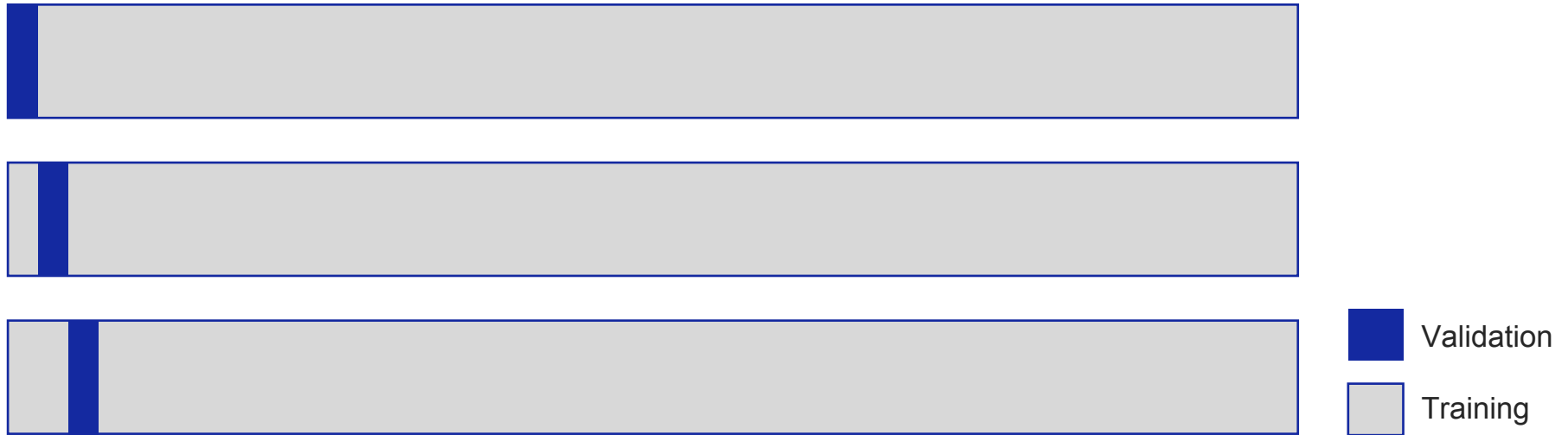
- 1) Split the data into a training set and a testing set.
- 2) Further subdivide the training set into a smaller training and a validation set.
- 3) Train the model with the smaller training set.
- 4) Evaluate the errors with the validation set.
- 5) Repeat from step 2) a few times.

- Cross-Validation method: k-Fold



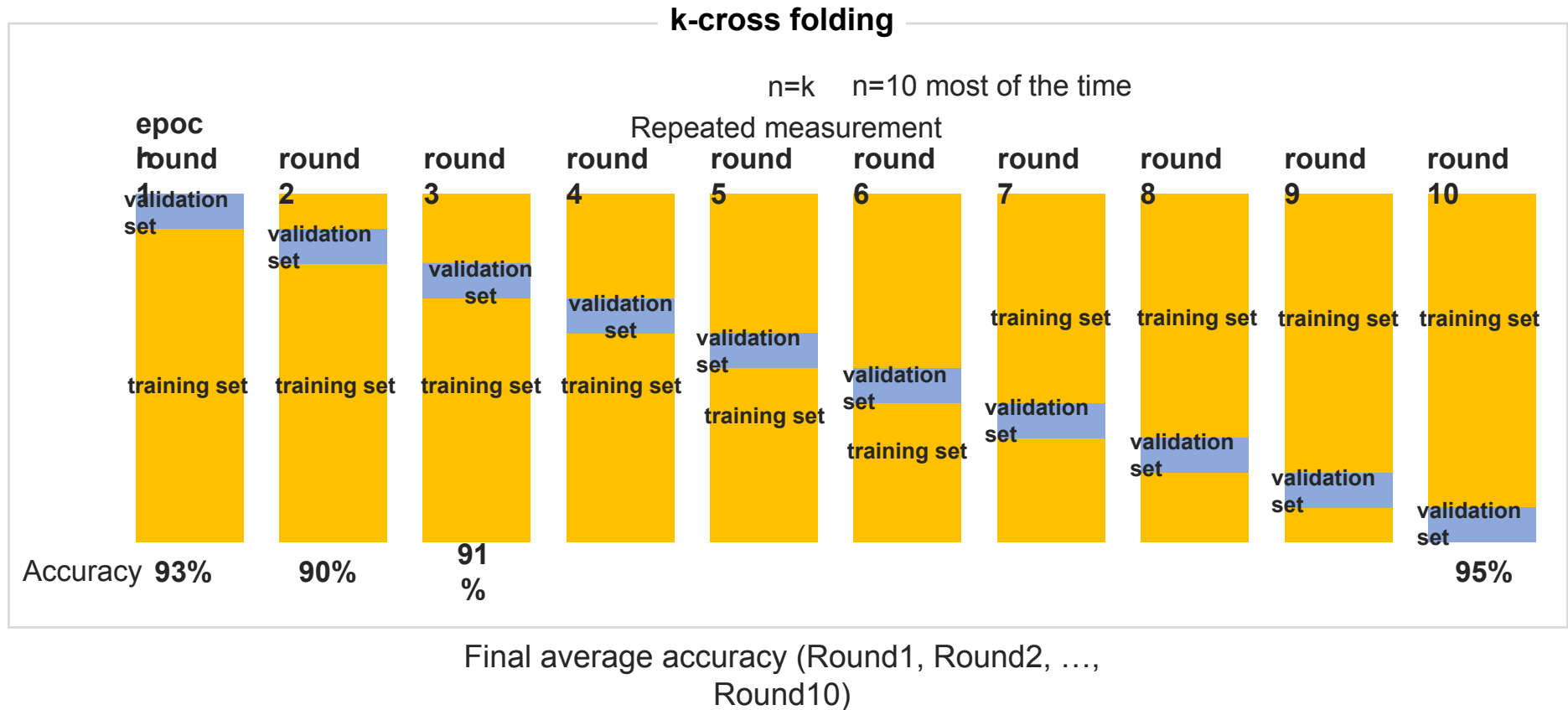
- Subdivide the training dataset into k equal parts. Then, apply sequentially.

- Cross-Validation method: Leave One Out (LOO)



- Leave only one observation for validation. Apply sequentially. More time consuming.

- Cross-Validation method: k-cross folding



Unit 1.

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Missing Value Processing

- Data cleansing for machine learning-based data analysis uses missing value and noise processing to eliminate discrepancies in collected data.
 - Missing value processing is done as follows.
 - First, import the iris data for quick examination.

```
In [1]: from sklearn.datasets import load_iris
```

```
In [2]: iris=load_iris()
```

```
In [3]: import pandas as pd
```

```
In [4]: iris['data']
```

```
Out[4]: array([[5.1, 3.5, 1.4, 0.2],  
               [4.9, 3. , 1.4, 0.2],  
               [4.7, 3.2, 1.3, 0.2],  
               [4.6, 3.1, 1.5, 0.2],  
               [5. , 3.6, 1.4, 0.2],  
               [5.4, 3.9, 1.7, 0.4],  
               [4.6, 3.4, 1.4, 0.3],  
               [5. , 3.4, 1.5, 0.2],  
               [4.4, 2.9, 1.4, 0.2],
```

- Missing value processing

```
In [7]: iris_df = pd.DataFrame(iris['data'],  
                                columns=['Sepal_Length', 'Sepal_Width', 'Petal_Length', 'Petal_Width'])
```

```
In [8]: iris_df.columns
```

```
Out[8]: Index(['Sepal_Length', 'Sepal_Width', 'Petal_Length', 'Petal_Width'], dtype='object')
```

```
In [9]: iris_df['Sepal_Length']
```

```
Out[9]: 0      5.1  
        1      4.9  
        2      4.7  
        3      4.6  
        4      5.0  
        ...  
       145     6.7  
       146     6.3  
       147     6.5  
       148     6.2  
       149     5.9  
        Name: Sepal_Length, Length: 150, dtype: float64
```

- Missing value processing

```
In [0]: import numpy as np
```

```
In [1]: np.mean(iris_df['Sepal_Length'])
```

```
Out[1]: 5.843333333333334
```

- Missing value processing

- 1) Ignore the record (row)

- In data classification, ignore the record if the class label is not distinguished.

Ex In the case of the iris data, ignore the fourth row as shown on the table below.

	x1	x2	x3	x4	y
	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	
5	5	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa

- 'Ignore the record' is extremely inefficient if missing values frequently occur.

- Missing value processing

- 2) Insert the missing value

- Enter a certain value like 'unknown' for missing value. Or enter the average value of data, such as the overall average value, median value, or class that belongs to the same record.

	x1	x2	x3	x4	y
	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	unknown
5	5	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa

- Missing value processing

- 2) Insert the missing value

- The average value of Sepal.Length for iris is 5.843 as provided earlier, so insert 5.843.

	x1	x2	x3	x4	y
	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4		3.1	1.5	0.2	setosa
5	5	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa

	x1	x2	x3	x4	y
	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	5.843	3.1	1.5	0.2	setosa
5	5	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa

- Missing value processing

- 3) Manual entry

- A person in charge (or an expert) should check the data and modify it into an appropriate value.
 - It requires a lot of time but provides high reliability.

- Missing value processing

- Identify the missing value from the table type of data below and make appropriate processing.
 - In python, the missing value is specified as np.nan or null value.
 - 'nan' is an abbreviation of Not a Number.

```
In [12]: data=[[1.0,2.0,3.0,4.0],[5.0,6.0,np.nan,8.0],[10.0,11.0,12.0,np.nan]]
```

```
In [13]: np.array(data)
```

```
Out[13]: array([[ 1.,  2.,  3.,  4.],  
                [ 5.,  6., nan,  8.],  
                [10., 11., 12., nan]])
```

•Missing value processing

```
In [14]: pd.DataFrame(np.array(data) , columns=['A','B','C','D'])
```

```
Out[14]:
```

	A	B	C	D
0	1.0	2.0	3.0	4.0
1	5.0	6.0	NaN	8.0
2	10.0	11.0	12.0	NaN

```
In [15]: df=pd.DataFrame(np.array(data) , columns=['A','B','C','D'])
```

- ▶ The omitted value is changed to NaN. In this case, it is not problematic due to the low number of data, but it is extremely inconvenient to manually find missing values from a huge data frame.

•Missing value processing

- `isnull()` returns the data frame with a Boolean, which shows if the cell has a numerical value (False) or is omitted with a numerical value (True). Then, `sum()` is used to obtain the number of omissions.
- It is mandatory to check the number of missing values when importing data.

```
In [16]: df.isnull()
```

```
Out[16]:
```

	A	B	C	D
0	False	False	False	False
1	False	False	True	False
2	False	False	False	True

```
In [17]: df.isnull().sum()
```

```
Out[17]: A    0  
         B    0  
         C    1  
         D    1  
         dtype: int64
```



Line 17

- The number of missing values can be counted.

- Removing the training sample or feature with missing value

- Use `dropna()` to completely delete a certain training sample (row) or feature (column).
- `help(df.dropna)` shows `axis=0` is default.

```
In [18]: df.dropna()
```

```
Out[18]:
```

	A	B	C	D
0	1.0	2.0	3.0	4.0



Line 18

- `axis=0` is default, so the row with the NaN value is deleted.

- Removing the training sample or feature with missing value

- When trying to reflect the deleted result immediately to the object, do not omit `inplace=True` option.

```
In [19]: df.dropna(inplace=True)
```

```
In [20]: df
```

```
Out[20]:
```

	A	B	C	D
0	1.0	2.0	3.0	4.0

- Removing the training sample or feature with missing value

- Delete the row with missing value.

```
In [21]: df=pd.DataFrame(np.array(data) , columns=['A','B','C','D'])
```

```
In [22]: df
```

```
Out[22]:
```

	A	B	C	D
0	1.0	2.0	3.0	4.0
1	5.0	6.0	NaN	8.0
2	10.0	11.0	12.0	NaN

- Removing the training sample or feature with missing value

```
In [23]: df.dropna(axis=1, inplace=True)
```

```
In [24]: df
```

```
Out[24]:
```

	A	B
0	1.0	2.0
1	5.0	6.0
2	10.0	11.0

- Removing the training sample or feature with missing value

- If all rows have NaN, use `how='all'` to delete.

```
In [25]: df=pd.DataFrame(np.array(data) , columns=['A','B','C','D'])
```

```
In [26]: df
```

```
Out[26]:
```

	A	B	C	D
0	1.0	2.0	3.0	4.0
1	5.0	6.0	NaN	8.0
2	10.0	11.0	12.0	NaN

- Removing the training sample or feature with missing value

```
In [27]: df.loc[3]=[np.nan,np.nan,np.nan,np.nan]
```

```
In [28]: df
```

```
Out[28]:
```

	A	B	C	D
0	1.0	2.0	3.0	4.0
1	5.0	6.0	NaN	8.0
2	10.0	11.0	12.0	NaN
3	NaN	NaN	NaN	NaN

- Removing the training sample or feature with missing value

```
In [29]: df.dropna(how='all', inplace=True)
```

```
In [30]: df
```

```
Out[30]:
```

	A	B	C	D
0	1.0	2.0	3.0	4.0
1	5.0	6.0	NaN	8.0
2	10.0	11.0	12.0	NaN

- Removing the training sample or feature with missing value

- When deleting 3 or more NaN values, use thresh.

```
In [31]: df=pd.DataFrame(np.array(data) , columns=['A','B','C','D'])  
df.loc[3]=[np.nan,np.nan,np.nan,3]
```

```
In [32]: df
```

```
Out[32]:
```

	A	B	C	D
0	1.0	2.0	3.0	4.0
1	5.0	6.0	NaN	8.0
2	10.0	11.0	12.0	NaN
3	NaN	NaN	NaN	3.0

- Removing the training sample or feature with missing value

```
In [33]: df.dropna(thresh=3, inplace=True)
```

```
In [34]: df
```

```
Out[34]:
```

	A	B	C	D
0	1.0	2.0	3.0	4.0
1	5.0	6.0	NaN	8.0
2	10.0	11.0	12.0	NaN

- Removing the training sample or feature with missing value
 - When deleting a row with NaN on a certain column, use `subject`.

```
In [35]: df
```

```
Out[35]:
```

	A	B	C	D
0	1.0	2.0	3.0	4.0
1	5.0	6.0	NaN	8.0
2	10.0	11.0	12.0	NaN

```
In [36]: df.dropna(subset=['C'], inplace=True)
```

```
In [37]: df
```

```
Out[37]:
```

	A	B	C	D
0	1.0	2.0	3.0	4.0
2	10.0	11.0	12.0	NaN

•Imputation

- It is sometimes hard to delete a training sample or a certain column because it loses too much useful data. If so, estimate missing values from other training samples in the dataset by kriging. The most commonly used method is to impute with average value, which is to change the missing value into the overall average of a certain column. In scikit-learn, use SimpleImputer class.

```
In [38]: from sklearn.impute import SimpleImputer
```

```
In [39]: df=pd.DataFrame(np.array(data) , columns=['A','B','C','D'])
```

```
In [40]: impt=SimpleImputer(missing_values=np.nan, strategy='mean')
```

- Impute with using df.values.

```
In [41]: df.values
```

```
Out[41]: array([[ 1.,  2.,  3.,  4.],  
               [ 5.,  6., nan,  8.],  
               [10., 11., 12., nan]])
```

•Imputation

```
In [42]: help(impt.fit)
```

```
Help on method fit in module sklearn.impute._base:
```

```
fit(X, y=None) method of sklearn.impute._base.SimpleImputer instance  
Fit the imputer on X.
```

```
Parameters
```

```
-----
```

```
X : {array-like, sparse matrix}, shape (n_samples, n_features)  
    Input data, where ``n_samples`` is the number of samples and  
    ``n_features`` is the number of features.
```

```
Returns
```

```
-----
```

```
self : SimpleImputer
```

•Imputation

```
In [43]: imputed=impt.fit(df.values)
```

```
In [44]: imputed.transform(df.values)
```

```
Out[44]: array([[ 1. ,  2. ,  3. ,  4. ],  
                [ 5. ,  6. ,  7.5,  8. ],  
                [10. , 11. , 12. ,  6. ]])
```

```
In [45]: (3+12)/2
```

```
Out[45]: 7.5
```



Line 45

- Check it is the average of the column.

- For strategy parameters, median and most_frequent can also be set.

- Review on the scikit-learn estimator API

- Impute the missing value of the dataset by using the SimpleImputer class of scikit-learn from the previous clause.
- The SimpleImputer class is a transformer class of scikit-learn that is used for data conversion.
- The two main methods for the estimator include fit and transform.
- Use the fit method to learn the model parameter in the training data.
- Use the transform method to convert the data into the learned parameter.
- The data array for conversion should be the same as the number of data features used in the model learning.

- **Categorical data processing**

- Data is generally classified into categorical and continuous scales depending on their features.
- In liberal arts and social science, questionnaires are mainly used to collect data.
- **Categorical scale**
 - It is a scale that can distinguish data into different categories and is classified into the nominal and ordinal scales.
- **Continuous scale**
 - It is a scale that divides linked data into the purpose of the survey. It is classified into interval and ratio scales.
- Actual datasets would include more than one categorical feature. As explained earlier, categorical data would classify sequential and non-sequential features. Ordinal scale can be referred to as sequential categorical scale that can array features with sequences.

•Categorical data processing

```
In [46]: df = pd.DataFrame([[ 'green', 'M', 10.1, 'class2'],  
                             [ 'red', 'L', 13.5, 'class1'],  
                             [ 'blue', 'XL', 15.3, 'class2']])
```

```
In [47]: df
```

```
Out[47]:
```

	0	1	2	3
0	green	M	10.1	class2
1	red	L	13.5	class1
2	blue	XL	15.3	class2

- ▶ The data on the table has features that have orders and do not have orders. The size is ordered, but the color is not ordered. Thus, the size is classified as the ordinal scale, while the color is the nominal scale.

- Categorical data processing

- Convert ordered data into numerical values. The reason why changing text data into numerical data is to allow a computer to process arithmetic operations.

```
In [48]: size_mapping = {'XL': 3,  
                        'L': 2,  
                        'M': 1}
```

```
In [49]: df.columns= ['color', 'size', 'price', 'classlabel']
```

```
In [50]: df
```

```
Out[50]:
```

	color	size	price	classlabel
0	green	M	10.1	class2
1	red	L	13.5	class1
2	blue	XL	15.3	class2

•Categorical data processing

```
In [51]: df['size']
```

```
Out[51]: 0    M  
         1    L  
         2   XL  
         Name: size, dtype: object
```

```
In [52]: df['size']=df['size'].map(size_mapping)
```

```
In [53]: df['size']
```

```
Out[53]: 0    1  
         1    2  
         2    3  
         Name: size, dtype: int64
```

```
In [54]: df
```

```
Out[54]:
```

	color	size	price	classlabel
0	green	1	10.1	class2
1	red	2	13.5	class1
2	blue	3	15.3	class2

- Class label encoding

- The class refers to the y value, a column with an actual value.
- Create a mapping to convert the class label from strings to integers.

```
In [55]: df['classlabel']
```

```
Out[55]: 0    class2  
         1    class1  
         2    class2  
         Name: classlabel, dtype: object
```

```
In [56]: df['classlabel'].value_counts()
```

```
Out[56]: class2    2  
         class1    1  
         Name: classlabel, dtype: int64
```

```
In [57]: np.unique(df['classlabel'])
```

```
Out[57]: array(['class1', 'class2'], dtype=object)
```

- Class label encoding

- 'enumerate' creates an object with an index.

```
In [58]: enumerate(np.unique(df['classlabel']))
```

```
Out[58]: <enumerate at 0x1b17cb40140>
```

```
In [59]: for idx, label in enumerate(np.unique(df['classlabel'])):  
         print(idx, label)
```

```
0 class1
```

```
1 class2
```

- 'enumerate' creates an object with an index.

```
In [60]: {label : idx for idx, label in enumerate(np.unique(df['classlabel']))}
```

```
Out[60]: {'class1': 0, 'class2': 1}
```

```
In [61]: class_mapping={label : idx for idx, label in enumerate(np.unique(df['classlabel']))}
```

- Class label encoding

- 'enumerate' creates an object with an index.

```
In [62]: df['classlabel'] = df['classlabel'].map(class_mapping)
df
```

```
Out[62]:
```

	color	size	price	classlabel
0	green	1	10.1	1
1	red	2	13.5	0
2	blue	3	15.3	1



Line 62

- Change the class label from strings to integers.

- Class label encoding

- Since the method in the previous slide is rather inconvenient, scikit-learn supports LabelEncoder for easy conversion.

```
In [63]: df = pd.DataFrame([['green', 'M', 10.1, 'class2'],  
                             ['red', 'L', 13.5, 'class1'],  
                             ['blue', 'XL', 15.3, 'class2']])  
df.columns= ['color', 'size', 'price', 'classlabel']
```

```
In [64]: df
```

```
Out[64]:
```

	color	size	price	classlabel
0	green	M	10.1	class2
1	red	L	13.5	class1
2	blue	XL	15.3	class2

- Class label encoding

```
In [65]: from sklearn.preprocessing import LabelEncoder
```

```
In [66]: enc=LabelEncoder()
```

```
In [67]: y=enc.fit_transform(df['classlabel'].values)
```

```
In [68]: y
```

```
Out[68]: array([1, 0, 1])
```

Unit 1.

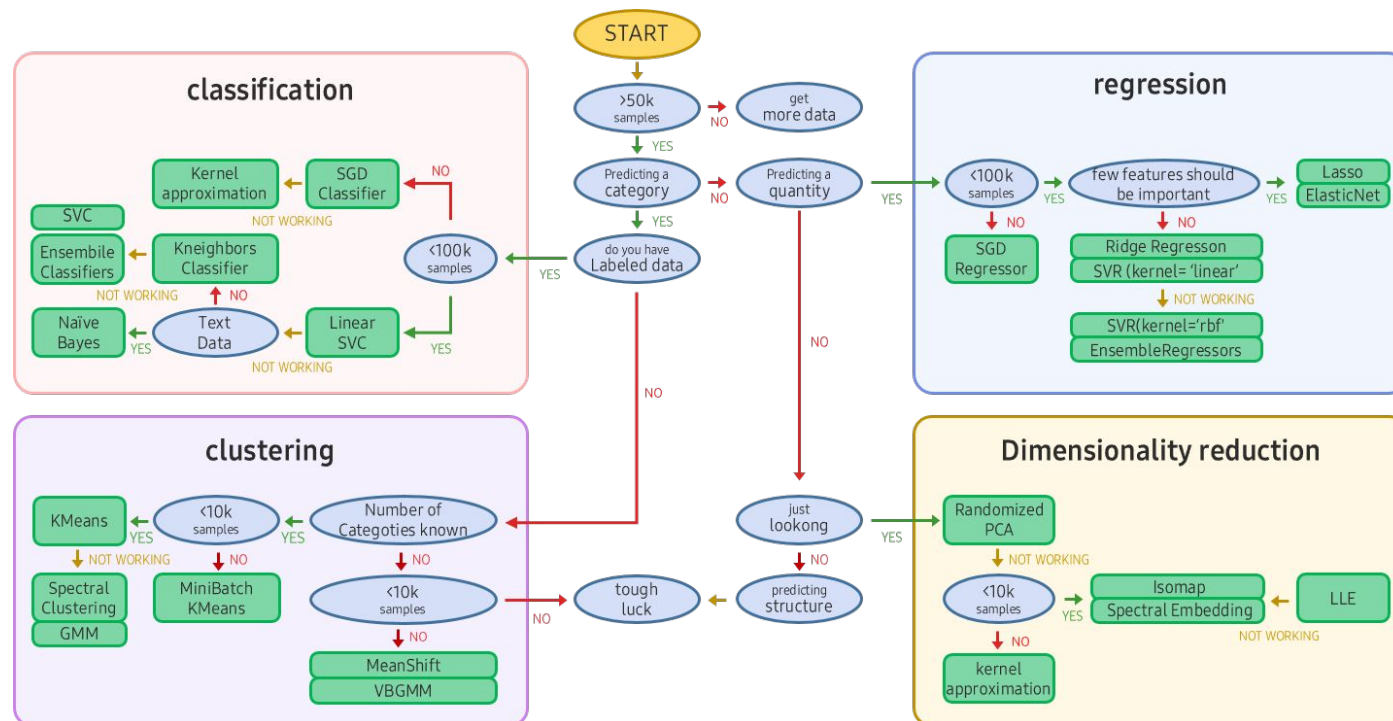
Machine Learning Based Data Analysis

- 1.1. What is Machine Learning?
- 1.2. Python scikit-learn Library for Machine Learning
- 1.3. Preparation and Division of Dataset
- 1.4. Data Pre-processing for Making a Good Training Dataset
- 1.5. Practicing to Find an Optimal Method to Solve Problems with scikit-learn

Finding an Optimal Method to Solve Problems with scikit-learn

•Practicing

- Use the following problem-solving methodology and consider which algorithm to apply. Perform pre-processing and overall process regarding the iris data and compare with the result code provided below.



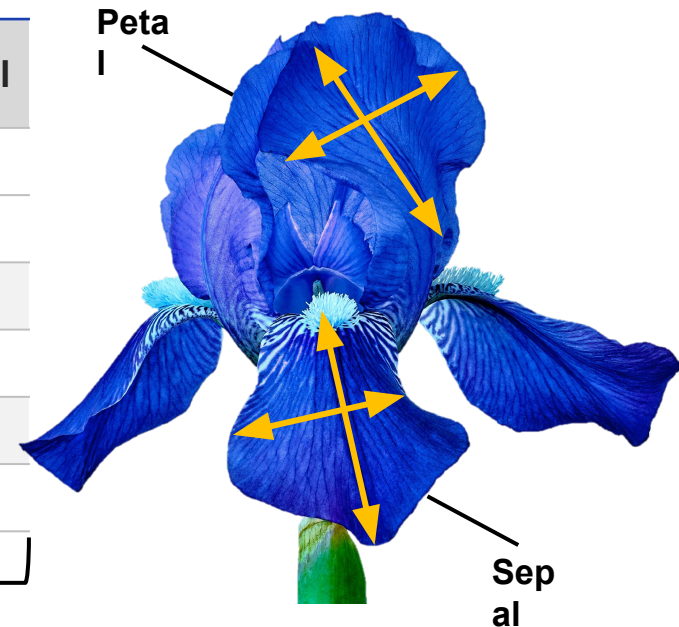
•Practicing

Sample
(Instance,
observation)

	Sepal length	Sepal width	Petal length	Petal width	Class label
1	5.1	3.5	1.4	0.2	Setosa
2	4.9	3.0	1.4	0.2	Setosa
...					
50	6.4	3.5	4.5	1.2	Versicolor
...					
150	5.9	3.0	5.0	1.8	Virginica

Feature
(Property,
measurement value,
dimension)

**Class
label**
(Target)



- Considerations in machine learning

- 1) Define the problem of the business and check for solutions or best alternative plans.
- 2) Check if it is possible to define it as a supervised or unsupervised problem.
- 3) Check which method to use for measuring model performance.
- 4) Check if the performance index is linked with the business objective and confirm if the project participants made an agreement regarding the performance.
 - In this practicing problem, define the problem as iris species (supervised) and suppose that the result is satisfactory if the performance predicts 85% or higher classification accuracy.

- Understanding the iris data

```
In [1]: from sklearn.datasets import load_iris  
data=load_iris()  
data.keys()
```

```
Out[1]: dict_keys(['data', 'target', 'frame', 'target_names', 'DESCR', 'feature_names', 'filename'])
```

- Domain knowledge of the iris data

- Data name: IRIS
- Number of data: 150
- Number of variables: 5
- Understanding variables

Sepal Length	Length information of the sepal
Sepal Width	Width information of the sepal
Petal Length	Length information of the petal
Petal Width	Width information of the petal
Species	Flower species, classified into setosa / versicolor / virginica

- Understanding the iris data

```
In [2]: print(data.DESCR)

.. _iris_dataset:

Iris plants dataset
-----

**Data Set Characteristics:**

: Number of Instances: 150 (50 in each of three classes)
: Number of Attributes: 4 numeric, predictive attributes and the class
: Attribute Information:
  - sepal length in cm
  - sepal width in cm
  - petal length in cm
  - petal width in cm
  - class:
    - Iris-Setosa
    - Iris-Versicolour
    - Iris-Virginica
```


- Understanding the iris data

```
:Summary Statistics:
```

```
=====
              Min  Max   Mean   SD   Class Correlation
=====
sepal length:  4.3  7.9   5.84   0.83    0.7826
sepal width:   2.0  4.4   3.05   0.43   -0.4194
petal length:   1.0  6.9   3.76   1.76    0.9490 (high!)
petal width:    0.1  2.5   1.20   0.76    0.9565 (high!)
=====
```

```
:Missing Attribute Values: None
```

```
:Class Distribution: 33.3% for each of 3 classes.
```

```
:Creator: R.A. Fisher
```

```
:Donor: Michael Marshall (MARSHALL%PLU@io.arc.nasa.gov)
```

```
:Date: July, 1988
```

The famous Iris database, first used by Sir R.A. Fisher. The dataset is taken from Fisher's paper. Note that it's the same as in R, but not as in the UCI Machine Learning Repository, which has two wrong data points.

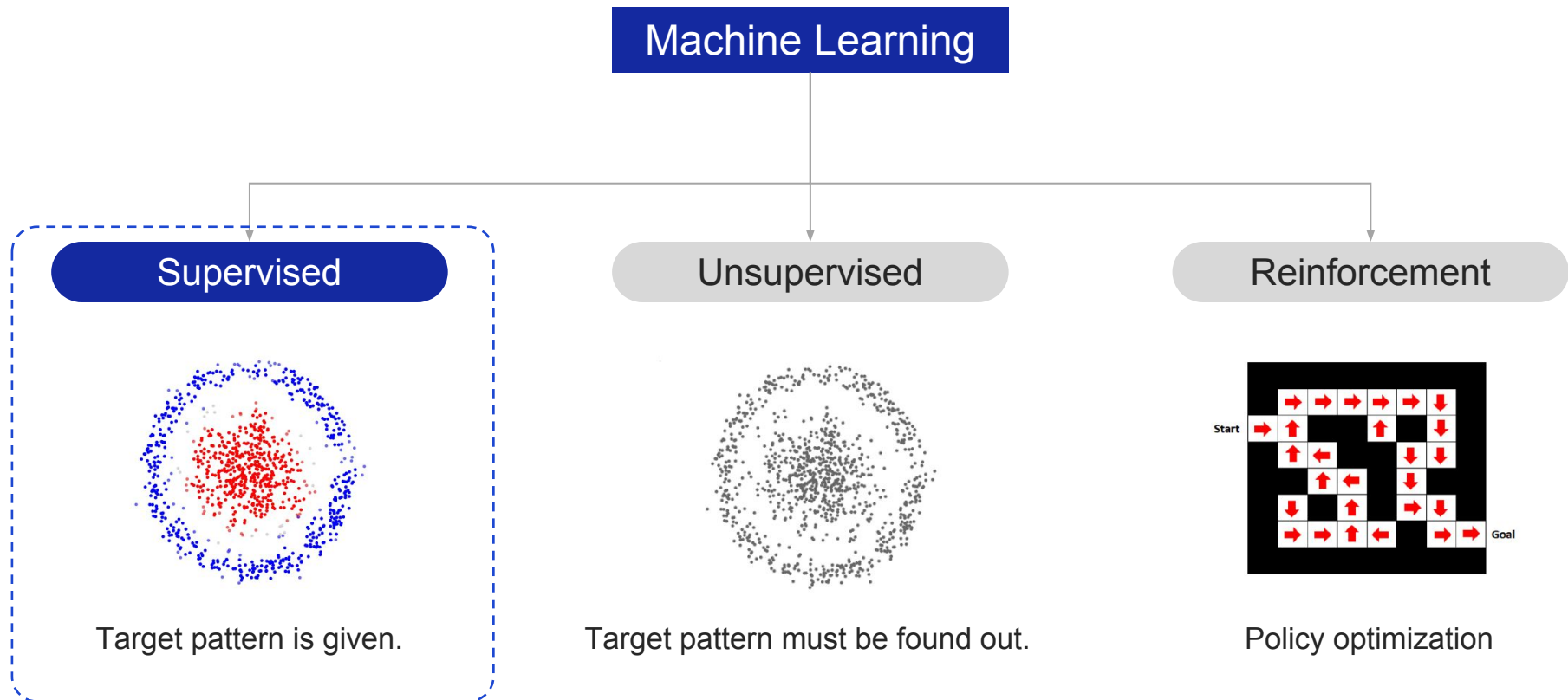
This is perhaps the best known database to be found in the pattern recognition literature. Fisher's paper is a classic in the field and is referenced frequently to this day. (See Duda & Hart, for example.) The data set contains 3 classes of 50 instances each, where each class refers to a type of iris plant. One class is linearly separable from the other 2; the latter are NOT linearly separable from each other.

Unit 2.

Application of Supervised Learning Model for Numerical Prediction

- 2.1. Training and Testing in Machine Learning
- 2.2. Linear Regression Basics
- 2.3. Linear Regression Diagnostics
- 2.4. Other Regression Types
- 2.5. Practicing the Supervised Learning Model for Numerical Prediction

Machine Learning Types

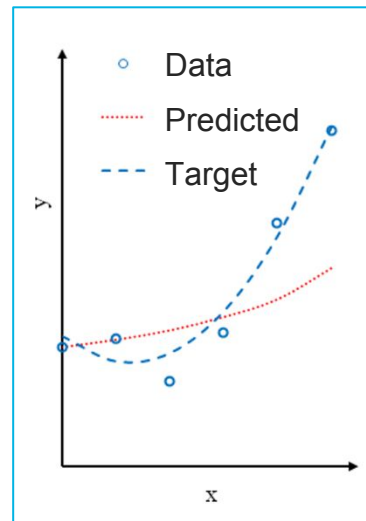


Training and Testing in Machine Learning

- What is learning (human)?
 - True learning is not just having a good memory.
 - Learned material must be put to test.
- Training and testing in machine learning
 - A trained model must be put to test to generalize.
 - Testing result is measured by errors.

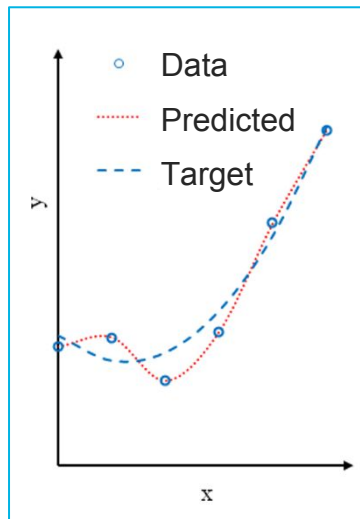
Error Types

- Bias error (Underfitting error)
 - Associated with simple/rigid/biased models.
 - Prediction cannot account for the detailed data pattern.
 - To lower this error type, increase the model complexity.



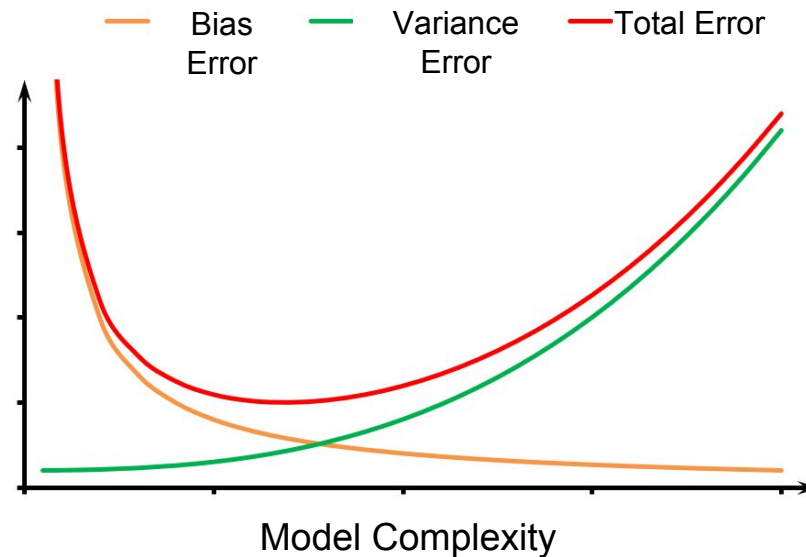
- Variance error (Overfitting error)

- Associated with models overly complex and sensitive to noise.
- Prediction performance is good while training but worsens when testing with a different dataset.
- To lower this error type, increase the amount of training data (*) or decrease the model complexity.



(*) By data augmentation method, for example.

- Total error

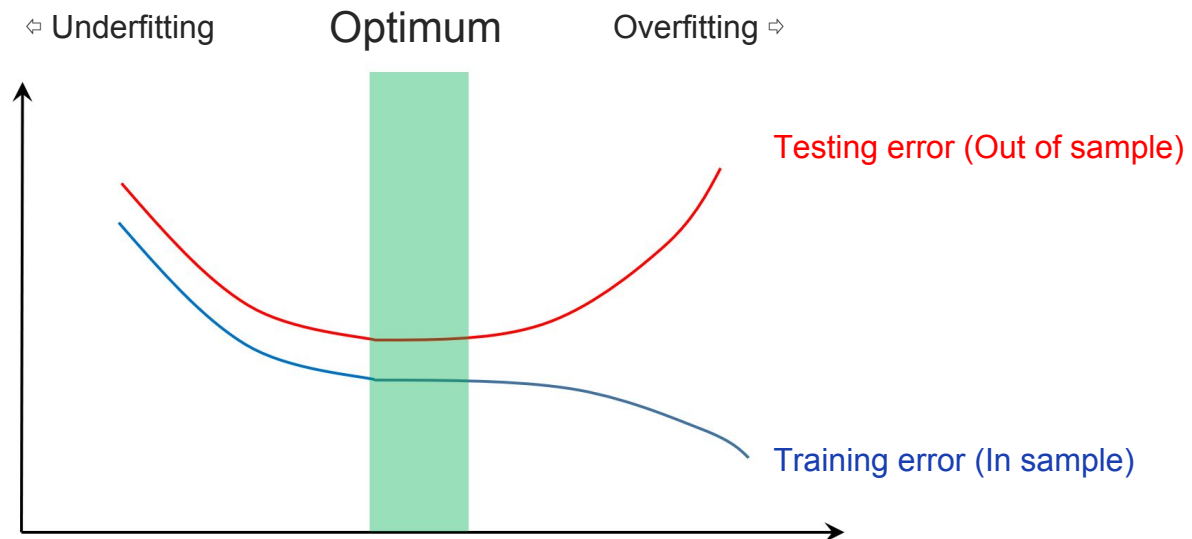


- The goal is to minimize the **Total error** = **Bias error** + **Variance error**.
- Just enough complexity is required to “optimize” the model.

Minimizing Errors

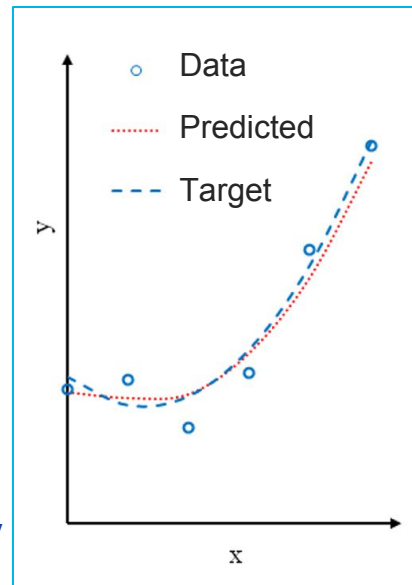
- Optimized machine learning model
 - Prediction performance should be good in both training and testing.
 - Given a machine learning algorithm, there is a model with just enough complexity (*).

(*) Complexity is usually controlled by the number of parameters.



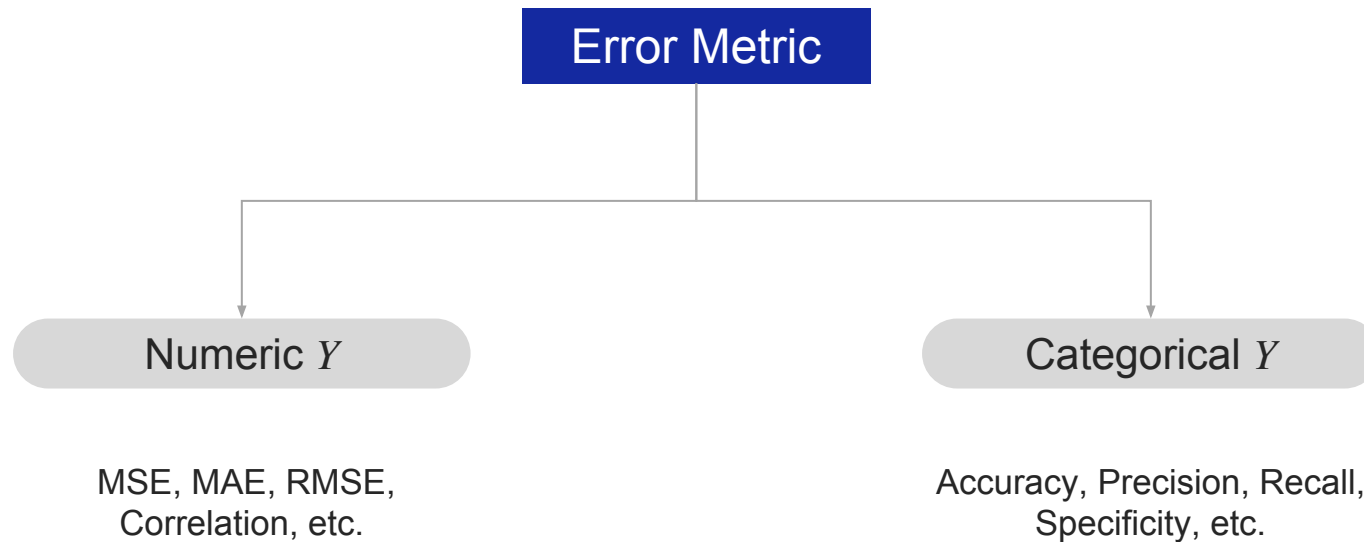
- Optimized machine learning model

- Prediction performance should be good in both training and testing.
- Given a machine learning algorithm, there is a model with just enough complexity (*).



(*) Complexity is usually controlled by the number of parameters.

Machine Learning Types

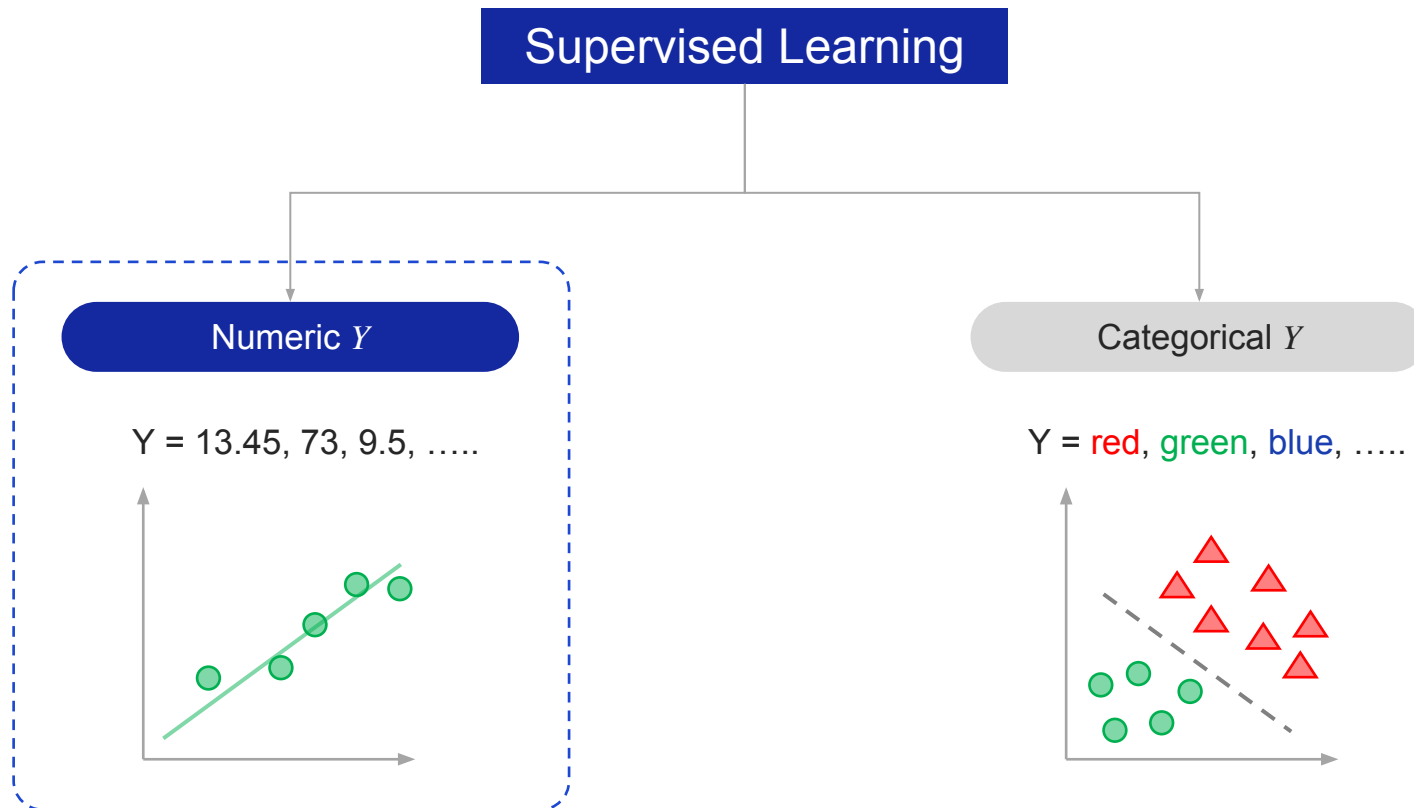


Unit 2.

Application of Supervised Learning Model for Numerical Prediction

- 2.1. Training and Testing in Machine Learning
- 2.2. Linear Regression Basics
- 2.3. Linear Regression Diagnostics
- 2.4. Other Regression Types
- 2.5. Practicing the Supervised Learning Model for Numerical Prediction

Linear Regression Basics



- About linear regression

- ▶ There is one or more explanatory variables: $X_1, X_2, \dots, X_k$
- ▶ There is one response variable: Y
- ▶ The variables X_i and Y are connected by a linear relation: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \epsilon$

- Purpose of linear regression

a) By modeling, find out which explanatory variables have the most impact on the response variable.

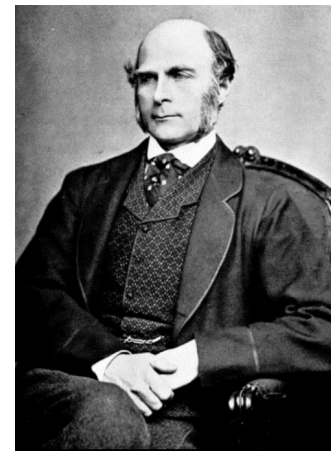
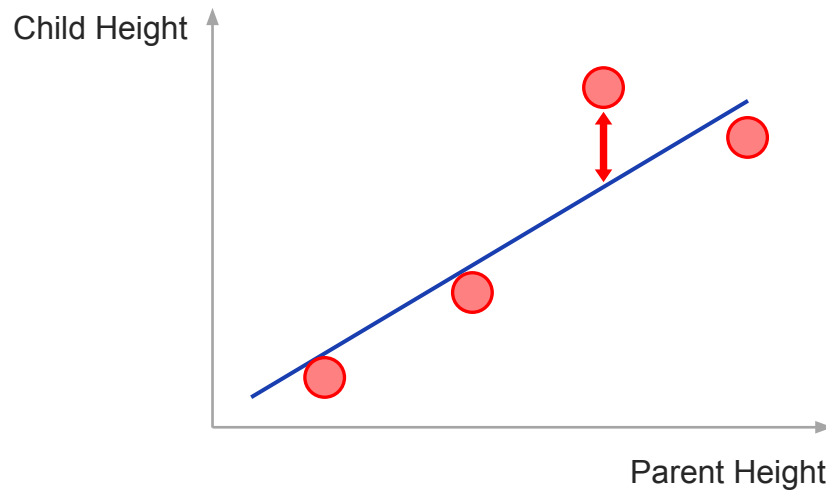
Ex If real estate price is the response variable Y , which are the most statistically meaningful explanatory variables? Area, location, age, distance to business center, etc.

b) Predict the response given the conditions for the explanatory variables.

Ex What is the price of a 10-year-old apartment with an area of 100 m^2 and located 3 km away from the business center? ← “predict” the value that is not open to the public yet.

- Historical background

- Term “regression” was coined by Francis Galton, 19th-century biologist.
- The heights of the descendants tend to regress towards the mean.



Francis Galton

- Pros

- Solid statistical and mathematical background
- Source of insights
- Fast training

- Cons

- Many assumptions: linearity, normality, independence of the explanatory variables, etc.
- Sensitive to outliers
- Prone to multi-collinearity

- Assumptions

- The response variable can be explained by a linear combination of the explanatory variables.
- There should be no multi-collinearity.
- Residuals should be normally distributed centered around 0.
- Residuals should be distributed with a constant variance.
- Residuals should be randomly distributed without a pattern.

} Residual analysis

- Linear model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \varepsilon$$

Response variable

Explanatory variables

- Variable values are given by data.

- Linear model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \varepsilon$$



Regression coefficients

- Regression coefficients are **model parameters**: capture the data patterns.

- Linear model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \varepsilon$$


$$E[\varepsilon] = 0$$

$$\text{Var}(\varepsilon) = \sigma_\varepsilon^2$$

- ▶ The error term ε should have zero mean and constant variance.

- Linear model

Ex

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \varepsilon$$

MPG N# of Cylinders H P Weight Auto or Manual

MPG can be **explained** by other variables.

- Interpreting the regression coefficients

$$\Delta Y = \beta_1 \Delta X_1 + \dots + \beta_i \Delta X_i + \dots + \beta_k \Delta X_k$$

- ▶ If $X_1, X_2, \dots, X_k$ change by $\Delta X_1, \Delta X_2, \dots, \Delta X_k$, then the change in Y is ΔY .

- Interpreting the regression coefficients

$$\Delta Y = \beta_1 \Delta X_1 + \cdots + \beta_i \Delta X_i + \cdots + \beta_k \Delta X_k$$


- ▶ β_i can be interpreted as the change in Y when the X_i is increased by a unit ($\Delta X_i=1$).

- Interpreting the regression coefficients

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \varepsilon$$

Intercept 0

- The intercept β_0 is the value of Y when **all the** $X_i = 0$. It's like a “baseline.”

- Interpreting the regression coefficients

Ex Wage survey of a company's employees

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

Wage
e

Experience
e

Qualification

- β_0 can be interpreted as the base wage when there is no experience or qualification.
- β_1 can be interpreted as the change in wage when the experience is increased by a unit.
- β_2 can be interpreted as the change in wage when the qualification is increased by a unit.

- Ordinary Least Squares (OLS) solution

$$y_j = \beta_0 + \beta_1 x_{j,1} + \beta_2 x_{j,2} + \cdots + \beta_K x_{j,k} + \varepsilon_j$$

$j \in [1, n]$

$n \gg k$
Over-determined!

- Now, we can write the linear relation in term of the actual data values.

- Ordinary Least Squares (OLS) solution

$$Y = X \beta + \varepsilon$$

- A compact notation using matrices

- Ordinary Least Squares (OLS) solution

$$\mathbf{Y} = \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

$$\mathbf{Y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

- A compact notation using matrices

- Ordinary Least Squares (OLS) solution

$$Y = \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

$$\mathbf{X} = \begin{bmatrix} 1 & x_{1,1} & \cdots & x_{1,k} \\ 1 & x_{2,1} & \cdots & x_{2,k} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n,1} & \cdots & x_{n,k} \end{bmatrix}$$

- A compact notation using matrices

- Ordinary Least Squares (OLS) solution

$$Y = X \beta + \varepsilon$$


$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}$$

- A compact notation using matrices

- Ordinary Least Squares (OLS) solution

$$Y = X \beta + \epsilon$$

$$\epsilon = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$


- A compact notation using matrices

- Ordinary Least Squares (OLS) solution

- As we have an overdetermined system of linear equations, the exact solution does not exist.
- We can minimize $|\boldsymbol{\varepsilon}|^2$ and get the “best” solution $\boldsymbol{\beta}$.
- The minimization condition for $|\boldsymbol{\varepsilon}|^2$ is given by the derivative:

$$\frac{d|\boldsymbol{\varepsilon}|^2}{d\boldsymbol{\beta}} = 0$$

- Ordinary Least Squares (OLS) solution

- The minimization condition for $|\boldsymbol{\varepsilon}|^2$ is given by the derivative that can be expanded as following:

$$\begin{aligned}\frac{d|\boldsymbol{\varepsilon}|^2}{d\boldsymbol{\beta}} &= \frac{d|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}|^2}{d\boldsymbol{\beta}} \\ &= \frac{d\{(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})\}}{d\boldsymbol{\beta}} \\ &= \frac{d\{\mathbf{Y}^T \mathbf{Y} - \boldsymbol{\beta}^T \mathbf{X}^T \mathbf{Y} - \mathbf{Y}^T \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\beta}^T \mathbf{X}^T \mathbf{X} \boldsymbol{\beta}\}}{d\boldsymbol{\beta}} \\ &= -2\mathbf{X}^T \mathbf{Y} + 2\mathbf{X}^T \mathbf{X} \boldsymbol{\beta} = 0\end{aligned}$$

- Ordinary Least Squares (OLS) solution

- The solution β from the previous slide is given by the following expression:

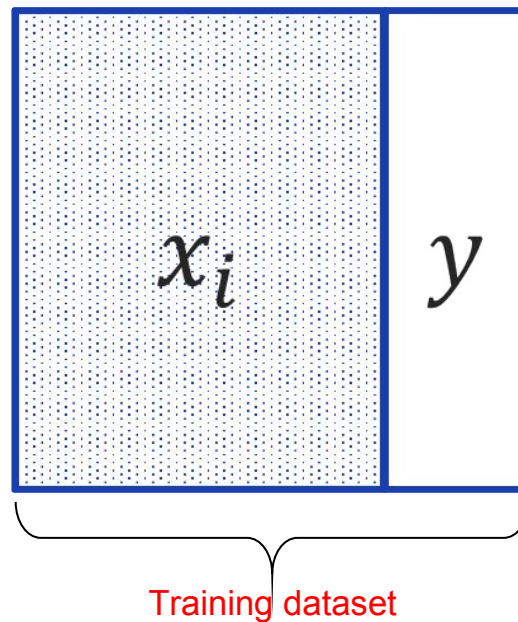
$$\beta = [(X^T X)^{-1} X^T] Y$$

Pseudo-inverse

- The matrix expression within the square parentheses is called “pseudo-inverse.”

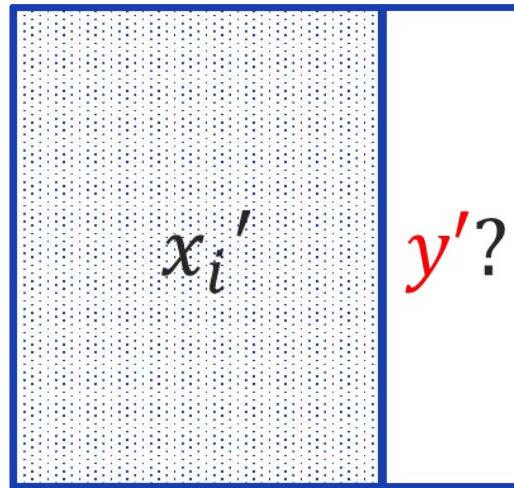
- Regression training and prediction (testing)

1) Training step: use the training dataset and get a set of model parameters $\{\beta_i\}$.



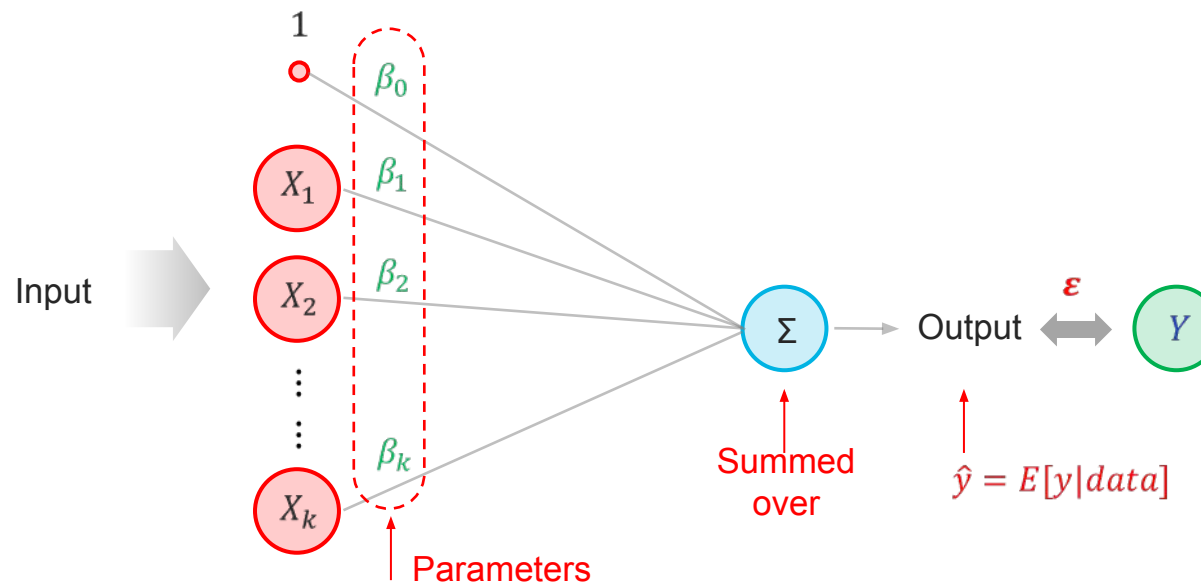
- Regression training and prediction (testing)

2) Prediction step: when a new set of $\{x_i'\}$ is given, calculate the value of y' , which was unknown.



- The predicted value of y' is denoted as $\hat{y}$, which is a conditional expectation $\hat{y} = E[y | \text{data}]$.
- Given the values $x_1', x_2', \dots, x_k'$, calculate $\hat{y} = \beta_0 + \beta_1 x_1' + \beta_2 x_2' + \dots + \beta_K x_k'$.

- Regression training and prediction (testing)



- Schematic view of the prediction step
- Notice some disagreement between the predicted $\hat{y}$ and the true y .

- Error metrics

$$MSE = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}$$

$$RMSE = \sqrt{MSE}$$

$$MAE = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{n}$$

$$MAPE = \frac{100}{n} \times \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

- Confidence interval (95%) when there is only one explanatory variable

$$[\hat{y} - \text{quantile}(0.975, n - 2) \sigma_{\hat{y}}, \hat{y} + \text{quantile}(0.975, n - 2) \sigma_{\hat{y}}]$$

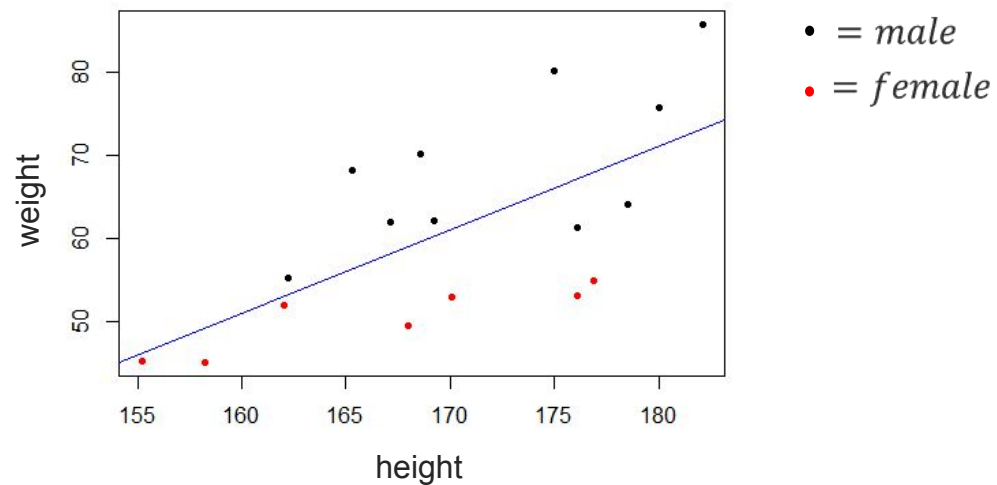
with

$$\sigma_{\hat{y}} = RMSE \times \sqrt{\left(\frac{1}{n} + \frac{(x' - \bar{x})^2}{\sum (x - \bar{x})^2} \right)}$$

- $\text{quantile}(\alpha, k)$ is Student-t quantile of degree of freedom k .

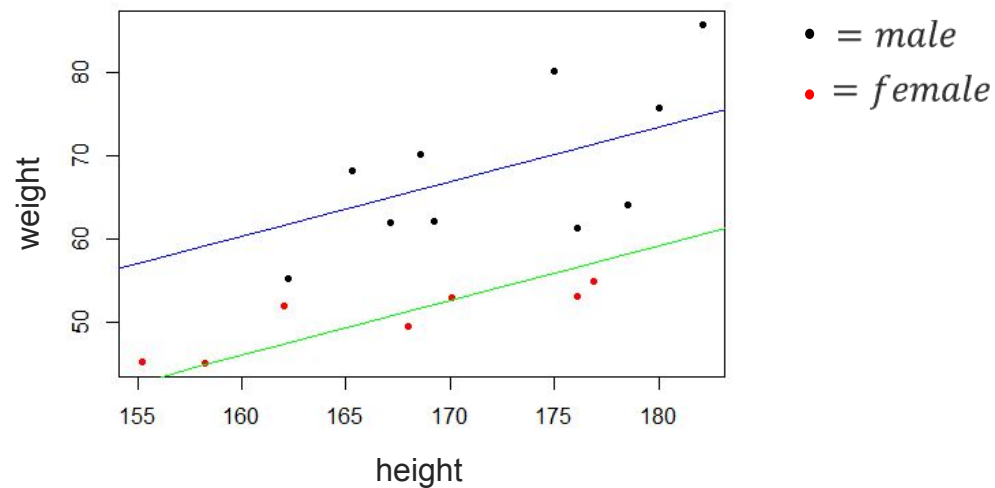
- Role of categorical variables

- Without categorical variable: $\text{weight} \sim \text{height}$



- Role of categorical variables

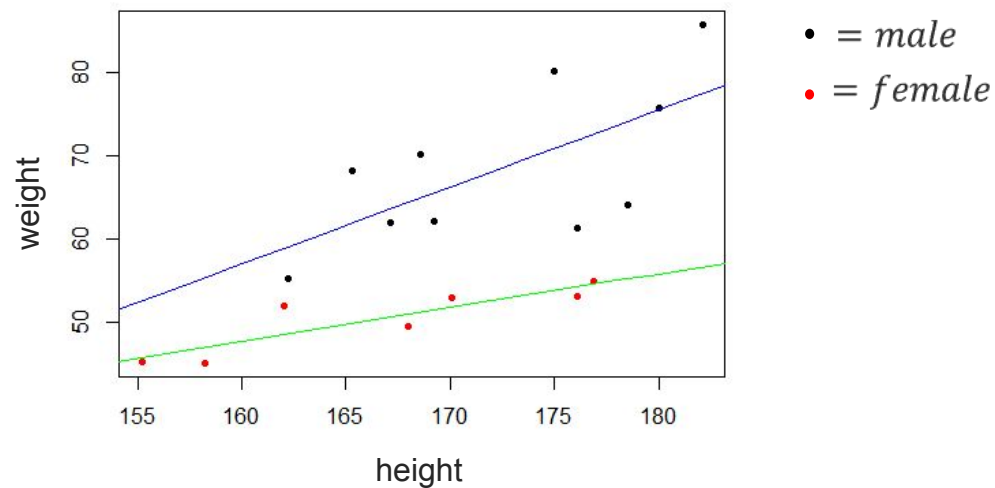
- With a categorical variable **gender**: $\text{weight} \sim \text{height} + \text{gender}$



- The categorical variable must be turned into dummy variable(s) first.
- Effectively raises or lowers the intercept.

- Role of categorical variables

- With a categorical variable **gender** that “interacts”: $\text{weight} \sim \text{height} + \text{gender} + \text{height} \times \text{gender}$



- Both the intercept and slope are dependent on the categorical variable.
- Further improves the error metrics.

Coding Exercise #0301



Follow practice steps on 'ex_0301.ipynb' file.

Unit 2.

Application of Supervised Learning Model for Numerical Prediction

- 2.1. Training and Testing in Machine Learning
- 2.2. Linear Regression Basics
- 2.3. Linear Regression Diagnostics
- 2.4. Other Regression Types
- 2.5. Practicing the Supervised Learning Model for Numerical Prediction

Linear Regression Diagnostics

- Linear regression diagnostic methods:
 - 1) Error metrics: MSE, RMSE, MAE, MAPE, etc.
 - 2) Coefficient of determination or “r-squared” R^2
 - 3) F-test for overall significance of the linear model
 - 4) t-test for significance of individual regression coefficients
 - 5) Correlation between Y and $\hat{Y}$
 - 6) Variance inflation factor (VIF)
- Modelling: optimization of the information criteria: AIC or BIC
- Residual and leverage analysis

- Error metrics

$$MSE = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}$$

$$RMSE = \sqrt{MSE}$$

$$MAE = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{n}$$

$$MAPE = \frac{100}{n} \times \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

- The smaller, the better!

- Coefficient of determination or R^2

$$R^2 = 1 - \frac{SSE}{SST}$$

$$\text{with } SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \text{ and } SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

- R^2 is bounded above and below: $0 < R^2 < 1$.
- R^2 close to one means that the response variable is well explained.
- As more explanatory variables are added, R^2 tends to increase spuriously: adjusted R^2 introduced.
- If there is only one explanatory variable X , then:

$$R^2 = \text{Cor}(X, Y)^2$$

- F-test for overall significance of the linear model

- Null hypothesis $H_0: \beta_1 = \beta_2 = \dots = \beta_K = 0$
- Alternate hypothesis H_1 : At least a β_i is non zero.

⇨ F-test statistic =
$$\frac{\text{Variance that can be explained}}{\text{Variance that cannot be explained}}$$

⇨ If the p-value is below a reference (say 0.05), then H_0 is rejected in favor of H_1 .

In this case, the linear model has an overall significance.

- t-test for significance of the individual regression coefficients

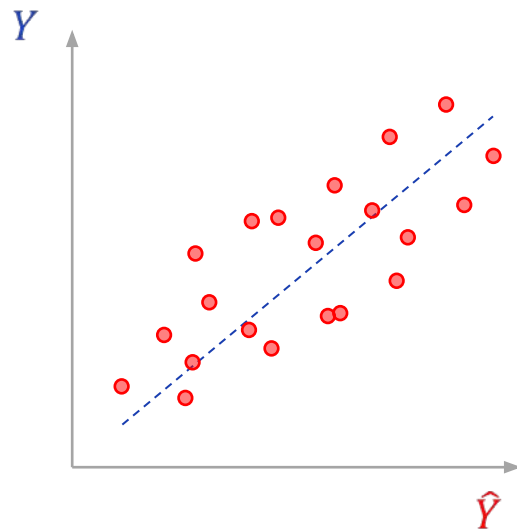
- Null hypothesis $H0: \beta_i=0$
- Alternate hypothesis $H1: \beta_i \neq 0$

⇨ t-test statistic =
$$\frac{\hat{\beta}_i}{\text{Standard error of } \beta_i}, \text{ where } \hat{\beta}_i = \text{estimated coefficient.}$$

⇨ If the p-value is below a reference (say 0.05), then $H0$ is rejected in favor of $H1$.

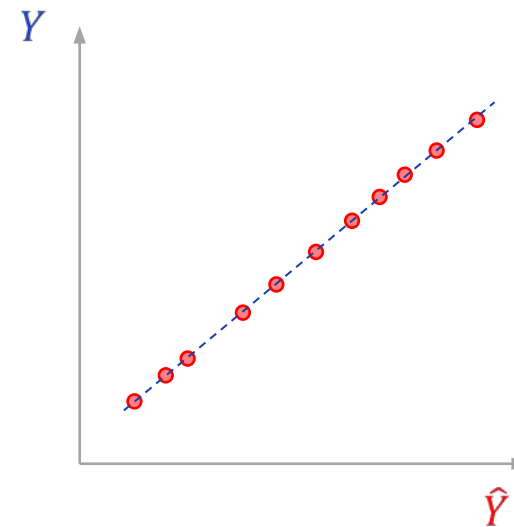
In this case, inclusion of the explanatory variable X_i is justified.

| Correlation between Y and $\hat{Y}$



Weak positive correlation

<



Strong positive correlation

- Variance inflation factor (VIF)

- Can measure the “seriousness” of multi-collinearity.
- In general, $VIF > 10$ is considered serious. However, this reference point is rather subjective.
- In case serious multi-collinearity is detected, model simplification is required.

- Variance inflation factor (VIF)

- ▶ VIF is calculated individually for each explanatory variable X_i .

1) Place X_i as the response and then regress using the expression:

$$X_i = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_{i-1} X_{i-1} + \beta_{i+1} X_{i+1} + \cdots + \varepsilon$$

2) Using the corresponding R_i^2 calculate VIF_i :

$$VIF_i = \frac{1}{1 - R_i^2}$$

- Information criteria and modelling

- Akaike information criteria (AIC) with p =number of parameters:

$$AIC = -2 \frac{\text{Log likelihood}}{n} + 2 \frac{p}{n}$$

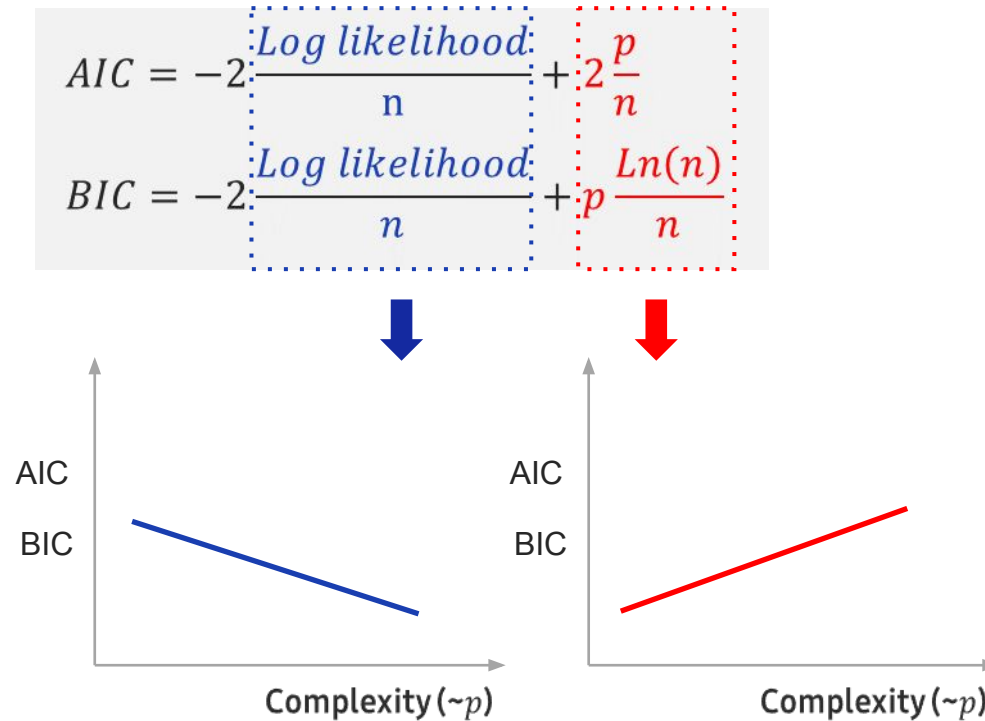
- Bayes information criteria (BIC) with p =number of parameters:

$$BIC = -2 \frac{\text{Log likelihood}}{n} + p \frac{\text{Ln}(n)}{n}$$

- In the expressions above, we have:

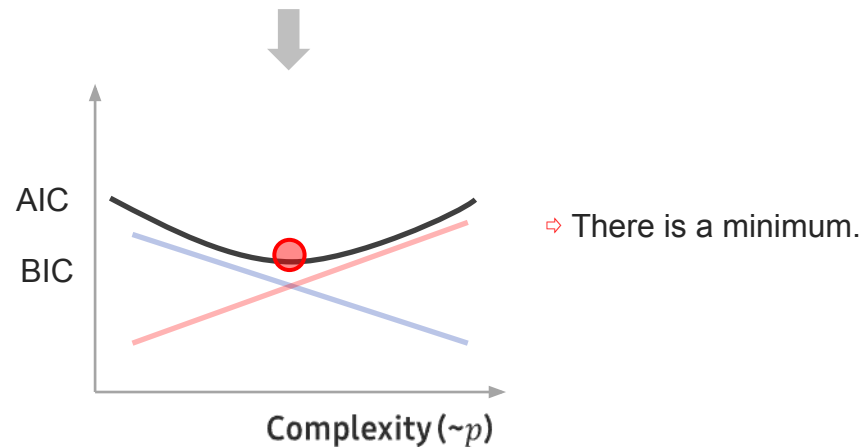
$$\text{Log Likelihood} = -\frac{n}{2} \left(1 + \text{Ln}(2\pi) + \text{Ln} \left(\frac{SSE}{n} \right) \right)$$

- Information criteria and modelling

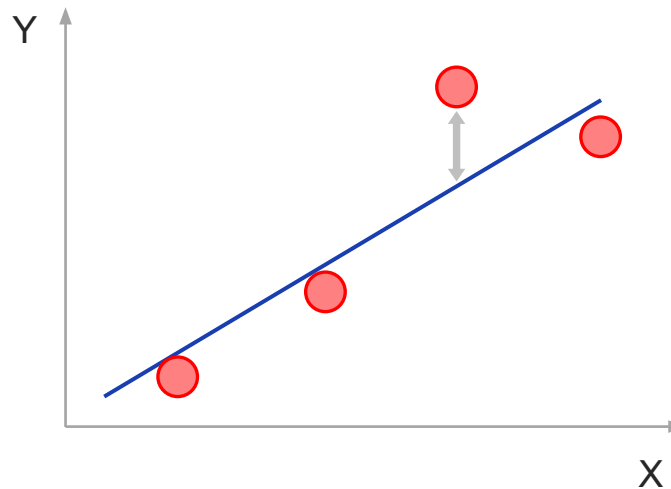


- Information criteria and modelling

$$\begin{aligned} AIC &= -2 \frac{\text{Log likelihood}}{n} + 2 \frac{p}{n} \\ BIC &= -2 \frac{\text{Log likelihood}}{n} + p \frac{\ln(n)}{n} \end{aligned}$$



- Residual analysis



- ▶ Residual is the difference between the predicted $\hat{y}$ and the real y .
- ▶ We can easily detect outliers in Y that deviate substantially from the main trend.

- Residual analysis

- Reasons for residual analysis:

- 1) To detect outliers in Y .
- 2) To verify the assumptions of linear regression.

Residuals should be normally distributed centered around 0.

Residuals should be distributed with a constant variance.

Residuals should be randomly distributed without a pattern.

- Leverage analysis

- Leverage of the i-th observation

$$\text{Leverage} = H_{ii} \quad , \quad \text{where } \tilde{H} = \tilde{X}(\tilde{X}^t \tilde{X})^{-1} \tilde{X}^t$$

- There is a “sum rule.”

$$\sum_{i=1}^n H_{ii} = p \quad \left\{ \begin{array}{l} \text{Mean leverage} \cong \frac{p}{n} \\ \text{Large leverage} > \frac{p}{n} \\ \text{Small leverage} < \frac{p}{n} \end{array} \right.$$

- Leverage tells how distant X is from the center => Detection of outliers in X .

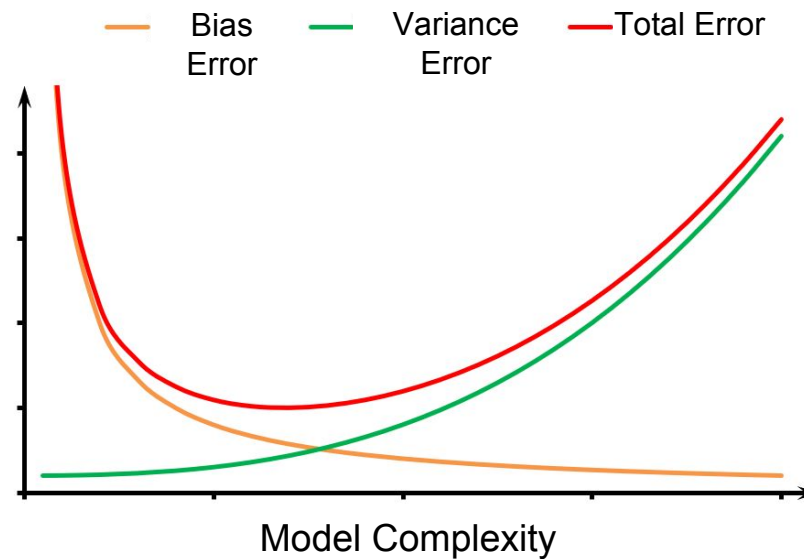
Unit 2.

Application of Supervised Learning Model for Numerical Prediction

- 2.1. Training and Testing in Machine Learning
- 2.2. Linear Regression Basics
- 2.3. Linear Regression Diagnostics
- 2.4. Other Regression Types
- 2.5. Practicing the Supervised Learning Model for Numerical Prediction

Regularized Regression

- Bias-Variance trade off



- Tradeoff relation between the **Bias error** and the **Variance error**.
- The goal should be to minimize the **Total error** = **Bias error** + **Variance error**.

- Ridge regression

- Useful when the usual linear regression overfits (bias error << variance error).
- We remember that the OLS solution consists in minimizing $|\varepsilon|^2$.
- In the Ridge regression, we minimize the following “loss function”:

$$L = |\varepsilon|^2 + \lambda \sum_{i=0}^k \beta_i^2$$

- In the loss function, $\lambda \sum_{i=0}^k \beta_i^2$ is included as penalty: “L2 regularization.”

- Ridge regression

- ▶ Positive and larger λ further constrains the coefficients β_i , decreasing the variance (overfitting) error.

$$L = |\varepsilon|^2 + \lambda \sum_{i=0}^k \beta_i^2$$

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \varepsilon$$

- ▶ However, too large λ can make the model too “biased.”
- ▶ Even when λ is large, the coefficients β_i never become exactly equal to zero.

- Lasso regression

- Useful when the usual linear regression overfits (bias error << variance error).
- In the Lasso regression, we minimize the following “loss function”:

$$L = |\boldsymbol{\varepsilon}|^2 + \lambda \sum_{i=0}^k |\beta_i|$$

- In the loss function, $\lambda \sum_{i=0}^k |\beta_i|$ is included as penalty: “L1 regularization.”
- Positive and larger λ further constraints the coefficients β_i decreasing the variance (overfitting) error.
- However, too large λ can make the model too “biased.”
- When λ is large, the coefficients β_i can become exactly equal to zero.

Polynomial Regression

- Polynomial regression

- Useful when the usual linear regression underfits (bias error >> variance error).
- We can model the relationship between X and Y using the polynomials:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon$$

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \varepsilon$$

- We notice that there is **only one** explanatory variable X .
- When the polynomial power is too high, overfitting may incur.

Poisson Regression

- Poisson regression

- Useful when we would like to model the response Y that represents counts or frequencies.

$$\text{Log}(\lambda) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_K X_K + \varepsilon$$

- We are assuming that Y follows the Poisson distribution.

$$P(y) = \frac{\lambda^y e^{-\lambda}}{y!}$$

a) Mean = λ

b) Variance = λ

c) Standard deviation = $\sqrt{\lambda}$

Unit 2.

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Practicing the Supervised Learning Model for Numerical Prediction

- Import modules.

```
In [ ]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import os
from sklearn.linear_model import LinearRegression
%matplotlib inline
```

Coding Exercise #0302



Follow practice steps on 'ex_0302.ipynb' file.

Coding Exercise #0303



Follow practice steps on 'ex_0303.ipynb' file.

Coding Exercise #0304



Follow practice steps on 'ex_0304.ipynb' file.

Coding Exercise #0305



Follow practice steps on 'ex_0305.ipynb' file.

Coding Exercise #0306



Follow practice steps on 'ex_0306.ipynb' file.



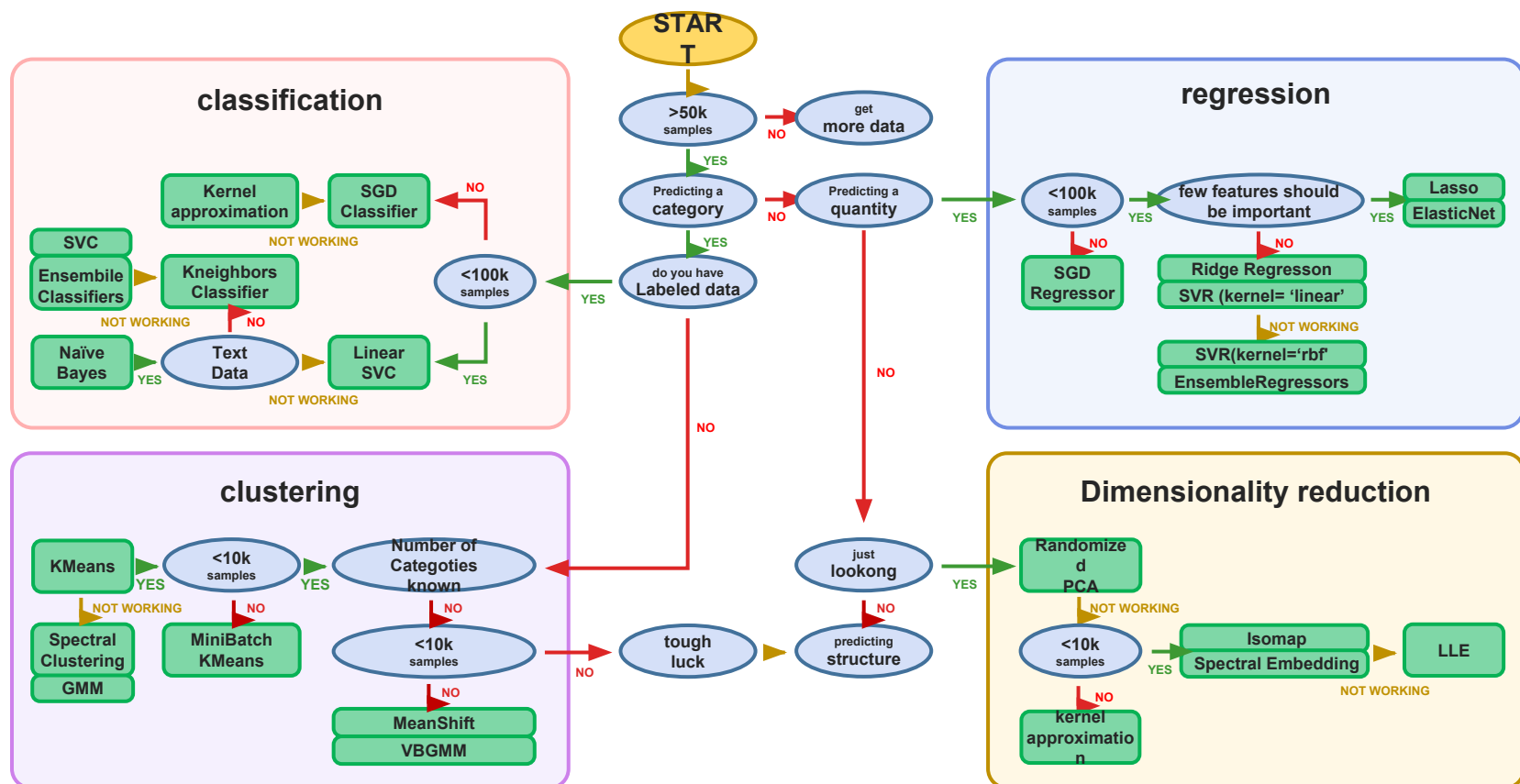
Unit 3.

Application of Supervised Learning Model for Classification

- 3.1. Training and Testing in Machine Learning
- 3.2. Logistic Regression Basics
- 3.3. Logistic Regression Performance Metrics

Training and Testing in Machine Learning

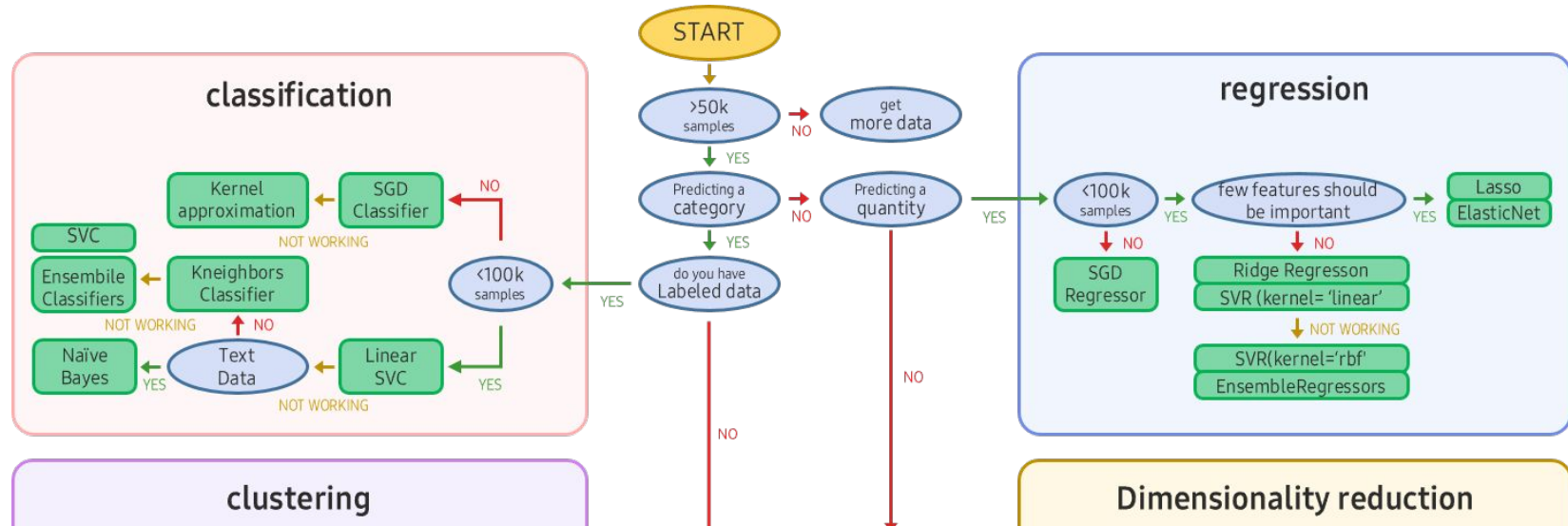
Select a machine learning algorithm suitable for the data.



■ Select a machine learning algorithm suitable for the data.

- Supervised learning is machine learning tasks that try to predict or classify the values of objective variables (or response variables) in unseen data as the training dataset has the objective variables (or response variables) to be predicted (Y).
- It is called supervised learning because the algorithm is told to predict a certain target.
- Supervised learning is mainly classified into classification and regression problems depending on the types of objective variables (or response variables).
- In other words, classification is applied if the objective variables (or response variables) are discrete or nominal, such as 'male/female,' 'spam mail/ham mail,' 'positive/neutral/negative,' 'Seoul/Busan/Gyeonggi/Gangwon,' etc. On the other hand, regression is applied if the objective variables (or response variables) are numerical, such as 0~10, -500~500, -3.5~3.5, $-\infty \sim \infty$, etc.
- When the given data does not have classification marks or objective variables (or response variables), and if it's not trying to predict target value, unsupervised learning is applied.
- Unsupervised learning is classified into clustering, association, dimension reduction, and others depending on the analysis purpose and methods. Further details will be explained in other chapters.

- Select a machine learning algorithm suitable for the data.



- As shown in the figure above, neural networks can be applied if there are correct answers and it is for classification purposes or has a lot of data. If not, it is possible to use a decision tree or SVC algorithm.
- Also, the Naïve Bayes algorithm can be used if the data is in text.
- The KNN method is used if it's not text data. It is also possible to use the ensemble method with a better performance.

| Machine learning for classification

- Classification is used if the objective variables (or response variables) can be classified into certain categories, such as discrete or nominal type. It is the most common and frequently found in machine learning-based data analysis.

- Machine learning algorithms for classification can be applied to extensive daily and business problems.

Ex Frequently used examples

- (1) Classifying spam mails
 - (2) Prediction of corporate bankruptcy
 - (3) Prediction of customer loss
 - (4) Classifying customers' credit rating
 - (5) Prediction of occurrence of a certain disease (e.g. cancer, heart disease, etc.)
 - (6) Prediction of customer reaction to a specific marketing event
 - (7) Prediction of customer's purchase
- The examples provided above are extremely few real-life applications; there are infinite fields of business that classification type machine learning is applied. This type of machine learning has been continuously applied with new R&D and business methods.

I Types of machine learning algorithm for classification (1/2)

- There are many different classification-type machine learning algorithms. Some of those can be used for regression problems. The following table provides descriptions of frequently used classification methods.

Type	Concept	Note
K-Nearest Neighbor	A classification method that uses the majority rule on the closest k objective variable values (or response variables) based on the distance between data coordination other than a certain data.	Lazy Learning
Naïve Bayes	A method based on Bayes' theorem that makes classification towards the higher probability by expressing the conditional probability of objective variables (or response variables) as a multiplication of the prior probability and likelihood function. All observed values are assumed to occur statistically independently from other observed values. (Referred to as a Naïve model since the assumption is given without confidence.)	Provability model (Bayes' theorem based conditional probability)
Logistic Regression	A method to estimate the probability of objective variables through the maximum likelihood estimation by assuming that the probability of the objective variable value being in a certain category is in the logistic function shape when the explanatory value is given.	Provability model (maximum likelihood estimation)
Decision Tree	A method to create classification rules by splitting the branches towards lower impurity or entropy in the order of variables most associated with objective variables.	Divide & Conquer

| Types of machine learning algorithm for classification (2/2)

- There are many different classification-type machine learning algorithms. Some of those can be used for regression problems. The following table provides descriptions of frequently used classification methods.

Type	Concept	Note
Artificial Neural Network	A method inspired by the human neuron network. Comprised of the input node, hidden node, and output node, this analysis method is used to solve complicated classification or black box value prediction problems.	Black box test
Support Vector Machine	A method to classify certain data by finding a plane that maximizes the margin between data in different categories.	Linear and non-linear (Kernel trick)
Random Forest	An ensemble method to decide the final classification result by making various decision trees based on given data and aggregating the predicted results of each decision tree through voting.	Ensemble model



Unit 3.

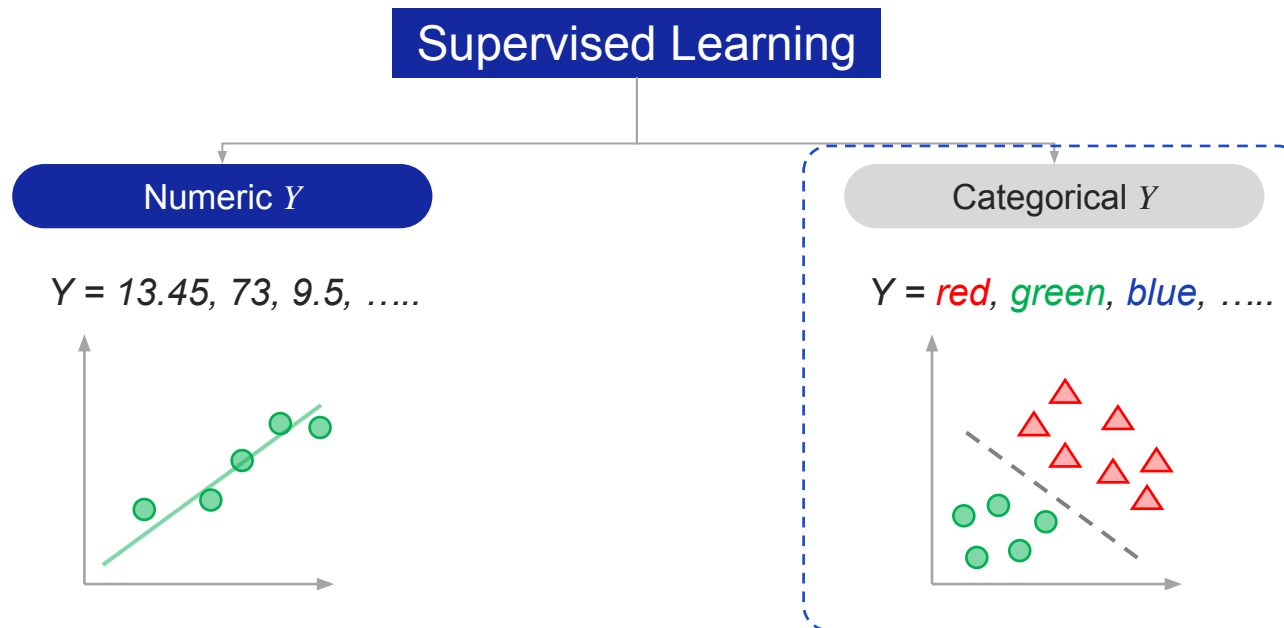
Application of Supervised Learning Model for Classification

- 3.1. Training and Testing in Machine Learning
- 3.2. Logistic Regression Basics
- 3.3. Logistic Regression Performance Metrics

Logistic Regression Basics

- What is logistic regression analysis?

- ▶ In contrast to general linear regression analysis used for numerical prediction, logistic regression analysis is used to classify the category of objective variables (y) to be predicted. In other words, what is being predicted is not the y value which is an objective variable. It is $P(Y=i)$, which is the probability of the objective variable y becoming a certain category (i).



- About linear regression

- ▶ There is one or more explanatory variables: $X_1, X_2, \dots, X_k$
- ▶ There is one response variable: Y
- ▶ The response variable Y is binary where possible values are $\{0,1\}$, $\{False, True\}$, $\{No, Yes\}$, etc.
- ▶ One of the most basic classification algorithms.

- Pros

- Simple and relatively easy to implement
- Source of intuitive insights
- Fast training

- Cons

- Not among the most accurate classification algorithms
- Assumes that the explanatory variables are independent without multi-collinearity

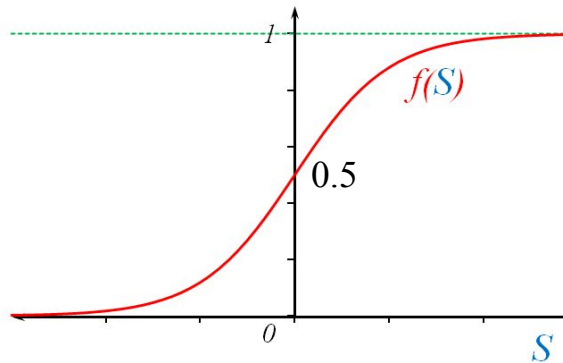
- About logistic regression

- ▶ The linear combinations of variables X_i is the so-called “Logit” denoted here as S .

$$S = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k$$

- ▶ The conditional probability of Y being equal to 1 is denoted as $P(Y = 1|\{x_i\})$.
- ▶ “Sigmoid” or “Logistic” function connects the probability with the logit.

$$f(S) = \frac{e^S}{1 + e^S}$$



- About logistic regression

- If $p = \text{probability of } Y=1$, we can define:

$$\text{Odds} = \frac{p}{1-p}$$

$$\text{Logit} = \text{Log}(\text{Odds}) = \text{Log}\left(\frac{p}{1-p}\right)$$

- The logistic function is the inverse of the logit (and vice versa):

$$S = \text{Log}\left(\frac{p}{1-p}\right)$$



$$p = \frac{e^S}{1 + e^S}$$

- About logistic regression

- ▶ For logistic regression, the categories of objective variable (Y) to be predicted are 0 and 1 (binomial logistic regression model). The possibility of the objective variable Y category becoming 1 is expressed as $p(Y=1) = P(Y)$. Start by expressing the description as a regression equation. The left side of the equation is commonly referred to as 'odds.'
- ▶ The 'odds' signify dividend rate, which categorizes into success/failure probabilities. If the success probability is high, it becomes 1; if it is low, it becomes 0. In other words, it is a success rate. Adding a log to odds becomes logit, and this function is logistic.

$$\frac{P(Y)}{1 - P(Y)} = \exp(\beta_0 + \beta_1 X)$$

- ▶ The left side of the equation is a ratio of probability; the right side is an exponential function that has a $(0, \infty)$ range. Add a log on both sides of the equation to give a range with $(-\infty, \infty)$ values on both.

$$\log\left(\frac{P(Y)}{1 - P(Y)}\right) = \beta_0 + \beta_1 X$$

- ▶ When looking at the equation in detail, the $\beta_0 + \beta_1 X$ on the right side is a linear model with a range $(-\infty, \infty)$; the left side also has a range $(-\infty, \infty)$. The $\log(P(Y))/(1-P(Y))$ on the left side of the equation is called the logit function.
- ▶ Or add 'exp' to both sides of the equation and arrange it regarding P(Y) to get the following equation.

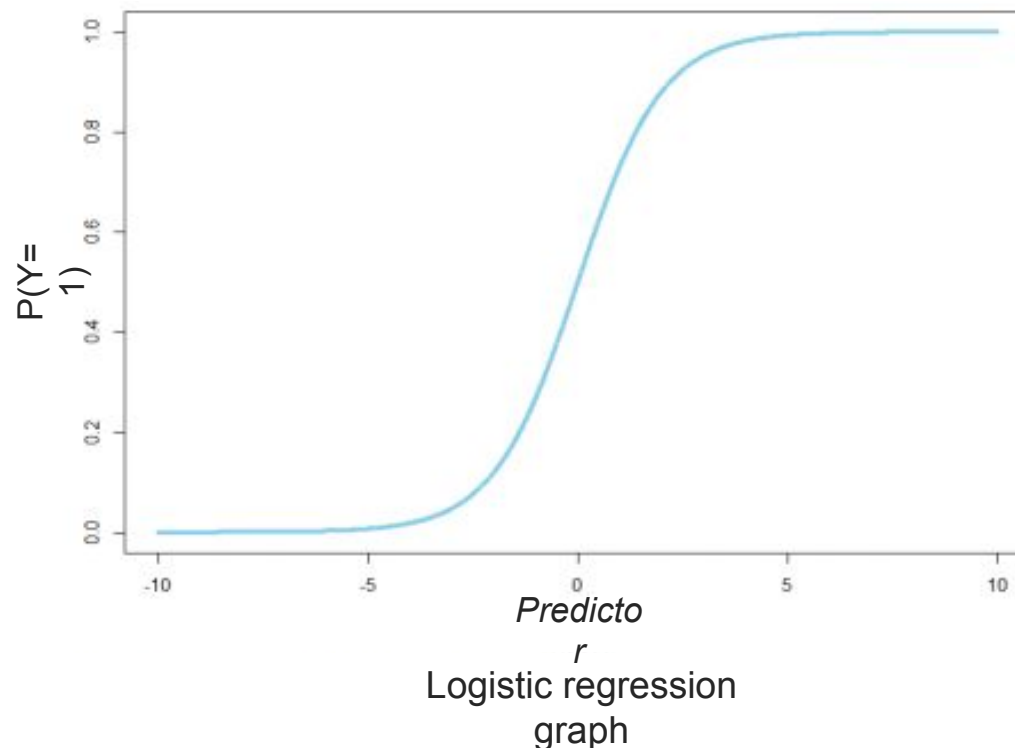
- About logistic regression

$$P(Y = 1) = \frac{e^{(\beta_0 + \beta_1 X)}}{1 + e^{(\beta_0 + \beta_1 X)}} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$$

- ▶ To sum up, logistic regression analysis is an algorithm that establishes a model by using the logistic function from the above equation and predicting parameters β_0, β_1 from the probability $P(Y=1) = P(Y)$ where objective variable Y having categorical value 1 from the training data. Maximum likelihood estimation is generally used for predicting parameters β_0 and β_1 . This is an analytical method that changes the formula. Since the direct calculation is difficult, estimation is done by giving a certain initial value and performing repeated calculations to adjust the value as a numerical calculation.

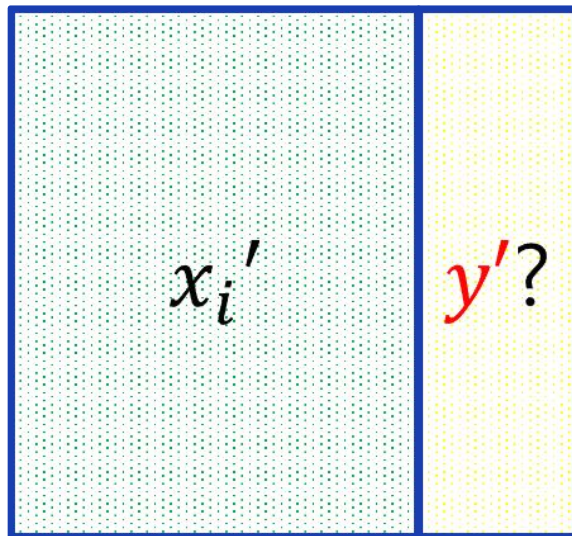
- About logistic regression

- ▶ The following figure shows a graph of the logistic function. As shown in the graph, the range of the X-axis is $-\infty$ to ∞ . The Y-axis is the probability of objective variable Y occurrence, which ranges between 0 and 1. Thus, the function $P(Y=1)$ shows an S-shaped curve from 0 to 1 as the x value increases.



- Logistic regression training and prediction (testing)

2) Prediction step: when a new set of $\{x_i'\}$ is given, calculate the value of y' , which was unknown.

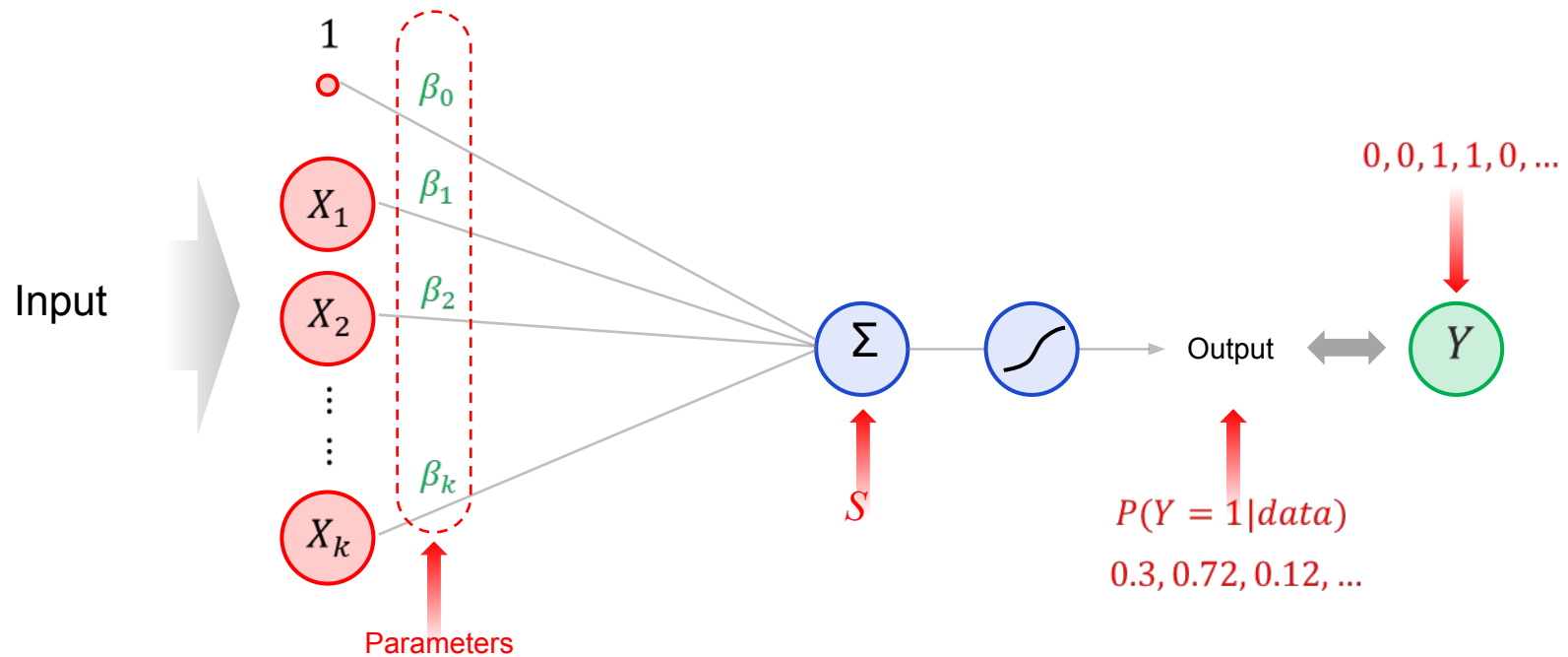


Compare the conditional probability with a cutoff.

$$P(Y' = 1|\{x_i'\}) > \text{Cutoff ?}$$

$$\Rightarrow \hat{y} = 1 \text{ or } 0$$

- Logistic regression training and prediction (testing)



- Training by gradient descent algorithm

- ▶ We can get the parameter set $\{\beta_i\}$ by minimizing a target function $L(\beta)$.

We get to minimize the difference between the real Y and the predicted $\hat{Y}$.

- ▶ We can think of $L(\beta)$ as a “loss” or “cost.”
- ▶ $L(\beta)$ is a function of the parameter set $\{\beta_i\}$, which can be represented by a vector β .

- Training by gradient descent algorithm

- ▶ $L(\beta)$ is the **negative** of logarithmic likelihood defined as following:

$$L(\beta) = \sum_{i=1}^n \text{Log} \left(1 + e^{-y_i \beta^T x_i} \right)$$

- ▶ In the above relation, x_i and y_i represent values given by the training dataset.
- ▶ Here, we assumed the conversion from $y_i = \{0,1\}$ to $y_i = \{-1, +1\}$.

- Training by gradient descent algorithm

- ▶ $L(\beta)$ is the **negative** of logarithmic likelihood defined as following:

$$L(\beta) = \sum_{i=1}^n \text{Log} \left(1 + e^{-y_i \beta^T x_i} \right)$$

$$x_i = \begin{bmatrix} 1 \\ x_{i,1} \\ x_{i,2} \\ \vdots \\ x_{i,k} \end{bmatrix}, \quad x_i^t = [1, x_{i,1}, x_{i,2}, \dots, x_{i,k}]$$

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}, \quad \beta^T = [\beta_0, \beta_1, \dots, \beta_k]$$

- Training by gradient descent algorithm

- ▶ The gradient of $L(\beta)$ is denoted as $\nabla L(\beta)$, which is also a function of the parameter set $\{\beta_i\}$ or β .
- ▶ The expression for the gradient $\nabla L(\beta)$ is:

$$\nabla L(\beta) = - \sum_{i=1}^N \frac{y_i x_i e^{-\beta^t x_i}}{1 + e^{-\beta^t x_i}}$$

- ▶ The growth of $L(\beta)$ is steepest along the direction of $\nabla L(\beta)$.
⇔ The descent of $L(\beta)$ is steepest along the direction of $-\nabla L(\beta)$.

- Training by gradient descent algorithm

- ▶ The expression for the gradient $\nabla L(\boldsymbol{\beta})$ is:

$$\nabla L(\boldsymbol{\beta}) = - \sum_{i=1}^N \frac{y_i \mathbf{x}_i e^{-\boldsymbol{\beta}^t \mathbf{x}_i}}{1 + e^{-\boldsymbol{\beta}^t \mathbf{x}_i}}$$

- ▶ The gradient $\nabla L(\boldsymbol{\beta})$ is obtained by calculating the partial derivatives of $L(\boldsymbol{\beta})$:

$$\nabla L(\boldsymbol{\beta}) = \begin{bmatrix} \frac{\partial L}{\partial \beta_0} \\ \frac{\partial L}{\partial \beta_1} \\ \vdots \\ \frac{\partial L}{\partial \beta_K} \end{bmatrix}$$

- Training by gradient descent algorithm

- ▶ $L(\beta)$ is minimized iteratively in small “steps” pushing β along the direction $-\nabla L(\beta)$.

- 1) β is randomly initialized.

- 2) Calculate the gradient $\nabla L(\beta)$.

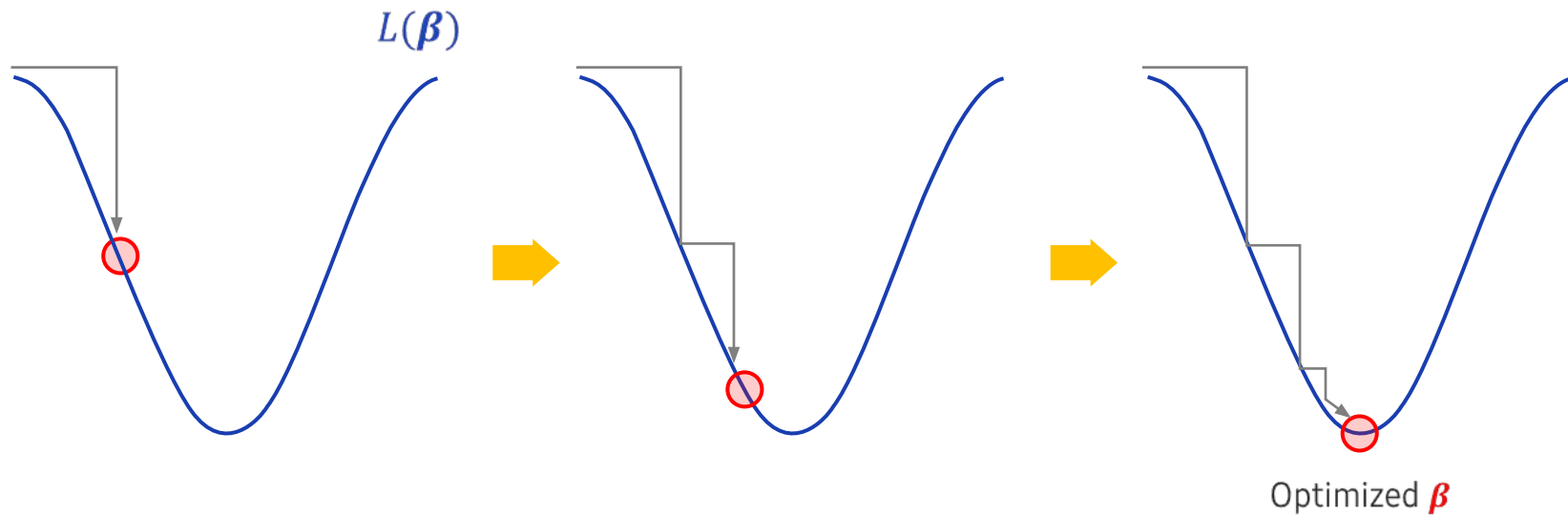
- 3) Update β by one step: $\beta \leftarrow \beta - \eta \nabla L(\beta)$.

Convergence speed is controlled by the “Learning rate” η .

- 4) Repeat from step 2) a fixed number of times (epochs).

- Training by gradient descent algorithm

- $L(\beta)$ is minimized iteratively in small “steps” pushing β along the direction $-\nabla L(\beta)$.



- Training by gradient descent algorithm

- The gradient function in Python

```
In [1]: def sigmoid(x):  
        s=1.0/(1.0 + np.exp(-x))  
        return s  
        def gradient(X,Y,beta):  
            z = np.dot(X.beta.T)*Y  
            ds = -Y*(1-sigmoid(z))*X  
            return ds.sum(axis=0)
```

- Training by gradient descent algorithm

- The gradient function in Python

```
In [2]: def train(self, input_X, input_Y, n_epochs):  
        ones_column = np.ones((input_X.shape[0],1))  
        X = np.concatenate((ones_column, input_X), axis=1)  
        Y = (2*input_Y - 1).reshape(-1,1)  
        for n in range(n_epochs):  
            self.beta = self.beta - self.rate*gradient(X,Y,self.beta)  
        return self.beta
```



Unit 3.

Application of Supervised Learning Model for Classification

- 3.1. Training and Testing in Machine Learning
- 3.2. Logistic Regression Basics
- 3.3. Logistic Regression Performance Metrics

Logistic Regression Performance Metrics

- Confusion matrix

- ▶ In the classification machine learning method, the most common method for evaluating the analysis model's result is the metric calculation, including classification accuracy, using a confusion matrix.
- ▶ The confusion matrix refers to the matrix that makes a crosstable of the predicted classification category from the analysis model and the actual classified category of data.

- Confusion matrix

		Predicted categorical value	
		Y	N
Actual categorical value	Y	O (TP: True Positive)	X (FN: False Negative)
	N	X (FP: False Positive)	O (TN: True Negative)

- Confusion matrix

- Confusion matrix can be made with a 2X2 crosstable, along with 3X3 and higher crosstables. For convenience, we will only use a 2X2 confusion matrix in this chapter.
- In the confusion matrix from the previous slide, the diagonally placed 'O' cases means the predicted and actual categorical values are the same. In other words, the classification machine learning predicted the results properly.
- On the other hand, if the predicted and actual categorical values differ, the machine learning model has made incorrect predictions.
- The categories that the analysis is mostly interested in are positive categories; the others are called negative categories. Depending on the accuracy of prediction (true or false) regarding positive and negative categories, the accurate classification of interested categories is called **TP (True Positive)**. The accurate classification of uninterested categories is called **TN (True Negative)**.
- The inaccurate classification of uninterested categories into interested categories is called **FP (False Positive)**. The inaccurate classification of interested categories into uninterested categories is called **FN (False Negative)**.
- There are various metrics based on different combinations of TP, TN, FP, and FN of the confusion matrix for evaluating the analysis result of classification machine learning methods.

- Metric

- Major metrics calculated from the confusion matrix include accuracy, error rate = $1 - \text{accuracy}$, sensitivity (also referred to as recall, hit ratio, TP rate, etc.), specificity, FP rate, precision, and others. Among those, accuracy, sensitivity, and precision are the most frequently used metrics.
- Also, there are F-Measure (or F1-Score) that combines sensitivity, precision, and Kappa Statistics, where the predicted and actual values of the analysis model are exactly the same. The calculation formulas and definitions of various metrics are in the following table.

•Metric

Metric	Calculation Formula	Definition
accuracy	$\frac{(TP+TN)}{(TP+TN+FP+FN)}$	Ratio of accurate prediction of actual classification category (Ratio of TP and TN from the entire prediction)
error rate	$\frac{(FP+FN)}{(TP+TN+FP+FN)}$	Ratio of inaccurate prediction of actual classification category (Identical to 1-accuracy)
sensitivity = TP Rate	$\frac{(TP)}{(TP+FN)}$	Ratio of accurate prediction to 'positive' from actual 'positive' categories (True Positive – also referred to as Recall, Hit Ratio, and TP Rate)
specificity	$\frac{(TN)}{(TN+FP)}$	Ratio of accurate prediction to 'negative' from actual 'negative' categories (True Negative)
FP Rate	$\frac{(FP)}{(TN+FP)}$	Ratio of inaccurate prediction to 'positive' from actual 'negative' categories = 1-specificity
precision	$\frac{(TP)}{(TP+FP)}$	Ratio of actual 'True Positive' from the ratio of predicted 'positive'
F-Measure (F1-Score)		Ranged between 0~1. It is the harmonic mean between precision and sensitivity (recall). If both precision and sensitivity are high, f-Measure also tends to have a larger value.
Kappa Statistic		The value after eliminating coincidental agreement between predicted and actual values of the model (Ranged between 0~1. When the value is closer to 1, the predicted and actual values of the model accurately coincide. The values do not coincide when the value gets closer to 0.)

- Metric

- Among the metrics from the previous slide, sensitivity signifies how well the actual 'positive' category is predicted 'positive.' The precision is the index showing the ratio of actual 'positive' from the predicted 'positive' categories. Thus, they are metrics that directly explain how well the classification machine learning analysis model classifies interested categorical values of objective variables.
- Sensitivity and precision are the most significant and frequently used metrics for classification machine learning results in real life.

- Confusion matrix

Ex			
		Actual 0	Actual 1
	Predicted 0	120	5
	Predicted 1	15	20

- Confusion matrix is a contingency table that counts the frequencies of the actual vs. predicted.

•Confusion matrix: Accuracy

Ex

	Actual 0	Actual 1
Predicted 0	120	5
Predicted 1	15	20

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}$$

- Accuracy is the ratio between the diagonal sum and the total sum.

•Confusion matrix: Sensitivity

Ex		Actual 0	Actual 1
	Predicted 0	120	5
	Predicted 1	15	20

$$\text{Sensitivity} = \frac{\text{Number of correctly predicted 1s}}{\text{Total number of 1s}}$$

•Confusion matrix: Specificity

Ex		Actual 0	Actual 1
	Predicted 0	120	5
	Predicted 1	15	20

$$\text{Specificity} = \frac{\text{Number of correctly predicted 0s}}{\text{Total number of 0s}}$$

•Confusion matrix: Precision

Ex		Actual 0	Actual 1
	Predicted 0	120	5
	Predicted 1	15	20

$$\text{Precision} = \frac{\text{Number of correctly predicted 1s}}{\text{Total number of 1s}}$$

- Note

- Accuracy alone is not sufficient for testing.

Ex If frauds constitute only 1% of all transactions, the accuracy of a fraud detection system (FDS) that predicts non-fraud in all transactions would be quite high at 99%.

However, such FDS would be useless because it misses the 1% that really matters.

- We should also consider metrics other than just the accuracy.

- Terminology

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}$$

$$\text{Sensitivity} = \frac{\text{Number of correctly predicted 1s}}{\text{Total number of 1s}}$$

$$\text{Specificity} = \frac{\text{Number of correctly predicted 0s}}{\text{Total number of 0s}}$$

$$\text{Precision} = \frac{\text{Number of correctly predicted 1s}}{\text{Total number of 1s}}$$

$$\text{Cohen's kappa } \kappa = \frac{\text{Accuracy} - p_e}{1 - p_e} \quad \text{where } p_e \text{ is the probability of correct prediction by chance.}$$

- Terminology

True Positive Rate =
Sensitivity

True Negative Rate =
Specificity

False Positive Rate = $\frac{\text{Number of actual 0s that are incorrectly predicted as 1}}{\text{Total number of 0s}} = 1 - \text{Specificity}$

False Negative Rate = $\frac{\text{Number of actual 1s that are incorrectly predicted as 0}}{\text{Total number of 1s}} = 1 - \text{Sensitivity}$

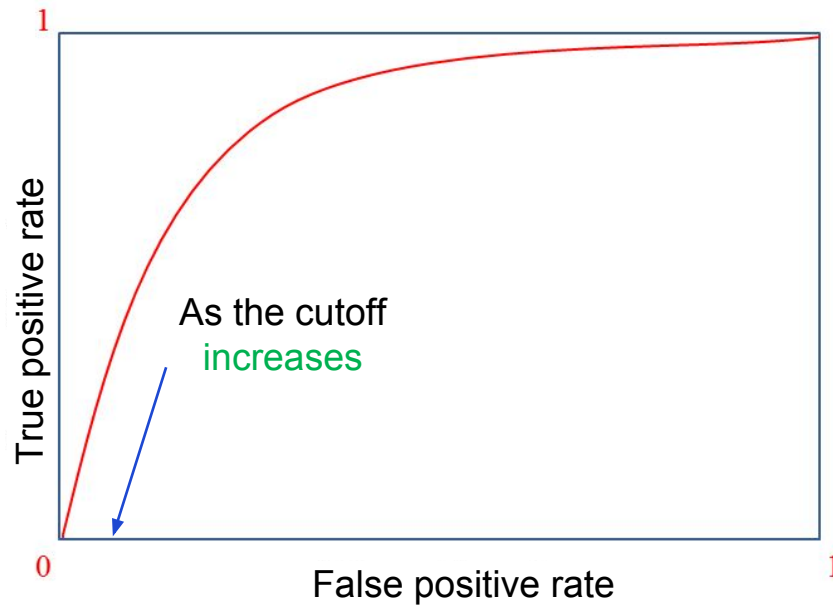
Positive Predicted Value =
Precision

- ROC curve



- ROC curve is a parametric plot with respect to the *cutoff* probability.

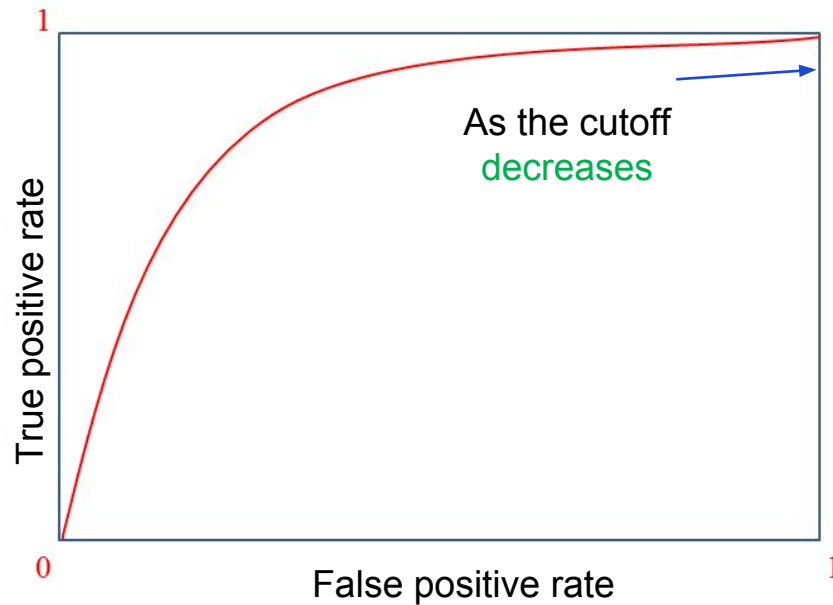
- ROC curve



As the cutoff increases (closer to 1)	
Performance Metric	Increase/ Decrease
True Positive Rate (Sensitivity)	↓
Specificity	↑
False Positive Rate (1-Specificity)	↓
Precision	↑

- ROC curve is a parametric plot with respect to the *cutoff* probability.

- ROC curve



As the cutoff decreases (closer to 0)	
Performance metric	Increase/ Decrease
True Positive Rate (Sensitivity)	↑
Specificity	↓
False Positive Rate (1-Specificity)	↑
Precision	↓

- ROC curve is a parametric plot with respect to the *cutoff* probability.

- ROC curve



- AUC stands for Area Under the Curve.
- AUC closer to 1 means good overall performance.

- Interpretation using Bayes' theorem

Ex There are 100 observations. In 6 cases, the actual response is 1.

In the rest of the 94 cases, the actual response is 0. It is known that for a given logistic regression model, the sensitivity = 0.92 and the specificity = 0.90.

For a new observation (only explanatory variables), this model predicts 1 as the response.

What is the probability of this prediction being correct?

a) We have $P(1) = 0.06$, $P(0) = 0.94$.

$$P(\text{Predicted } 1 \mid \text{Actual } 1) = 0.92 \Leftrightarrow \text{"Sensitivity"}$$

$$P(\text{Predicted } 0 \mid \text{Actual } 0) = 0.90 \Leftrightarrow \text{"Specificity"}$$

$$\text{We can also derive } P(\text{Predicted } 1 \mid \text{Actual } 0) = 1 - P(\text{Predicted } 0 \mid \text{Actual } 0) = 0.1$$

- Interpretation using Bayes' theorem

Ex There are 100 observations. In 6 cases, the actual response is 1.

In the rest of the 94 cases, the actual response is 0. It is known that for a given logistic regression model, the sensitivity = 0.92 and the specificity = 0.90.

For a new observation (only explanatory variables), this model predicts 1 as the response.

What is the probability of this prediction being correct?

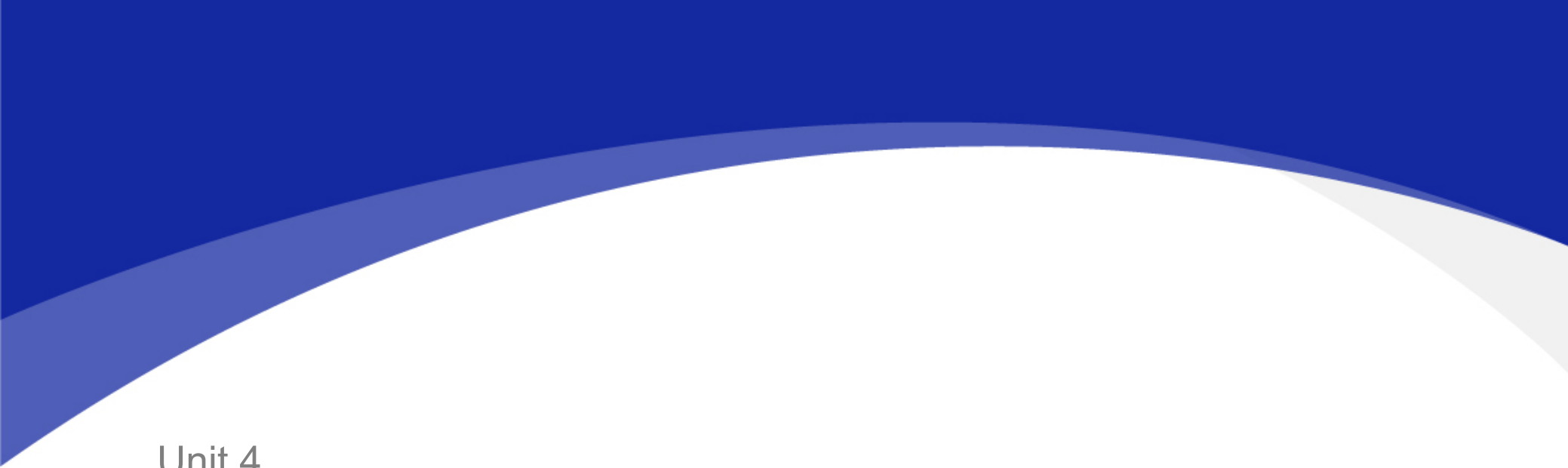
b) The answer we are seeking is given by

$$\begin{aligned} P(\text{Actual } 1 | \text{Predicted } 1) &= \frac{P(\text{Predicted } 1 | \text{Actual } 1)P(1)}{P(\text{Predicted } 1 | \text{Actual } 1)P(1) + P(\text{Predicted } 1 | \text{Actual } 0)P(0)} \\ &= \frac{0.92 \times 0.06}{0.92 \times 0.06 + 0.1 \times 0.94} \cong \mathbf{0.37} \end{aligned}$$

Coding Exercise #0307



Follow practice steps on 'ex_0307.ipynb' file.



Unit 4.

Decision Tree

- 4.1. Tree Algorithm

Overview of Decision Tree

- Definition

- A classification model that analyzes data collected from the past and expresses the patterns found (characteristics of each category) as a combination of features

- Purpose

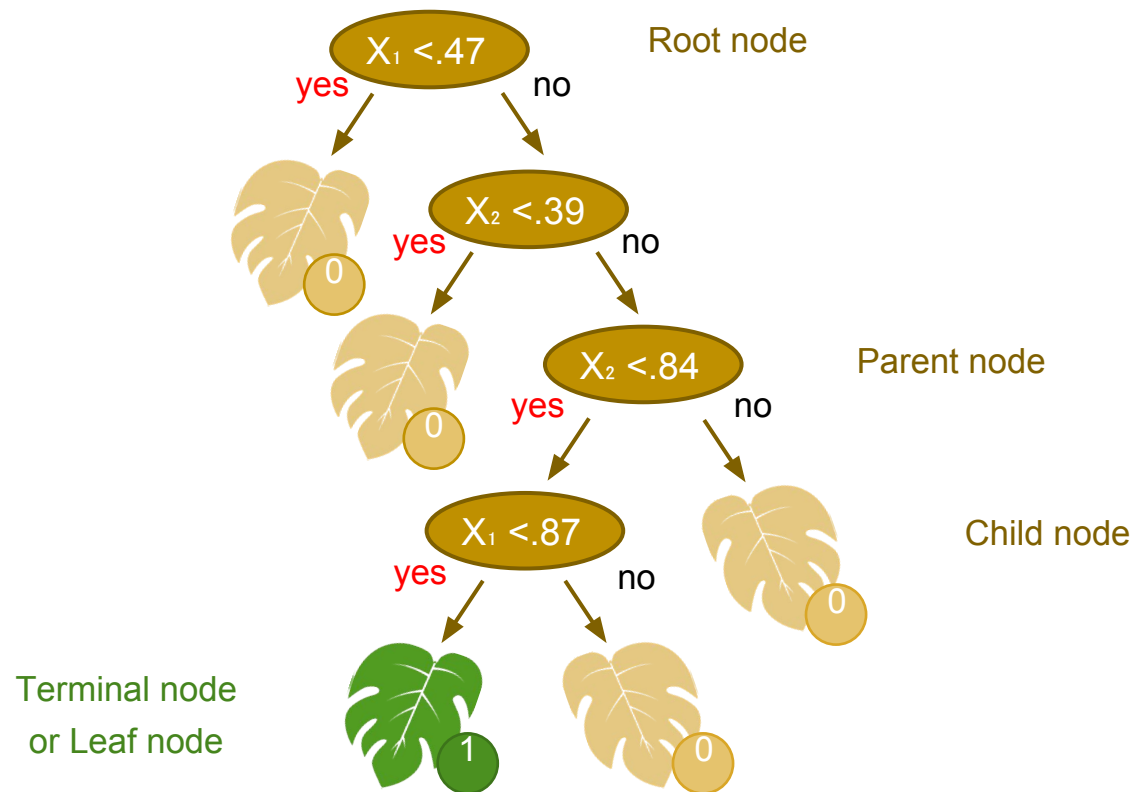
- Classification of unseen data and prediction of categorical values
- Extraction of generalized knowledge in a tree structure from the data

- Classification depending on the objective variable types

- Categorical variable: Classification Tree
- Continuous variable: Regression Tree

Composition

- Node, Branch, Depth



How to Construct a Decision Tree Model

Construction of a decision tree

Construction of a decision tree	Making a decision tree by designating an appropriate split criterion and stopping rules according to the purpose and data structure of analysis
Branching	Removing branches that have a high risk of error rate or inappropriate rules
Validity evaluation	Evaluation of the decision tree through cross-validation using the gain chart, risk chart, or test data
Interpretation and prediction	Interpretation of the decision tree and setting a prediction model

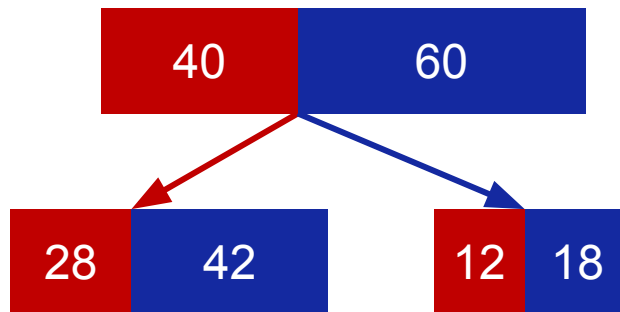
Split Criterion of the Decision Tree

Analysis process of a decision tree

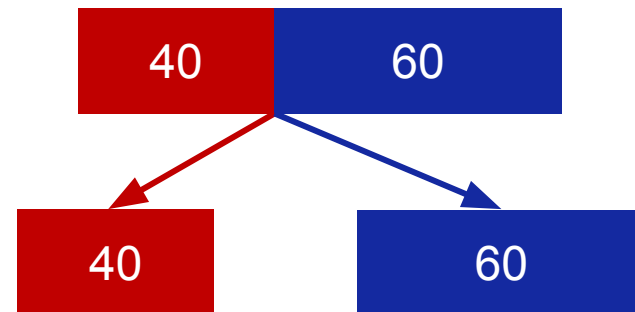
- Repetitive splitting: Repetitive splitting of the dimensional space of independent variables using training data
- Branching: Branching with evaluation data

Split criterion

- Create the classification tree so that the purity of the child node is greater than that of the parent node.



**No
Improvement**



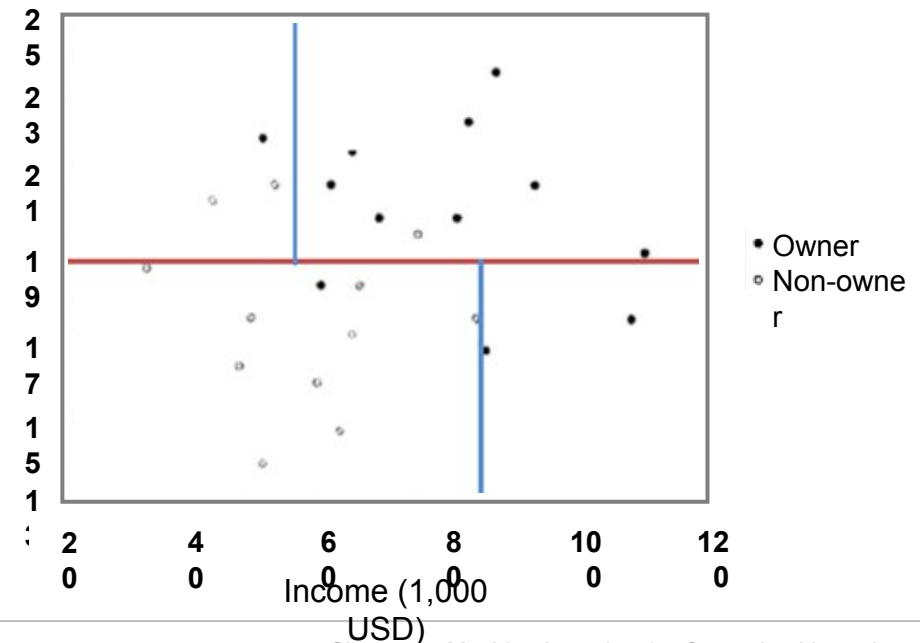
**Perfect
Split**

Repetitive splitting process

Purpose

- Divide the entire space into rectangles and make each as pure or homogeneous as possible.
- Definition of 'pure':
 - Divide the area into pure or homogenous rectangular spaces as much as possible
 - All variables in the final rectangle belong to the same group.

Size of the housing site
(1,000 ft^2)



■ Repetitive splitting process

- 1) Select x_i , one of the variables, and the x_i value (s_i as a split criterion) is designated to split the p-dimension space into two.
- 2) $x_i \rightarrow \{x_i \leq s_i\} \cup \{x_i > s_i\}$
- 3) Select a variable again and split it in the same way.
- 4) Repeat the process until it reaches desired purity.

Split Criterion

| Discrete objective variables

- Chi-squared statistic – p-value: Creates child nodes with a predictor variable with the lowest p-value and the optimal partitioning
- Gini index: Selects child nodes with a predictor variable that reduces the Gini index and the optimal partitioning
- Entropy measure: Creates child nodes with a predictor variable with the lowest entropy measure and the optimal partitioning

Split Criterion

Continuous objective variables

- F statistic in ANOVA: Creates child nodes with a predictor variable with the lowest p-value and the optimal partitioning
- Variance reduction: Creates child nodes with the optimal partitioning that maximizes variance reduction

Selection of algorithms and classification variables

	Discrete objective variables	Continuous objective variables
CHAID (multi space partitioning)	Chi-squared statistic	ANOVA F statistic
CART (binary space partitioning)	Gini index	Variance reduction
C4.5	Entropy measure	

Impurity measure

Gini index

- ▶ Selects child nodes with a predictor variable that reduces the Gini index and the optimal partitioning.
- ▶ If the T dataset is split into k categories and the category performance ratios are $p_1, \dots, p_k$, it is expressed as the following equation.

$$Gini(T) = 1 - \sum_{l=1}^k p_l^2$$

high impurity(diversity), low purity



$$GI = 1 - (3/8)^2 - (3/8)^2 - (1/8)^2 - (3/8)^2 = .69$$

low impurity(diversity), high purity



$$GI = 1 - (6/7)^2 - (1/7)^2 = .24$$

| Entropy measure

- ▶ In thermodynamics, entropy measures the degree of disorder.
- ▶ Creates child nodes with a predictor variable with the lowest entropy measure and the optimal partitioning.
- ▶ If the T dataset is split into k categories and the category performance ratios are $p_1, \dots, p_k$, it is expressed as the following equation.

$$Entropy(T) = - \sum_{l=1}^k p_l \log_2 p_l$$

Ex If 4 categories consist of ratios of 0.25, 0.25, 0.25, 0.25 (T0):

$$Entropy(T_0) = -(0.25 \log_2 0.25) * 4 = 1.39$$

Ex If 4 categories consists of ratios of 0.5, 0.25, 0.25, 0 (T1):

$$Entropy(T_1) = -(0.5 \log_2 0.5 + 0.25 \log_2 0.25 + 0.25 \log_2 0.25) = 1.04$$

Stopping Criteria

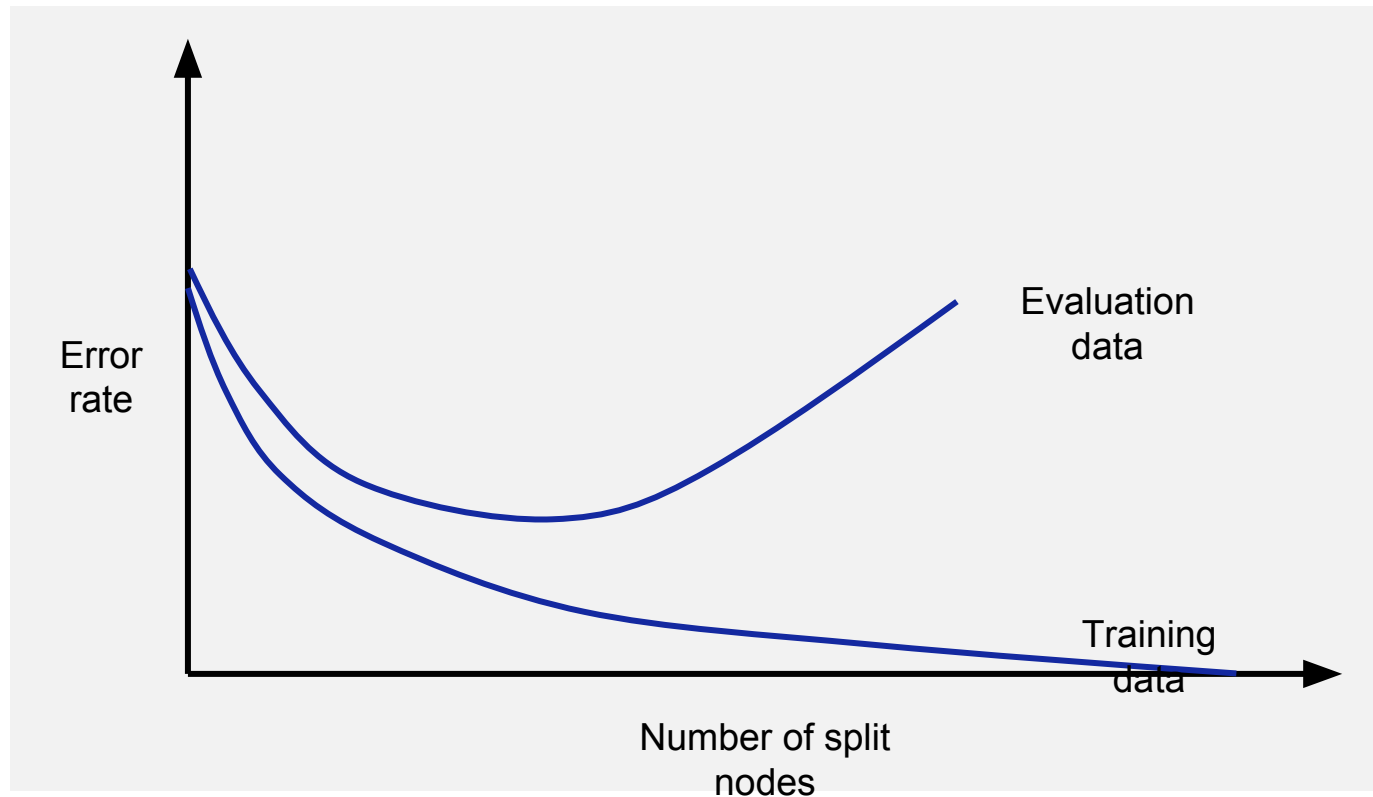
- | A rule to designate the current node as a terminal node without further splitting
 - Designates the depth of the decision tree
 - Designates the minimum number of records in the terminal node

Branching Criteria

- | Application of test data
 - Application of the test data to the constructed model
 - Reviewing the predictive value of the constructed model through test data
 - Removing the branches that have a high risk of error rate or inappropriate rule of inference
- | By an expert
 - An expert reviewing the validity of rules suggested in the constructed model
 - Removing rules without validity

Overfitting Problem

Overfitting problem graph



Pros

- Creation of understandable rules (can be expressed with SQL)
- Useful in classification prediction
- Able to work with both continuous and discrete variables
- Shows a more relatively significant variable

Cons

- Not suitable to predict continuous variable values
- Unable to perform time series analysis
- Not stable

Tree Algorithm

- Pros

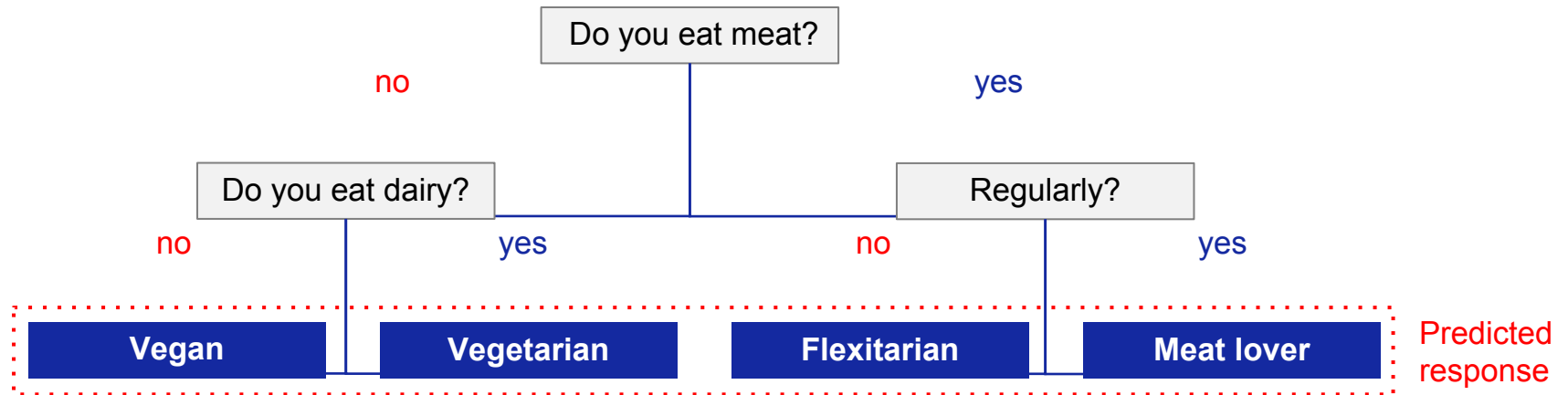
- Intuitive and easy to understand
- No assumptions about the variables
- No need to scale or normalize data
- Not that sensitive to the outliers

- Cons

- Not that powerful in the most basic form
- Prone to overfitting. Thus, “pruning” is often required.

| Classification Tree

Ex



- Training step creates an inverted tree structure as above.
- Conditions are evaluated at the nodes and then branch out.
- Each leaf node corresponds to a **region** in the configurational space.

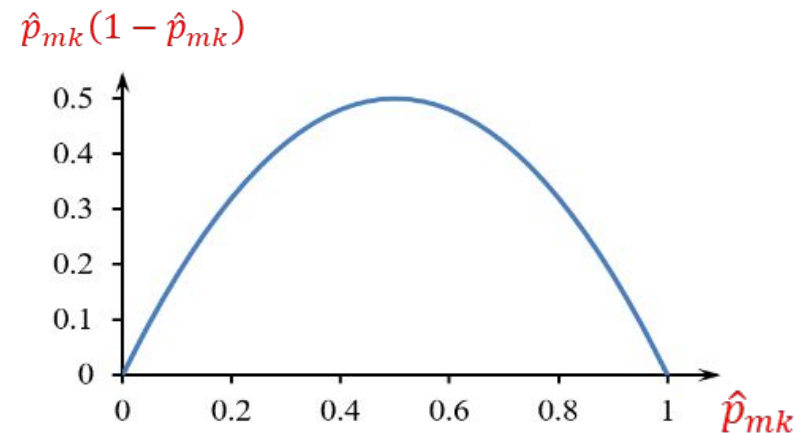
Classification Tree

- The tree structure is trained by minimizing the Gini impurity (or entropy).

$$G_m = \sum_{k=1}^K \hat{p}_{mk}(1 - \hat{p}_{mk}) = 1 - \sum_{k=1}^K \hat{p}_{mk}^2$$

or

$$Entropy_m = - \sum_{k=1}^K \hat{p}_{mk} \text{Log}(\hat{p}_{mk})$$



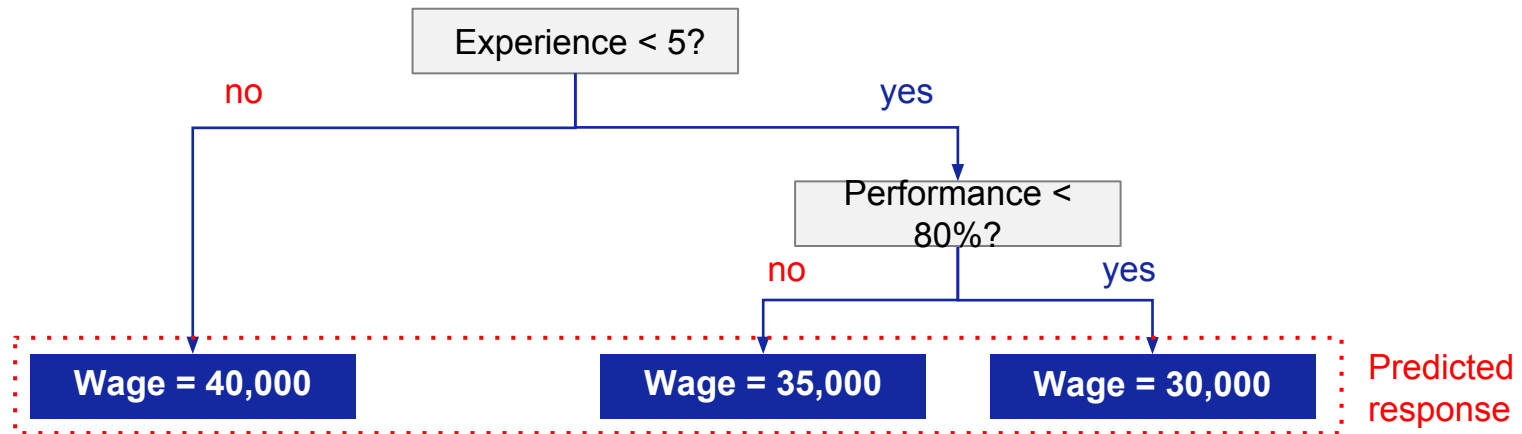
- G_m is the Gini impurity in the leaf node m .
 - $Entropy_m$ is the entropy in the leaf node m .
 - Here, $\hat{p}_{mk}$ is the proportion of the class k in the leaf node m .
 - K is the total number of possible classes.
 - The class with the largest proportion is the prediction at that leaf node.
- } The smaller, the better.

| Classification Tree: Procedure

- a) Make a basic tree.
- b) Prune branches that do not provide better performance.
Pruning can be done during the cross-validation step.
- c) Predict with the optimized tree.

| Regression Tree

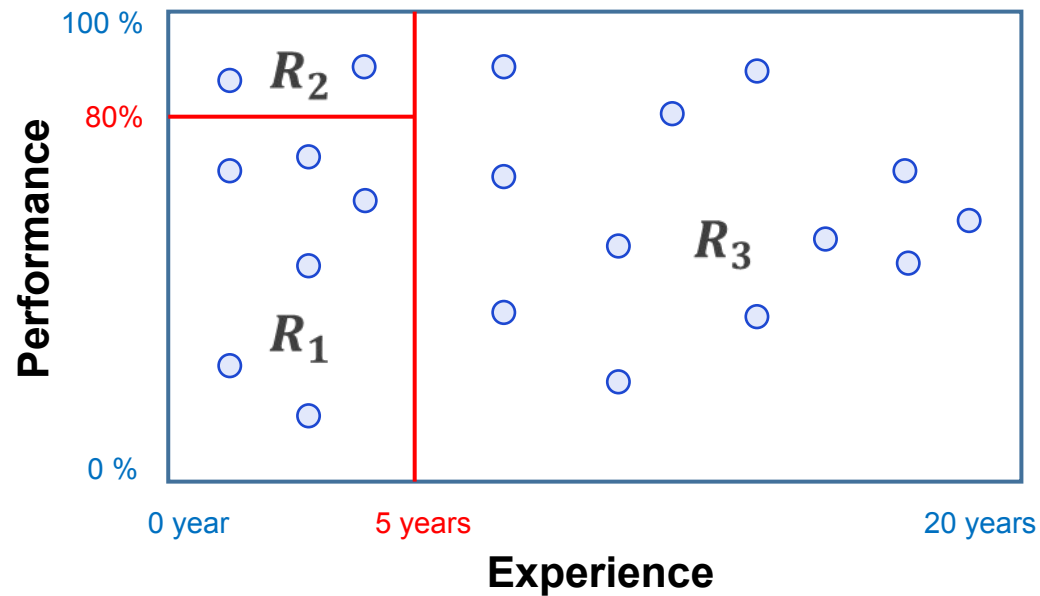
Ex



- Predicts numeric values rather than categories or classes.
- Conditions are evaluated at the nodes and then branch out.
- Each leaf node corresponds to a **region** in the configurational space.

| Regression Tree

Ex



- Each leaf node corresponds to a **region** in the configurational space.

Regression Tree

- ▶ The configurational space is split into regions: $\{R_1, R_2, \dots, R_J\}$.
- ▶ The tree structure is trained by minimizing the RSS (residual sum of squares):

$$RSS = \sum_{j=1}^J \sum_{i \in R_j} (y_i - \hat{y}_{R_j})^2$$

- J is the total number of leaf nodes.
- The predicted value in the j -th region $\hat{y}_{R_j}$ is given by the average in that region.
- For the observations belonging to the region R_j , the same predicted response $\hat{y}_{R_j}$ is assigned analogous to the classification Tree.

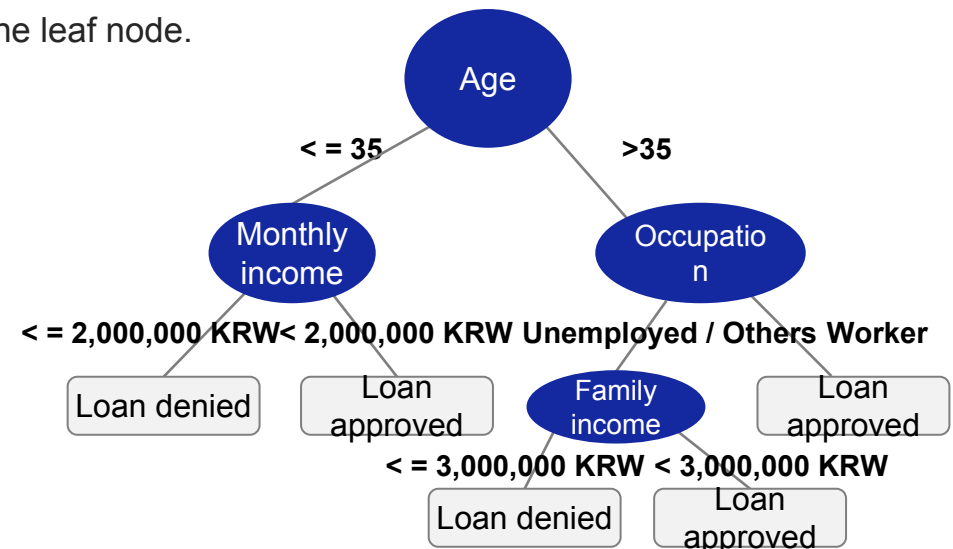
| scikit-learn DecisionTreeClassifier/Regressor Hyperparameters:

Hyperparameter	Explanation
max_depth	The maximum depth of a tree
min_samples_leaf	The minimum number of sample points required to be at a leaf node
min_samples_split	The minimum number of sample points required to split an internal node
max_features	The number of features to consider when looking for the best split
max_leaf_nodes	The maximum number of leaf nodes in the tree

- Need to be tuned for optimized performance.
- More information can be found at:
Classifier: <https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeClassifier.html>
Regressor: <https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeRegressor.html>

Decision Tree

- Decision tree refers to a modeling technique as the shape of a tree branching out from root to leaf nodes. It depends on reference values of independent variables (explanatory or input variables) that affect the classification or prediction of objective variables.
- In the decision tree, each node is split in the form of if-then depending on explanatory variables' characteristics or reference values. When following the tree structure, it is possible to easily understand how the attribute value of data is classified into the category.
- The figure provided below is a typical form of the decision tree. From the example, 'age' is the root. It can be inferred that 'age' is the most significant variable when deciding the loan approval.
- The squared shape node at the end of each branch is the leaf node.



Split Criteria of Decision Tree

- Decision tree gives a question to each node and separates data by branching according to the response.
- To evaluate how well the data is separated, a specific criterion is required. In general, impurity is the evaluation criterion. The impurity becomes higher as various classifications are mixed in the node; it is the lowest when there's only one classification.
- Thus, if each node's impurity is low after node splitting, we can say that the decision tree is well classified.
- Impurity measures include Gini impurity and entropy.

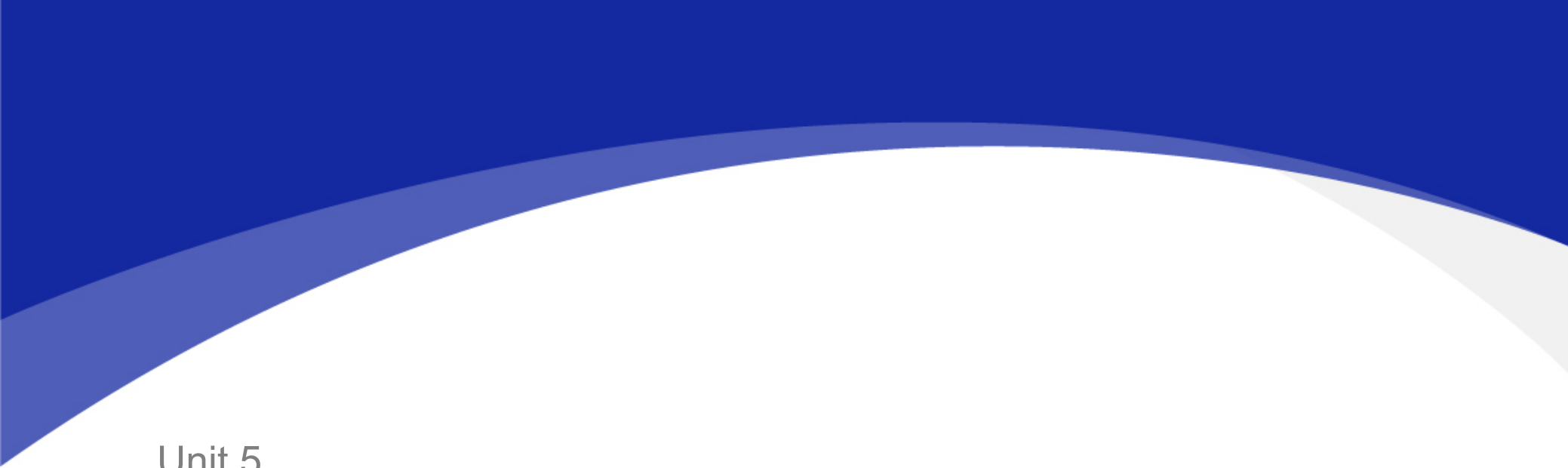
$$\text{Gini } Coff. = 1 - \sum_{i=1}^K p_i^2, \quad \text{Entropy } Coff. = - \sum_{i=1}^K p_i \log_2 p_i$$

- The impurity index is 0 when $p_i = 0$ or $p_i = 1$, and it gets the largest when $p_i = \frac{1}{2}$, thus making a parabola. In other words, the impurity index is the lowest when there's a certain classification in the node or the node is completely free from any classification. In contrast, the impurity becomes the largest when many classifications are found in the same node.

Coding Exercise #0308



Follow practice steps on 'ex_0308.ipynb' file.



Unit 5.

Naïve Bayes Algorithm

- 5.1. Naïve Bayes Algorithm

Naïve Bayes Algorithm

- About Naïve Bayes algorithm
 - It is a straightforward application of Bayes' theorem.
- Pros
 - Intuitive and simple
 - Not that sensitive to the noise and outliers
 - Fast
- Cons
 - Assumes that the features are independent, which may not be strictly true
 - Not among the best-performing algorithms

- About Naïve Bayes algorithm

- Bayes' theorem:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

- Now we take $A=Class$ and $B=Data$, then:

$$P_{post}(Class) = P(Class|Data) = \frac{P(Data|Class)P_{prior}(Class)}{P(Data)}$$

- We are more interested in comparing relative probabilities:

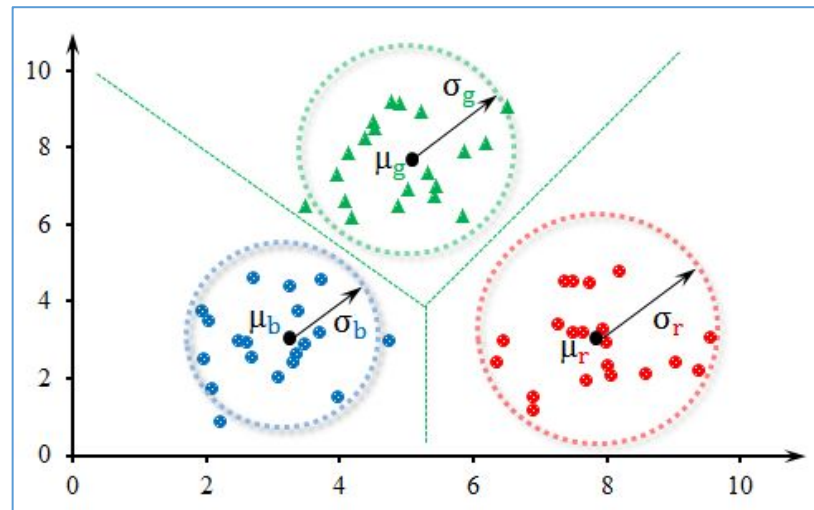
$$P_{post}(Class) \propto P(Data|Class)P_{prior}(Class)$$

- About Naïve Bayes algorithm

- We can approximate $P(Data | Class)$ by a Gaussian (normal distribution):

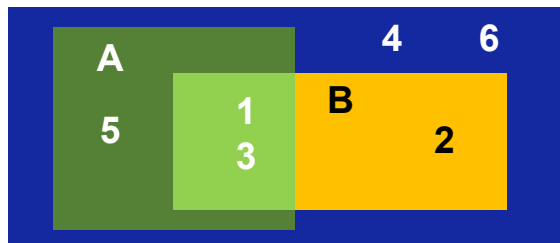
$$P_{post}(Class) \propto \frac{1}{\sqrt{2\pi}\sigma_j} \exp\left(-\frac{1}{2\sigma_j^2}(x - \mu_j)^2\right) P_{prior}(Class)$$

where the parameters μ_j and σ_j are “learned” from the training data.



- Random variable

- ▶ The variable whose values are unknown until the outcome
- ▶ Independent events
 - If the probability of the simultaneous occurrence of two cases is identical to the multiplication of the probabilities of each event to occur, then the two events are independent of each other.
 - $P(A \cap B) = P(A)P(B)$
- ▶ Dice



Conditional probability

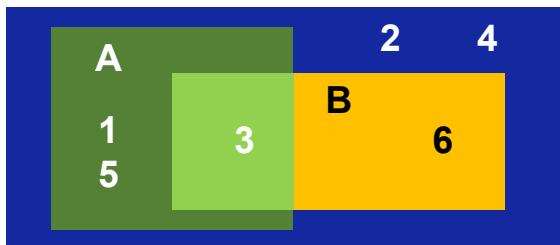
- Events of B (1,2,3) when the A events (1,3,5) occur
- 0.6666667



		Total	Event A	Event B	Probability A	Probability B
Event A	Odd number	1	1	1	0.5	0.5
Event B	Less than 3	2		2		
		3	3	3		
		4				
		5	5			
		6				

- Random variable

- The variable whose values are unknown until the outcome
- Independent events
 - Not affected
- Dice



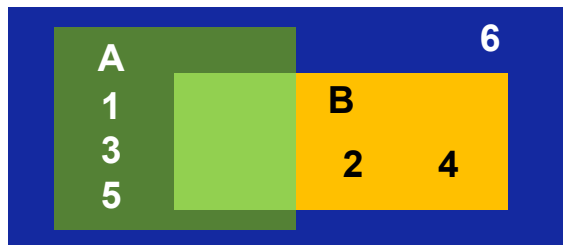
		Total	Event A	Event B	Probability A	Probability B
Event A	Odd number	1	1		0.5	0.333
Event B	Multiple of 3	2				
		3	3	3		
		4				
		5	5	6		
		6				

Conditional probability

- Events of B (3, 6) when the A events (1,3,5) occur
- 0.33333333

- Random variable

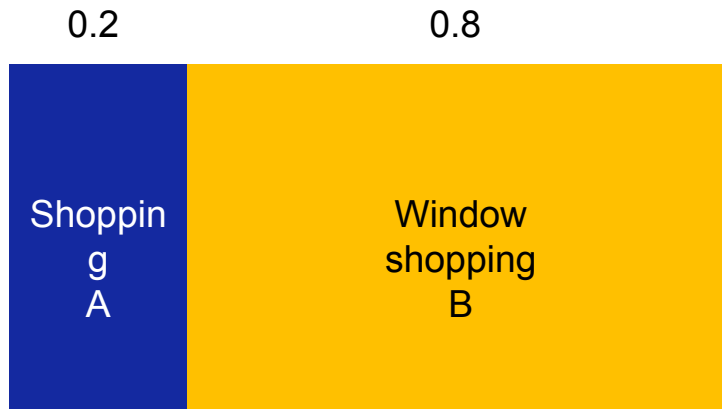
- The variable whose values are unknown until the outcome
- Exclusive events
 - Intersection is the null set.
- Dice



- Introduction to Bayesian statistics

1) Differentiate groups of people for 'shopping' and 'window shopping' through Bayesian interpretation.

- Setting prior probability for different types
- Prior probability: Probability before getting certain information



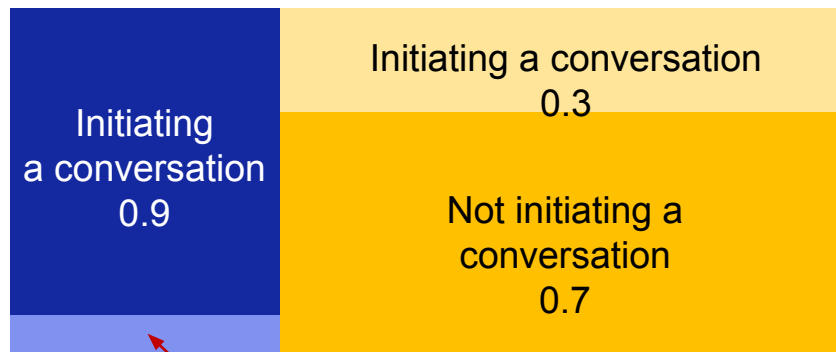
- Introduction to Bayesian statistics

2) Set the conditional probability of 'initiating a conversation' for each type.

	Probability of initiating a conversation	Probability of not initiating a conversation	Total
Shopping	0.9	0.1	1
Window shopping	0.3	0.7	1

Shopping
0.2

Window shopping
0.8



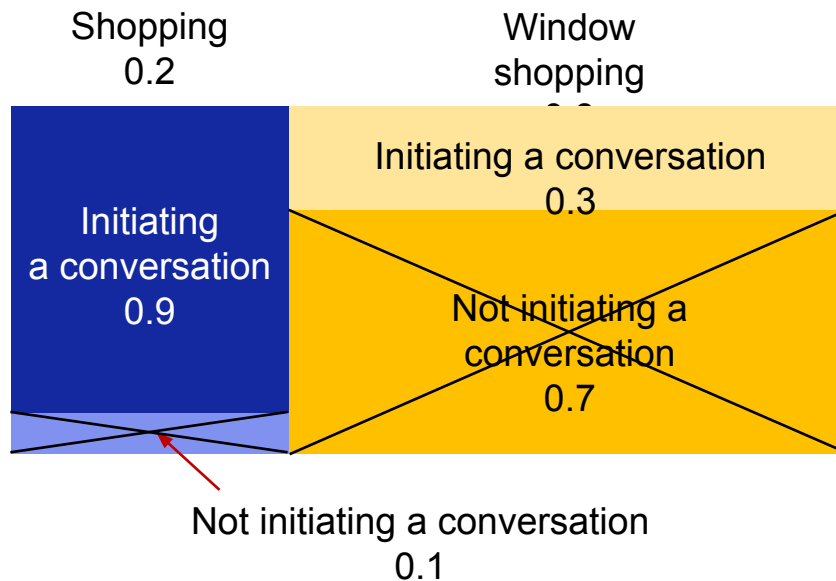
Not initiating a conversation
0.1

- The probability of each section is the area of each rectangle. (0.18, 0.02, 0.24, 0.56)

- Introduction to Bayesian statistics

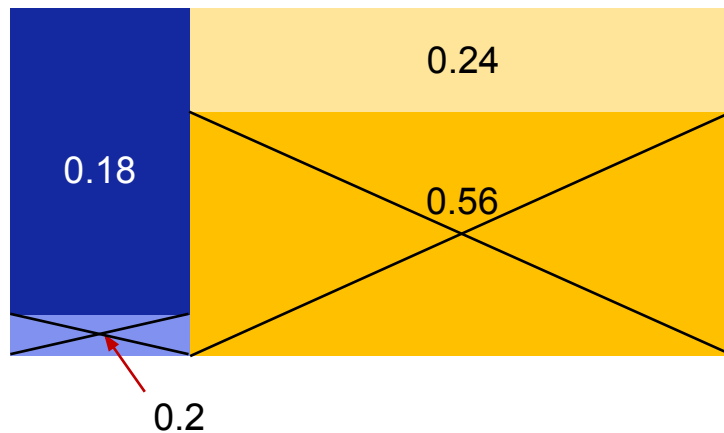
3) Remove 'impossible event' from the observations.

- The customer initiated a conversation (additional information).



- Introduction to Bayesian statistics

4) Calculate Bayesian probability of the shopping group.



	Shopping	Window shopping
Normalizati on	3/7	4/7
	43%	

Result
Initiating a
conversation

Cause
A certain group

- The result is 0.18:0.24, which is 3:4.
- Thus, it can be estimated that the probability of the customer who initiated a conversation being in the shopping group is 3/7 (43%). It is either called Bayesian statistics or posterior probability.
- Chooses if a person who initiated a conversation is in the shopping or window shopping group based on probability.

- Introduction to Bayesian statistics

- The probability of a customer who initiated a conversation making a purchase is now different.
- Bayesian updating for customer type

Final summary The prior probability of being a shopping group 0.2
Observing to initiate a conversation
The posterior probability of being a shopping group $3/7 \rightarrow 0.428571$

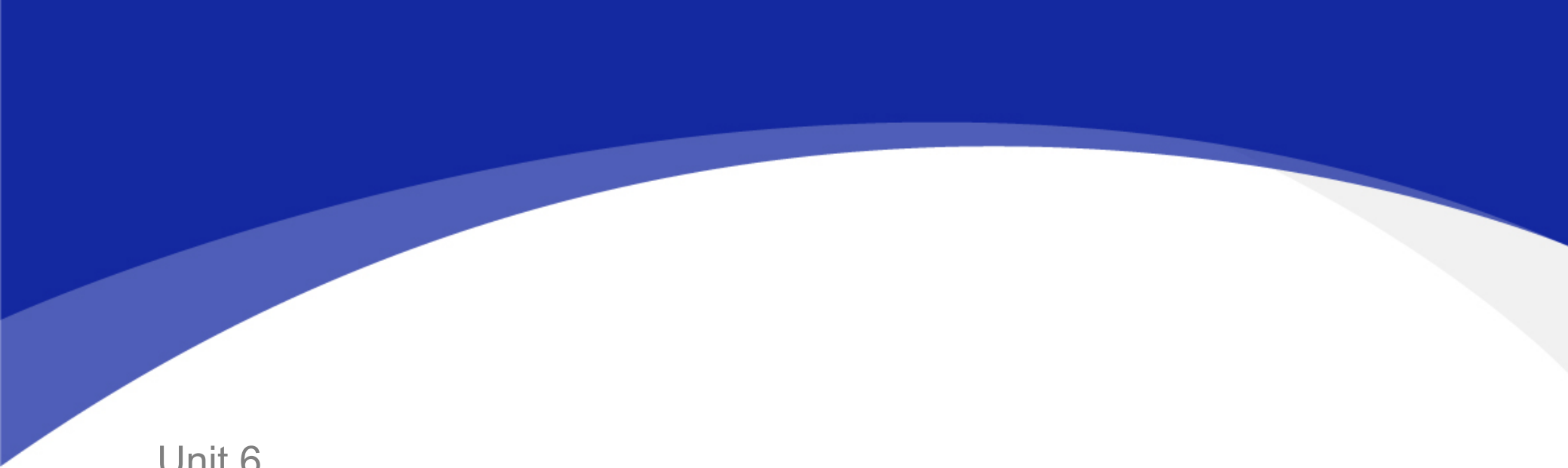
A diagram illustrating the Bayesian update process. It shows a flow from a prior probability of 0.2 to a posterior probability of 0.428571. The transition is labeled with the fraction 3/7, representing the likelihood of observing a conversation given the customer is in the shopping group. An arrow points from the 0.2 to the 0.428571, with the 3/7 positioned above the arrow.

- The customer wouldn't be in the shopping group for sure, but the probability is doubled.
- **Bayesian estimation** is 'performing Bayesian updating to the posterior probability based on the behavioral observation of the prior probability.'
- Such an estimation method is called Bayesian statistics.
- **Bayes' theorem**
 - Obtaining the posterior probability based on the prior probability

Coding Exercise #0309



Follow practice steps on 'ex_0309.ipynb' file.



Unit 6.

KNN Algorithm

- 6.1. KNN Algorithm

KNN Algorithm

- About KNN (K-Nearest Neighbors)
 - One of the simplest algorithms
 - Prediction is based on the k nearest neighboring points.
 - There are classification and regression variants.
 - Classification: prediction decided by the majority class of the nearest neighbors.
 - Regression: prediction given by the average of the nearest neighbors.

- About KNN (K-Nearest Neighbors)

- KNN is a classification and prediction technique for a dataset with the unknown category of the objective variable. It designates the most similar category of the surrounding dataset.
- For KNN, specific criteria are required for measuring the 'analogy' between the certain dataset and the surrounding one and how many datasets will be used for the final classification of the objective variable category.

1) Analogy measurement

- There are many ways to measure the analogy between data. In general, analogy calculation is done by inverting the squared Euclidean distance between two points or using the Pearson correlation coefficient. If the points are discrete variables, use the Jaccard coefficient.

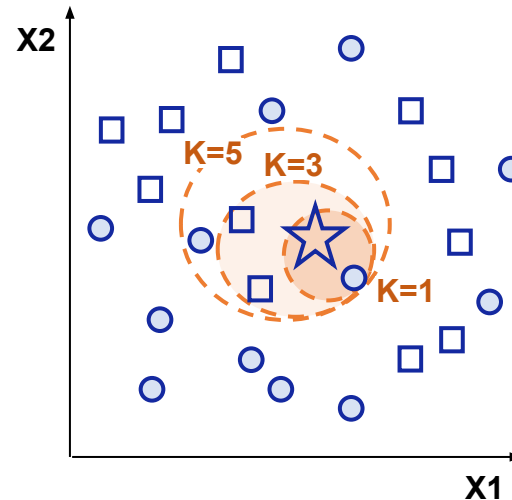
2) Objective variable classification criteria

- K in the KNN refers to the number of surrounding data points being used for classifying objective variables of the specific data point after measuring the analogy between the specific data point and other surrounding datasets.

Ex Take movie recommendations as an example. It recommends a movie that is similar to the customer's favorite movie and that the customer hasn't watched yet. When considering only one movie that is the most similar to the customer's favorite, it is '1-nearest neighbor.' When considering three similar movies, it would be '3-nearest neighbor.'

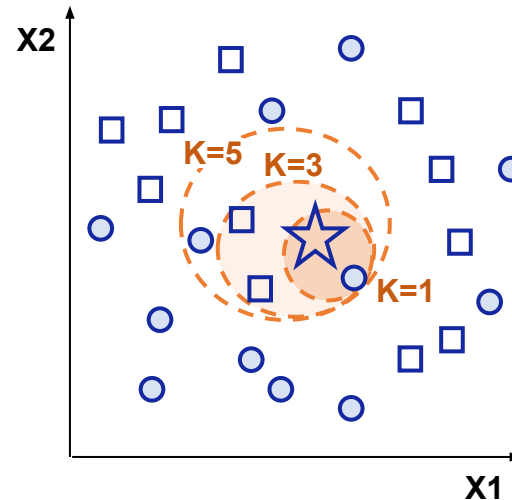
- Likewise, K-nearest neighbor is a technique to determine a new category according to the principle of majority rule. It finds k surrounding data points similar to the specific data point and considers the classification category of the specific data point.

- About KNN (K-Nearest Neighbors)



- ▶ The figure above conceptualizes the change in objective variables depending on the K value setting in the K-nearest neighbor method. The '☆' point is the data value that needs to be classified; the points shaped in a square and circle are other data points present around.
- ▶ When setting K=1, since the closest data point to the star is a circle, the objective variable of '☆' will be classified as '○.' On the other hand, if K=3, three points closest to '☆' will be considered, including two squares and one circle. In this case, '☆' will be classified as '□.' Also, if K=5, since there are more circles around '☆,' it will be again classified as '○.'

- About KNN (K-Nearest Neighbors)



- ▶ Thus, different K values significantly change the K-nearest neighbor's predicted result of the objective variable category. This is why setting an appropriate K value is important in the K-nearest neighbor method.
- ▶ However, there are no clear theoretical or statistical standards for an appropriate K value. Different K values are usually set for repeated tests, and the final K value is set once it shows the optimal classification performance. Generally, a random initial K value is designated between $K=3$ and $K=9$, which is then used to test the classification performance with training and evaluation data to select the optimal K value.

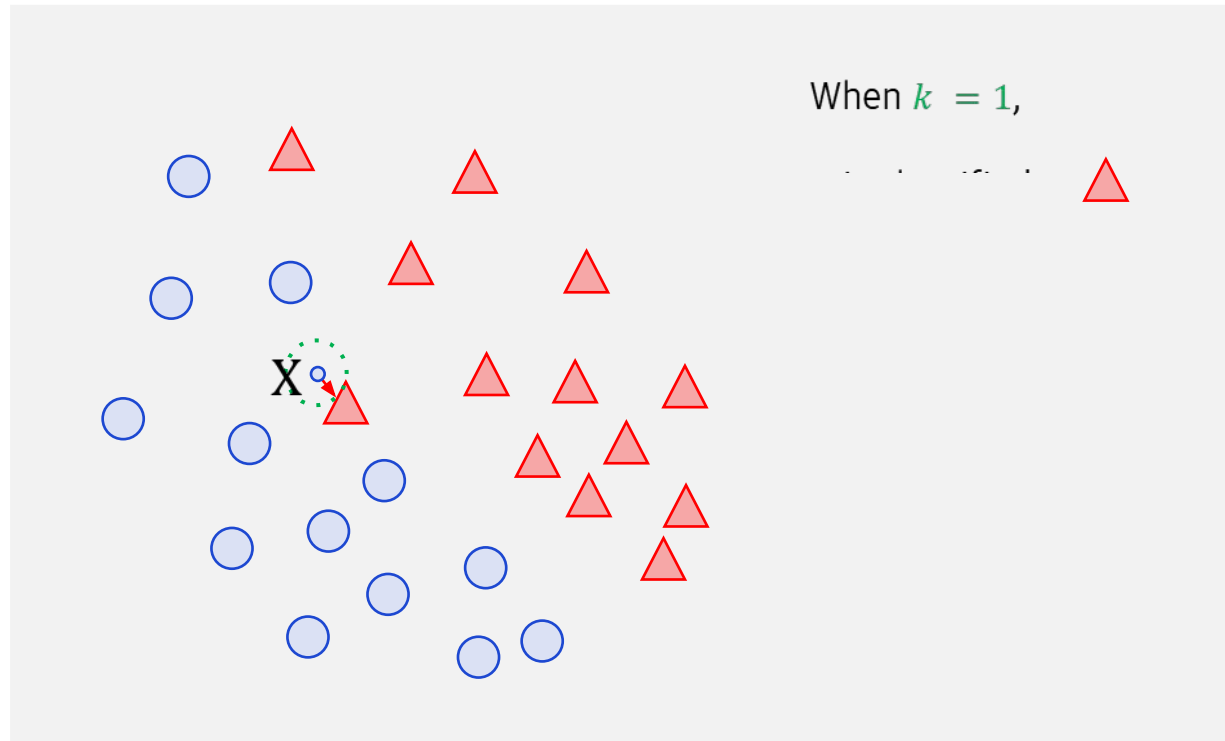
- Pros

- Simple and intuitive
- No model parameters to calculate. So, there is no training step.

- Cons

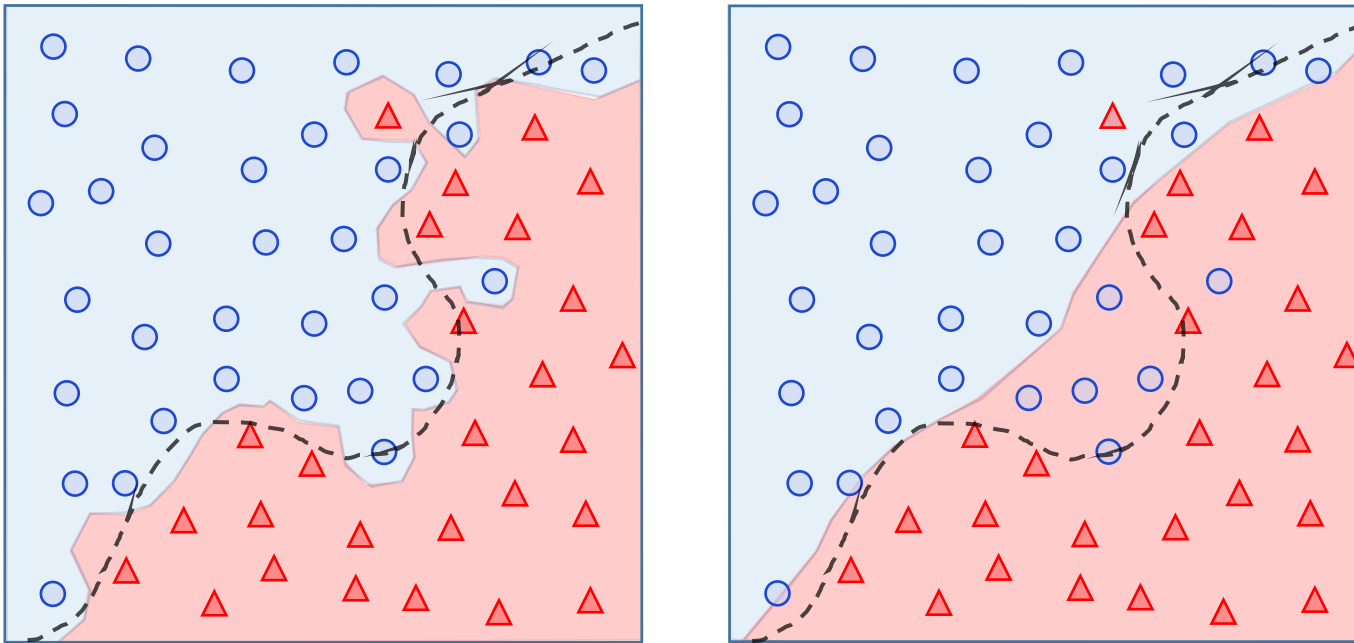
- Since there is no “model,” little insight can be extracted.
- No model parameters that store the learned pattern. The training dataset is required for prediction.
- Prediction is not efficient ⇨ “Lazy algorithm.”

- KNN classification algorithm



- KNN classification algorithm

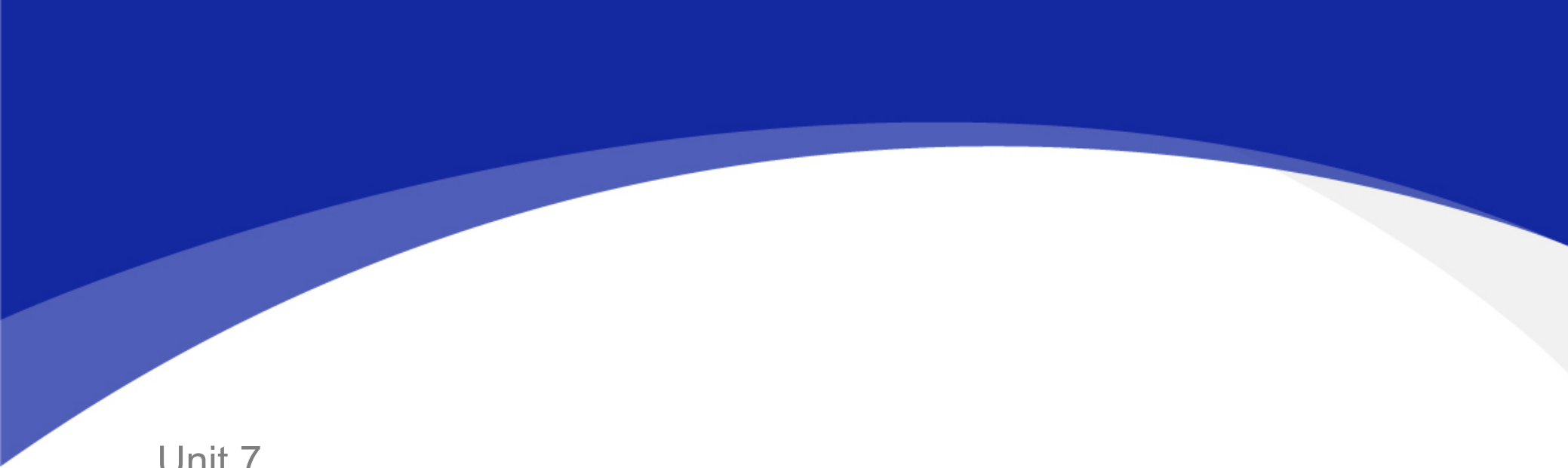
- When k is too small, overfitting may happen.



Coding Exercise #0310



Follow practice steps on 'ex_0310.ipynb' file.



Unit 7.

SVM Algorithm

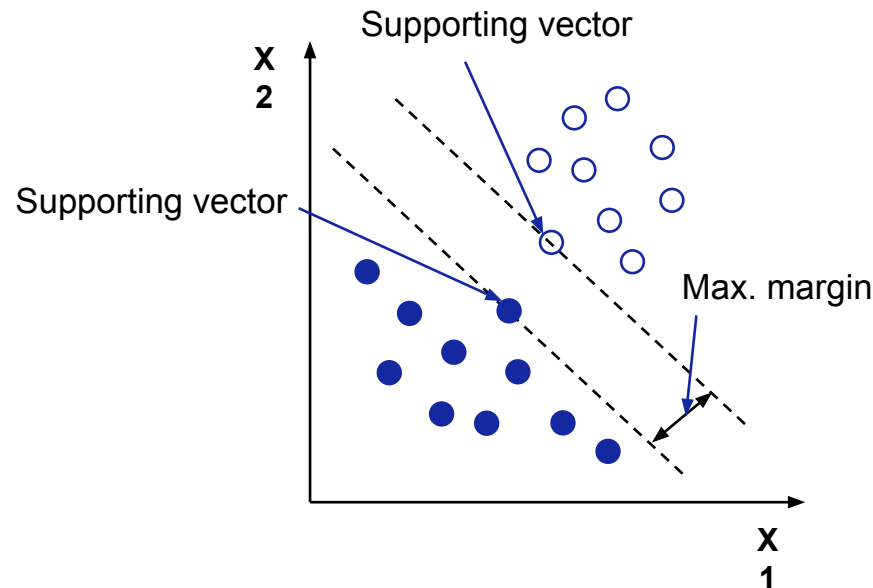
- 7.1. SVM Algorithm

SVM Algorithm

- About SVM (Support Vector Machine)
 - Enhanced classification accuracy by maximizing the margin
 - Effective non-linear classification boundary by the “kernel” transformation

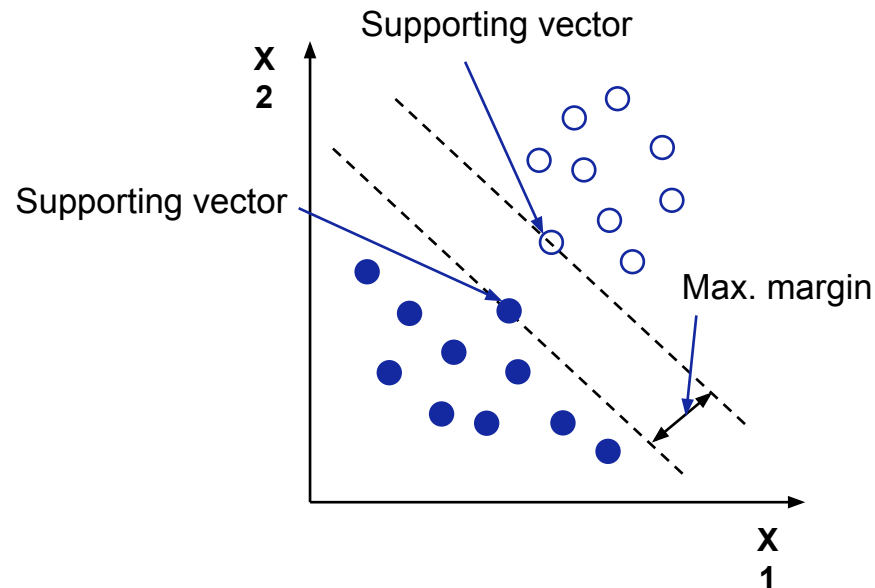
- About SVM (Support Vector Machine)

- Support vector is a model that classifies data by finding the line (or hyperplane) where the distance (margin) between data in different categories becomes maximum.
- As shown in the figure below, the support vector machine model finds the hyperplane that splits the data into two different categories with maximum margin to classify data.
- There would be many lines or planes that split data into two different categories. However, points that go over the classification boundary can occur in evaluation data or unknown future data as the points near the boundary slightly change.



- About SVM (Support Vector Machine)

- To minimize such a possibility, a line (or hyperplane) making the maximum margin between different data categories should be found. In other words, the aim is to find a hyperplane that can induce maximum differentiation for the best possible generalization to classify and predict the future data, not the current training data.
- The point in the category closest to the boundary line is called a support vector, and each classification should have at least one support vector.



- About SVM (Support Vector Machine)

- However, it is not always possible to classify all data linearly. Sometimes, data classification should be done in a curve or a more complex non-linear classification plane.
- If so, use the Kernel trick method for mapping the given data into a higher dimension and find a hyperplane that can classify the data in the converted dimension.
- Instead of converting the data to a higher dimension, the Kernel trick method uses the Kernel function that returns a similar value to when performing internal calculations between vectors in the higher dimension. This can result in a non-linear classification without moving data to a higher dimension.
- The major Kernel functions include polynomial, Gaussian, and sigmoid kernels.

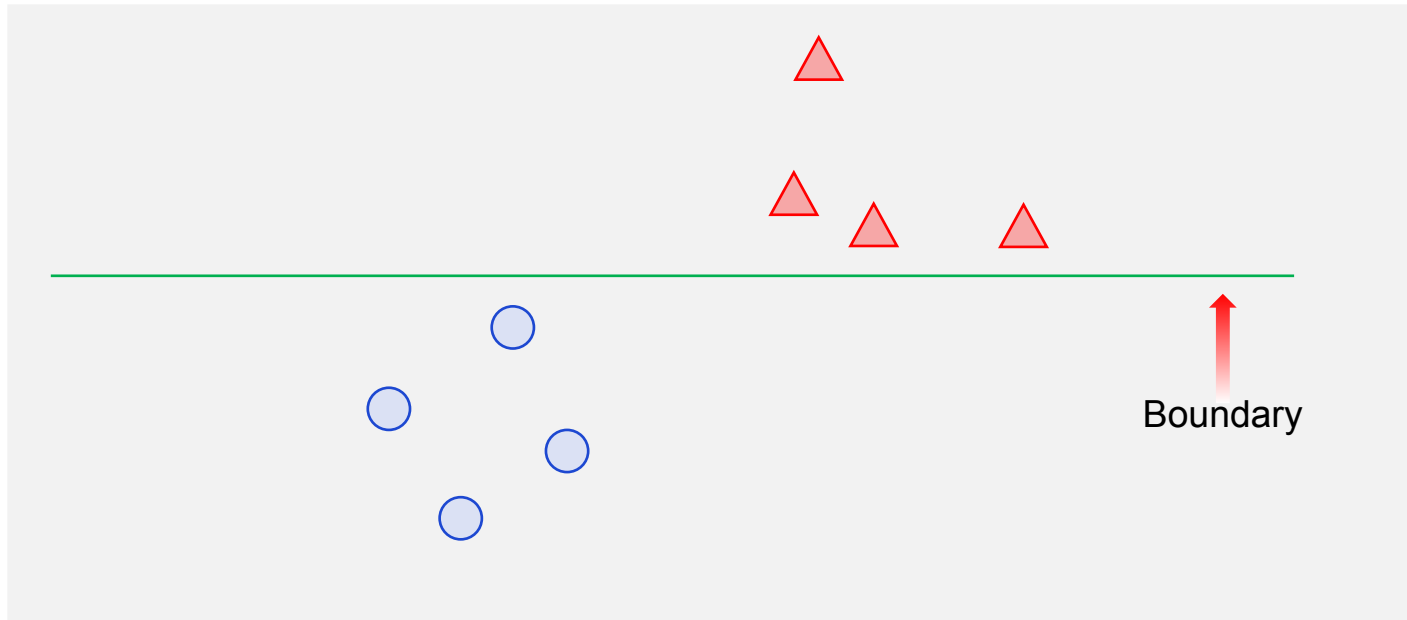
- Pros

- Not very sensitive to the outliers.
- Performance is good.

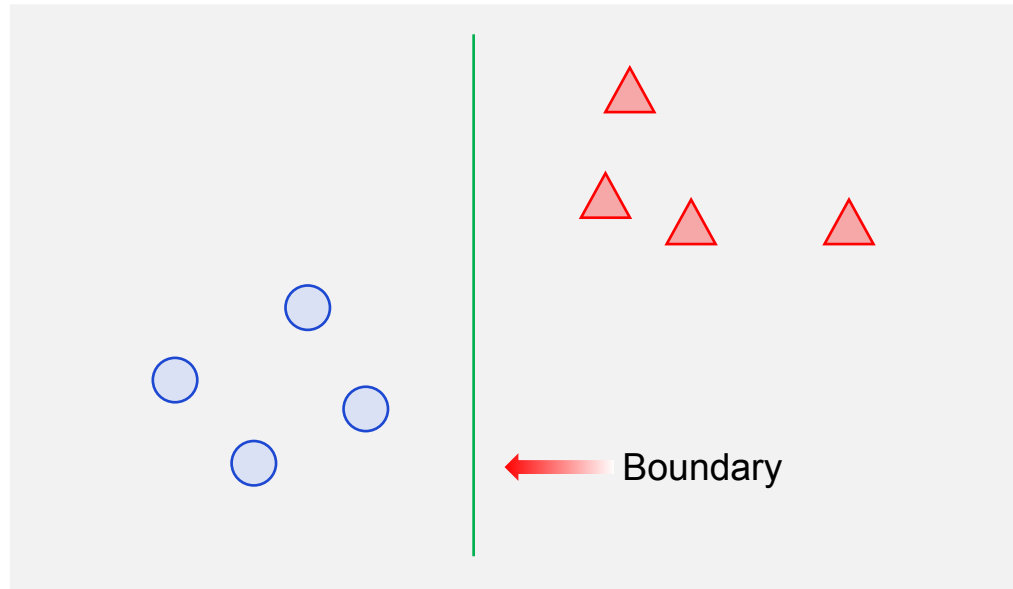
- Cons

- Training is relatively slow. Performs poorly for large data.
- The kernel and the hyperparameter set should be carefully optimized.
- Not much insight can be gained.

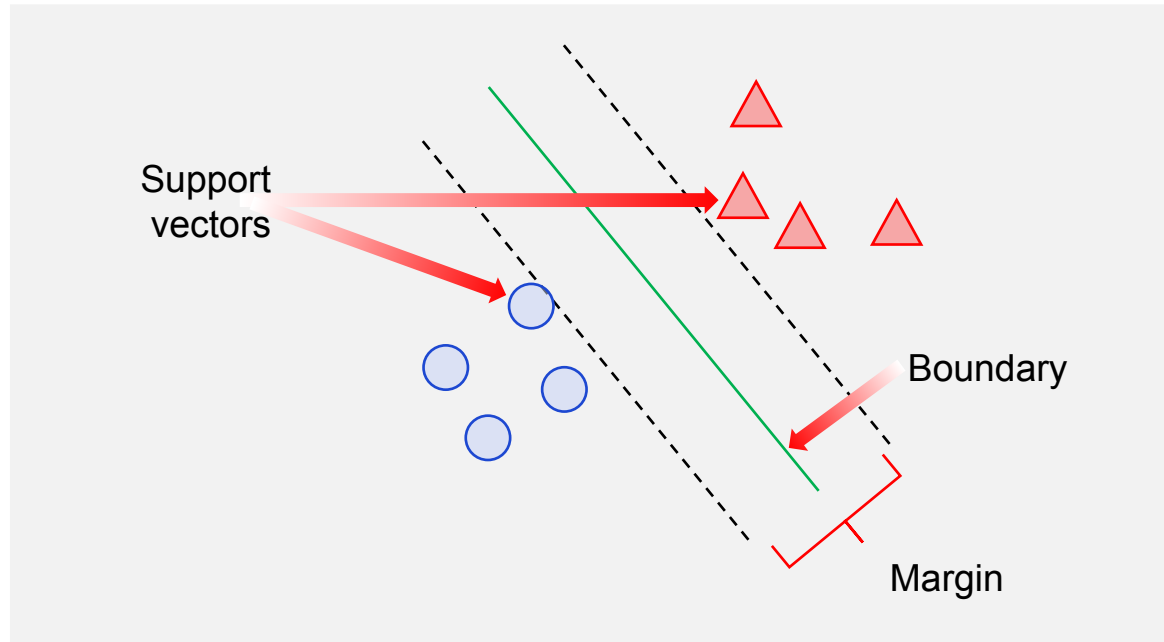
- SVM classification algorithm



- SVM classification algorithm



- SVM classification algorithm



- Hyperplane

- For a k dimensional configurational space, the hyperplane has the dimension $k-1$.

Ex For a two dimensional space, a hyperplane is a bisecting line that can be parametrized as:

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0$$

The two dimensional space is subdivided into two:

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 > 0$$

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 < 0$$

An observation would belong to either one of them $\Rightarrow$ binary classification!

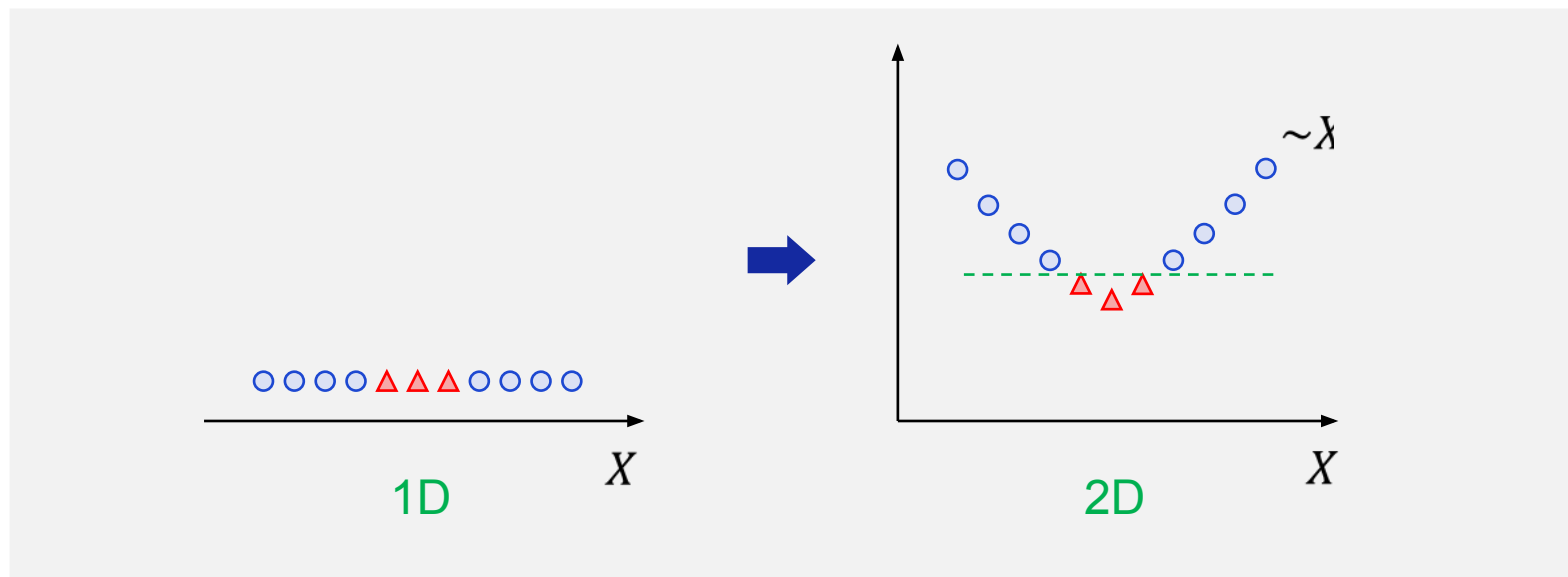
- Margin optimization (for the binary y)

- ▶ The boundary margin is maximized.
- ▶ If a clear margin is not possible to establish, an error is allowed within a limit.
- ▶ Target is to maximize M by optimizing the parameters $\beta_0, \beta_1, \dots, \beta_k$ subject to the constraints:
 - **Constraint 1:** $\sum_{j=1}^k \beta_j^2 = 1$
 - **Constraint 2:** $y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_k x_{ik}) \geq M(1 - \varepsilon_i)$, $i = 1, 2, \dots, n$
 y_i is either -1 or +1.
 - **Constraint 3:** $\varepsilon_i \geq 0$ and $\sum_{i=1}^k \varepsilon_i \leq C$
 $\Leftrightarrow C$ is a hyperparameter related to the miss-classification errors.

- Kernel

- Mapping to a higher dimension using the “kernel” functions.
- Kernel functions introduce an effective non-linear classification boundary.

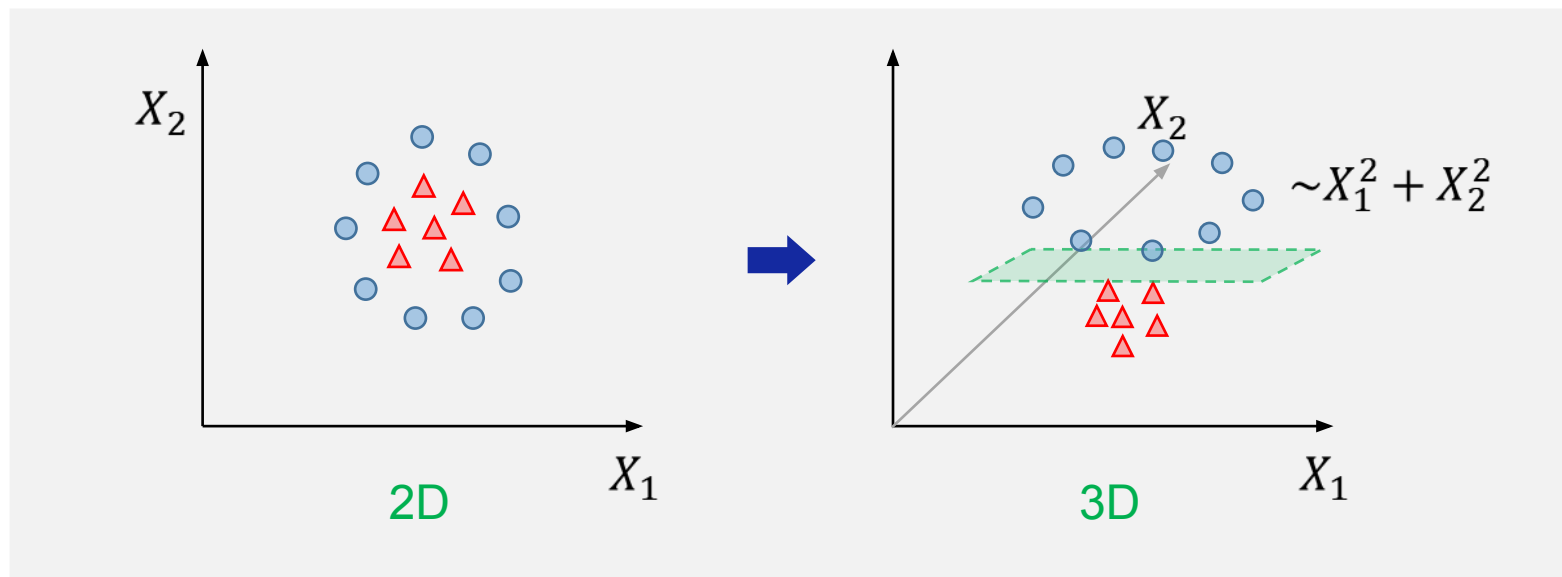
Ex Polynomial kernel



•Kernel

- Mapping to a higher dimension using the “kernel” functions.
- Kernel functions introduce an effective non-linear classification boundary.

Ex Polynomial kernel



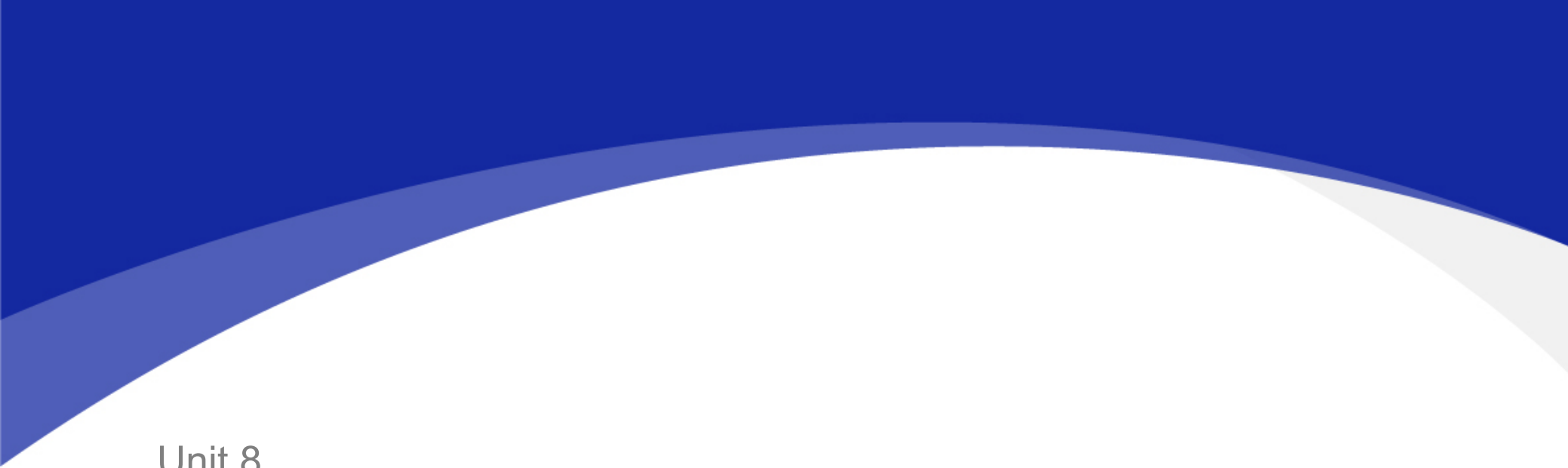
•Kernel

- ▶ Effective mapping to a higher dimension by giving the inner product of two vectors $\mathbf{x}_1$ and $\mathbf{x}_2$.
 - Linear: $K(\mathbf{x}_1, \mathbf{x}_2) = \mathbf{x}_1^t \mathbf{x}_2$
 - Polynomial: $K(\mathbf{x}_1, \mathbf{x}_2) = (\mathbf{x}_1^t \mathbf{x}_2 + c)^d$, where $c > 0$
 - Sigmoid: $K(\mathbf{x}_1, \mathbf{x}_2) = \tanh(a(\mathbf{x}_1^t \mathbf{x}_2) + b)$, where $a, b > 0$
 - Radial function basis (rbf): $K(\mathbf{x}_1, \mathbf{x}_2) = \exp(-\gamma \|\mathbf{x}_1 - \mathbf{x}_2\|^2)$, where $\gamma > 0$

Coding Exercise #0311



Follow practice steps on 'ex_0311.ipynb' file.



Unit 8.

Ensemble Algorithm

- 8.1. The Concept of Ensemble Algorithm and Voting
- 8.2. Bagging & Random Forest
- 8.3. Boosting

Ensemble Algorithms

- About ensemble algorithms
 - Strong predictive model based on the weaker learners.
 - **Voting type:**
 - A collection of basic learners that “vote”
 - An ensemble of different kinds of learners **Ex** Combine Tree, KNN, SVM, etc.
 - **Bagging type:**
 - A collection of independent weak learners that “vote”
 - An ensemble of the same kind of weak learners **Ex** Random Forest
 - **Boosting type:**
 - A series of weak learners that adaptatively learn and predict **Ex** AdaBoost, GBM, XGBoost, etc.
 - The series is grown by adding new learners multiplied by the “boosting weights.”

- About ensemble algorithms

- Making a more suitable decision-making by appropriately combining multiple opinions obtained by different experts
- Majority Voting – Analysis using many different classification models with one identical training set
- Bagging (different training set)
 - bootstrap + aggregating
 - Bootstrap + aggregating
 - Bagging (random sampling with replacement (random bootstrap))
- Ex** Random Forest
- Boosting
 - Retain 50% of incorrectly classified data or use weight for sample selection.

- bootstrap

- Estimation of unknown statistics
- Easy and effective estimation method with the unknown distribution of the model parameter sample
- Process of recalculating the statistics and model for each sample through additional sampling with replacement from the current sample
 - No assumption is required such that the parameters or sample statistics should be in a normal distribution.
- bootstrap sample – Aggregation of observed data (sampling with replacement obtained from the record value and dependent variable)

Contents to be learned in the future

- Resampling – Involves permutation under-sampling without replacement
- Bootstrap aggregation – Making a result from aggregating predicted values obtained from different bootstrap samples

- What is ensemble learning?

- The term 'ensemble' can be defined as follows:
 - In statistical mechanics, an ensemble of a system refers to the collection of equivalent systems.
- In other words, it is an assembly of similar groups.
- Instead of expecting performance results from a single model, ensemble learning draws a better result using the collective intelligence of different models, such as averaging out many different single models or making a decision based on the majority vote.
- There are many different ensemble methods using collective intelligence.
 - Voting – Drawing results through voting
 - Bagging – Bootstrap aggregating (duplicated creation of various samples)
 - Boosting – Weighting by supplementing previous errors
 - Stacking – A meta-model based on different models
- There can be other more different methods since ensemble learning applies a certain technique/methodology. Yet, the four methods listed above are the most representative ensemble techniques in the sklearn library.

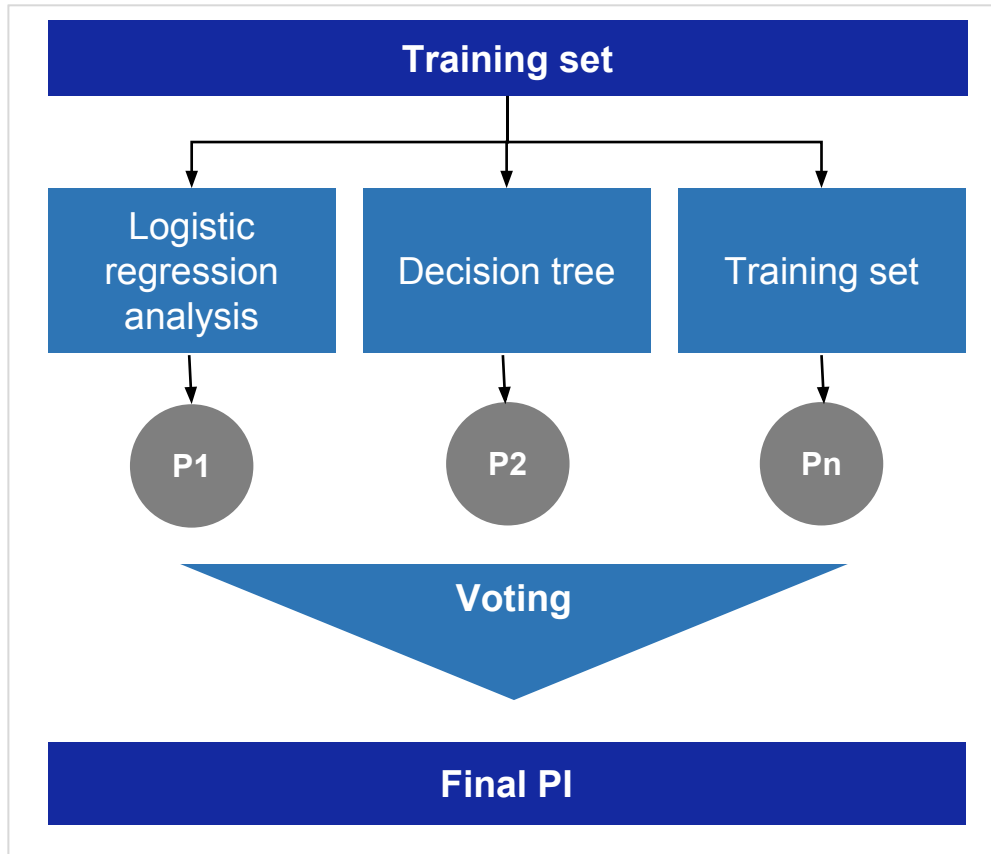
Voting Ensemble

- Voting ensemble
 - Can be applied to classification and regression.
 - The learners that form an ensemble should be of different kinds for a good performance.
 - Two voting methods for the classifier:
 - Hard: predicted class label is given by the majority rule voting.
 - Soft: predicted class label is given by the **argmax** of the sum of the predicted probabilities.

- Voting

- As the word itself, voting makes a decision through votes. Voting is similar to bagging as it uses a voting method, but they are highly differentiated from each other as follows:
 - Voting: Combines different algorithm models.
 - Bagging: Uses different sample combinations within the same algorithm.
- Voting selects final results by having final voting on results deduced by different algorithms.
- Voting is classified into hard voting and a soft voting.
 - Hard voting: Decides the final value of the result through voting.
 - Soft voting: Draws the final value by adding all the probability values of getting the final result and then calculating every probability of the final result.

•Voting



- Use a single training set.
- Use different classification models.
- Predictive value
 - hard voting
 - soft voting
- If the predictive values of each analysis model are different from voting, choose the result with the most values.
 - **Predict the highest class by averaging out the prediction of individual classifier.**

- Voting

- Hard Voting

- Taking classification as an example. Suppose the predictive values for classification are 1,0,0,1,1 since 1 has three votes and 0 has 2 votes. In that case, 1 becomes the final predictive value in the hard voting method.

- Soft Voting

- Soft voting method calculates the average value of each probability and then determines the one with the highest probability.
 - Suppose the probability of getting class 0 is (0.4, 0.9, 0.9, 0.4, 0.4), and the probability of getting class 1 is (0.6, 0.1, 0.1, 0.6, 0.6). The final probability of getting class 0 is $(0.4+0.9+0.9+0.4+0.4) / 5 = 0.6$; the final probability of getting class 1 is $(0.6+0.1+0.1+0.6+0.6) / 5 = 0.4$. Therefore, the selected final value is different from the result of the hard voting above.
 - In general, using the soft voting method is considered more reasonable than the hard voting method in competitions because the soft voting method provides a much better actual performance result.

- Voting ensemble in scikit-learn

- To import the voting ensemble as class:

```
from sklearn.ensemble import VotingClassifier    # For classification
from sklearn.ensemble import VotingRegressor    # For regression
```

- To instantiate an object that implements voting ensemble:

```
Ex myKNN = KNeighborsClassifier(n_neighbors = 3)
    myLL = LogisticRegression()
    myVotingEnsemble=VotingClassifier(estimators=[('lr',myLL),('knn',myKNN)],voting='hard')
```

- Voting ensemble in scikit-learn

- We can train and predict just like any other estimator:

```
Ex myVotingEnsemble.fit(X_train, Y_train)
    myVotingEnsemble.predict(X_test)
```

- scikit-learn VotingClassifier/Regressor Hyperparameters:

Hyperparameter	Explanation
estimators	The list of basic learner objects
voting	Either 'soft' or 'hard' (for classifier only)

- More information can be found at:

Classifier: <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.VotingClassifier.html>

Regressor: <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.VotingRegressor.html>

Unit 8.

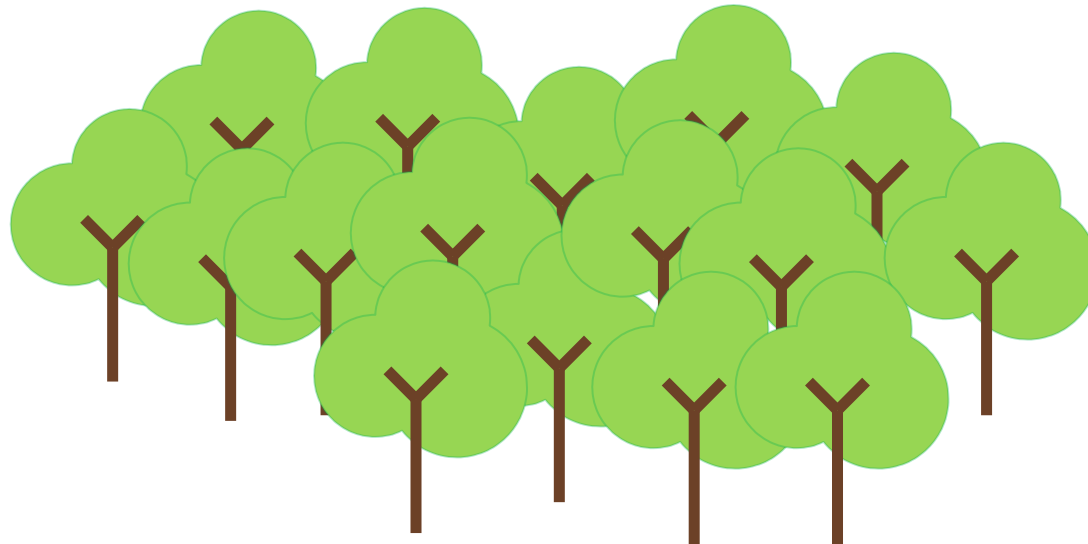
Ensemble Algorithm

- 8.1. The Concept of Ensemble Algorithm and Voting
- 8.2. Bagging & Random Forest
- 8.3. Boosting

Bagging Ensemble: Random Forest

- About Random Forest
 - An ensemble algorithm based on trees
 - Can be applied to classification and regression
- Pros
 - Powerful
 - Few assumptions
 - Little or no concern about the overfitting problem
- Cons
 - Training is time consuming.

- Random Forest algorithm



- Randomly chosen trees form a forest
- Prediction is decided by the majority vote.

- Random Forest algorithm

- 1) Make trees with randomly selected variables and observations.
- 2) Keep only those with the lowest Gini impurity (or entropy).
- 3) Repeat from step 1) a given number of times.
- 4) Using the trees gathered during the training step, we can make predictions by majority vote.

- scikit-learn RandomForestClassifier/Regressor Hyperparameters:

Hyperparameter	Explanation
n_estimators	The number of trees in the forest
max_depth	The maximum depth of a tree
min_samples_leaf	The minimum number of sample points required to be at a leaf node
min_samples_split	The minimum number of sample points required to split an internal node
max_features	The number of features to consider when looking for the best split

- › Except for “n_estimators,” the rest are analogous to those of DecisionTreeClassifier/Regressor.
- › More information can be found at:

Classifier: <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html>

Regressor: <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html>

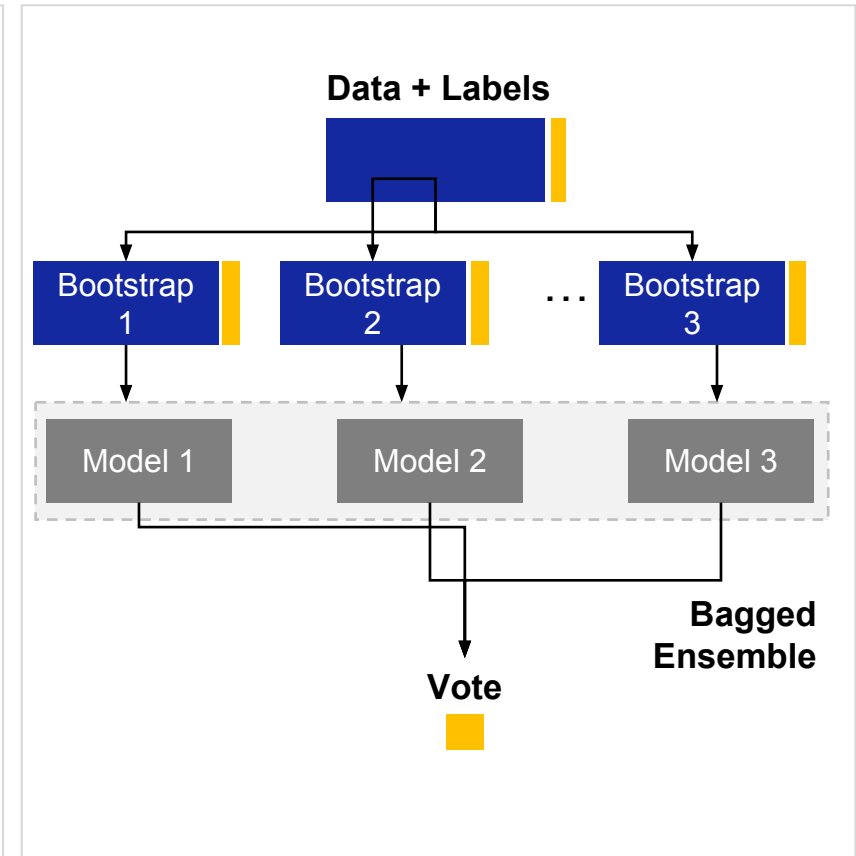
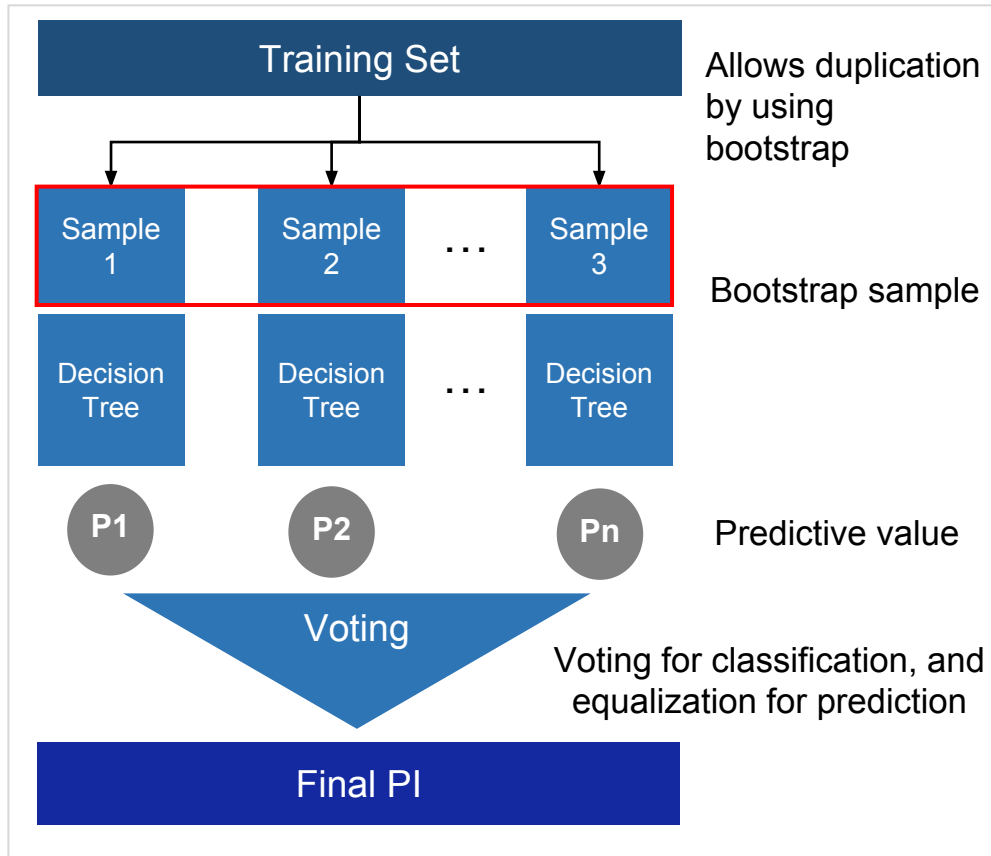
- Bagging

- Bagging-based ensemble method

- Ex** Random Forest algorithm

- Easy to use since it is well constructed in the Sklearn library
 - Relatively quick performance speed
 - High performance
 - The ensemble method has been widely used as it raises the performance level and is easy to use. The bagging-based ensemble method is commonly found in the high-ranked solutions in Kaggle.

- Bagging



- Bagging

- Bagging is an abbreviation of Bootstrap Aggregating.
- Bootstrap = Sample
- Aggregating = Adding up
- Bootstrap refers to a method that allows overlapping of different datasets for sampling and splitting.

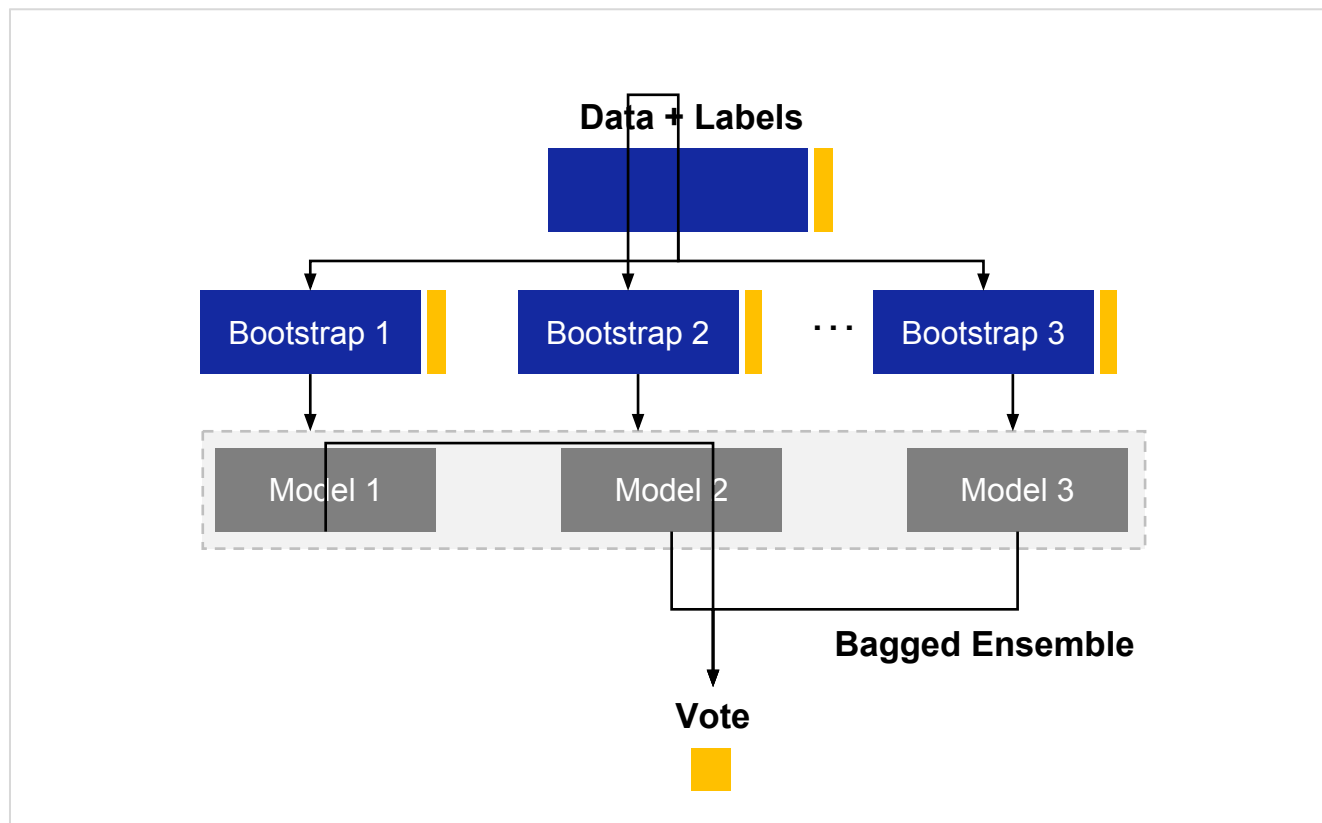
- Bagging

Ex Random Forest is a typical bagging method algorithm.

- It creates multiple decision trees and performs sampling of different datasets while allowing overlapped datasets.
- If the dataset consists of [1, 2, 3, 4, 5]:
 - Group 1 = [1, 2, 3]
 - Group 2 = [1, 3, 4]
 - Group 3 = [2, 3, 5]
- This is the bootstrap method. In the classification problem, voting is done on each tree trained with different sampling for the final prediction result.
- In regression problems, the average of each obtained value is calculated.

- Bagging

- ▶ “**Bagging**”: Bootstrap **AGG**regat**ING**



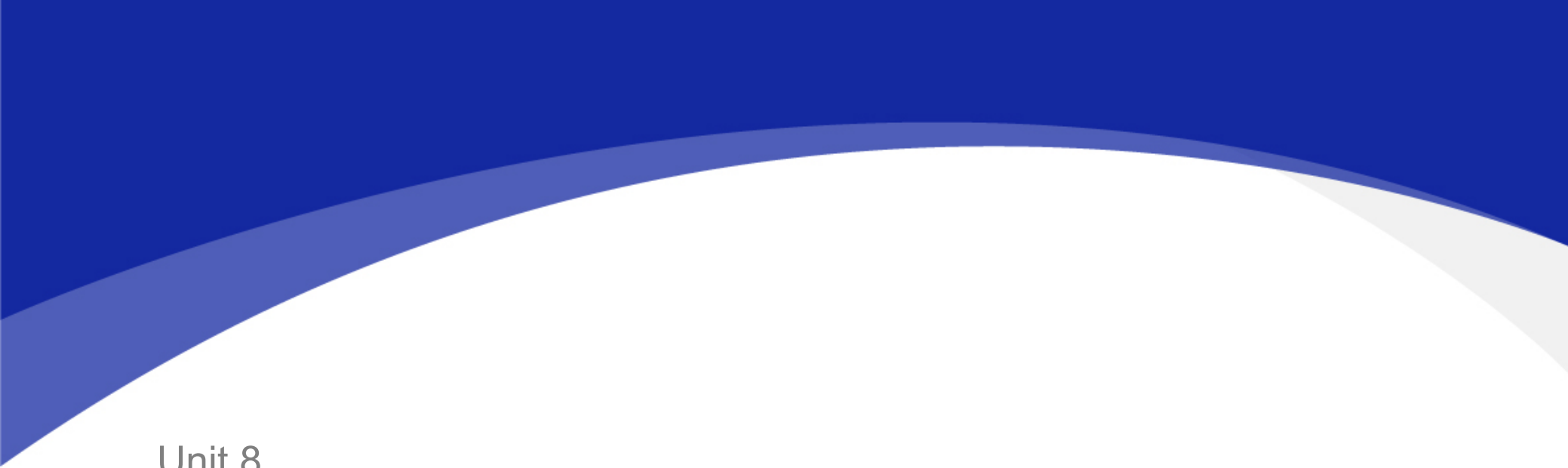
- Differences Between Bagging and Voting

- The greatest difference between the bagging and voting methods is whether to use multiple single algorithms or apply various algorithms to the same sample dataset.
- In general, the bagging method is to train a single algorithm with different sampling datasets and perform voting. It has relatively better usability than the voting method because it uses a single algorithm. Thus, what is important is the hyperparameter of the single algorithm.
- Taking the Random Forest algorithm as an example again, it is suitable to obtain the baseline score as it requires simple hyperparameter setting such as how many trees to use (`n_estimators`), maximum depth (`max_depth`), and the minimum number of samples for splitting (`min_samples_leaf`).

- Advantages of the Bagging Ensemble

- When using the bagging method, it can reduce variance compared to making a prediction with a single model. There are three major training errors in a model: variance, noise, and bias. (Of course, more significant factors include overfitting/underfitting and many different preprocessing issues, but assume they are already being set.)
- The ensemble method reduces variance, thus enhancing the performance of the final result.

- `sklearn.ensemble.BaggingClassifier/BaggingRegressor`
 - The sklearn library package provides the wrapper class called `BaggingClassifier/BaggingRegressor`.
 - When designating the base algorithm to the `base_estimator` parameter, the `BaggingClassifier/BaggingRegressor` performs bagging ensemble.



Unit 8.

Ensemble Algorithm

- 8.1. The Concept of Ensemble Algorithm and Voting
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Boosting Ensemble: AdaBoost

- About AdaBoost
 - A sequence of weak learners such as trees
 - A weighted ensemble of weak learners
 - One set of weights that increase the importance of the better-performing learners
 - One set of weights that increase the importance of the wrongly classified observations
 - Similar pros and cons as the Random Forest

- AdaBoost classification algorithm

- 1) For the first step ($m=1$), equal weight is assigned to the observations:

- 2) For the boost sequence $m=1, \dots, M$:

- a) Train the learner $G_m(x)$ using observations weighted by $w_i^{(m)}$.

- b) Calculate the error ratio:
$$\varepsilon_m = \frac{\sum_{i=1}^n w_i^{(m)} I(y_i \neq G_m(x_i))}{\sum_{i=1}^n w_i^{(m)}}$$

$\Rightarrow I(y_i \neq G_m(x_i))$ gives 1 for an incorrect prediction, else 0.

$\Rightarrow 0 \leq \varepsilon_m \leq 1$

- AdaBoost classification algorithm

$$\alpha_m = \frac{1}{2} \log \left(\frac{1 - \varepsilon_m}{\varepsilon_m} \right)$$

⇒ As $\varepsilon_m \rightarrow 0$, α_m is a large positive number. The learner is given more importance!

⇒ As $\varepsilon_m \cong 0.5$, $\alpha_m \cong 0$.

⇒ As $\varepsilon_m \rightarrow 1$, α_m is a large negative number.

d) For the next step, the weights of the **wrongly** predicted observation are rescaled by a factor e^{α_m} .

This can be compactly expressed as $w_i^{(m+1)} = w_i^{(m)} \times e^{\alpha_m \cdot I(y_i \neq G_m(x_i))}$, where $i = 1, \dots, n$

⇒ In the next sequence step, the wrongly predicted observations receive heavier weight.

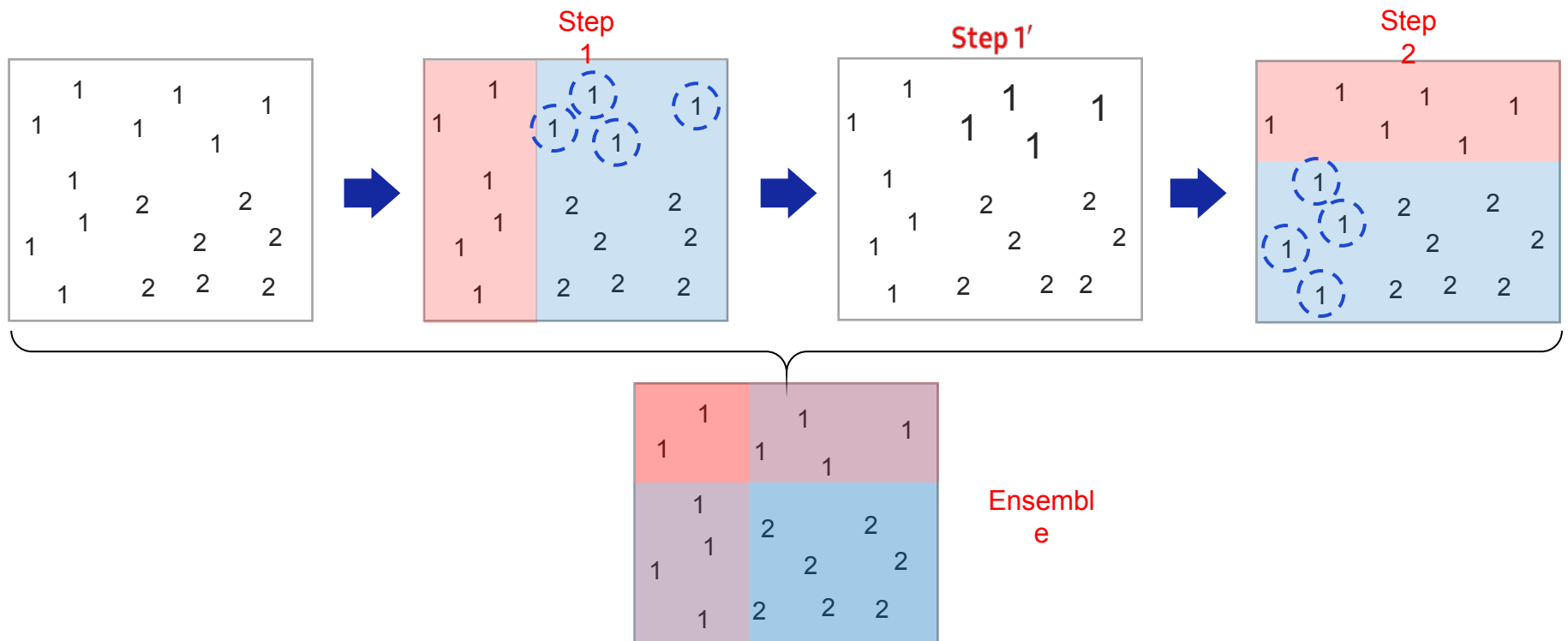
- AdaBoost classification algorithm

4) The ensemble classifier is made up of a linear combination of weak learners:

$$G_{ensemble}(\mathbf{x}) = \text{sign} \left(\sum_{m=1}^M \alpha_m G_m(\mathbf{x}) \right)$$

5) For a new testing condition $\mathbf{x}'$, we can predict y' by $G_{ensemble}(\mathbf{x}')$.

- AdaBoost classification algorithm



- scikit-learn AdaBoostClassifier/Regressor Hyperparameters

Hyperparameter	Explanation
base_estimator	The base estimator with which the boosted ensemble is built
n_estimators	The maximum number of estimators at which boosting is terminated
learning_rate	The rate by which the contribution of each learner is shrunk
algorithm	Either 'SAMME' or 'SAMME.R'

- “base_estimator” is by default `None`, which means `DecisionTreeClassifier(max_depth=1)`.

- More information can be found at:

Classifier: <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.AdaBoostClassifier.html>

Regressor: <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.AdaBoostRegressor.html>

Boosting Ensemble: GBM & XGBoost

- Gradient Boosting Machine (GBM)
 - ▶ GBM is also made up of a sequence of weak learners similar to AdaBoost.
 - ▶ The disagreement between the predicted $\hat{y}$ and the true y can be represented by a “loss” function.
 - ▶ In the sequence of weak learners, the learner at step m , $G_m(\mathbf{x})$ can be obtained by moving the previous step learner $G_{m-1}(\mathbf{x})$ towards the direction that decreases the loss as defined above.
 - ▶ These updating movements are restricted to a set of base learners.

- scikit-learn GradientBoostingClassifier/Regressor Hyperparameters

Hyperparameter	Explanation
loss	The loss function
n_estimators	The number of boosting steps (weak learners)
learning_rate	The contribution of each weak learner
subsample	The fraction of data that will be used by individual weak learner

- Need to be tuned for optimized performance.
- More information can be found at:

Classifier: <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.GradientBoostingClassifier.html>

Regressor: <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.GradientBoostingRegressor.html>

- Extreme Gradient Boosting (XGBoost)
 - Improves upon GBM in the execution speed.
 - More resistant to the overfitting than GBM.
 - Not included in the scikit-learn library. Requires installation of the “xgboost” library.

- XGBClassifier/Regressor Hyperparameters

Hyperparameter	Explanation
booster	gbtree or gblinear
n_estimators	The number of boosting steps (weak learners)
learning_rate	The contribution of each weak learner
subsample	The fraction of data that will be used by individual weak learner

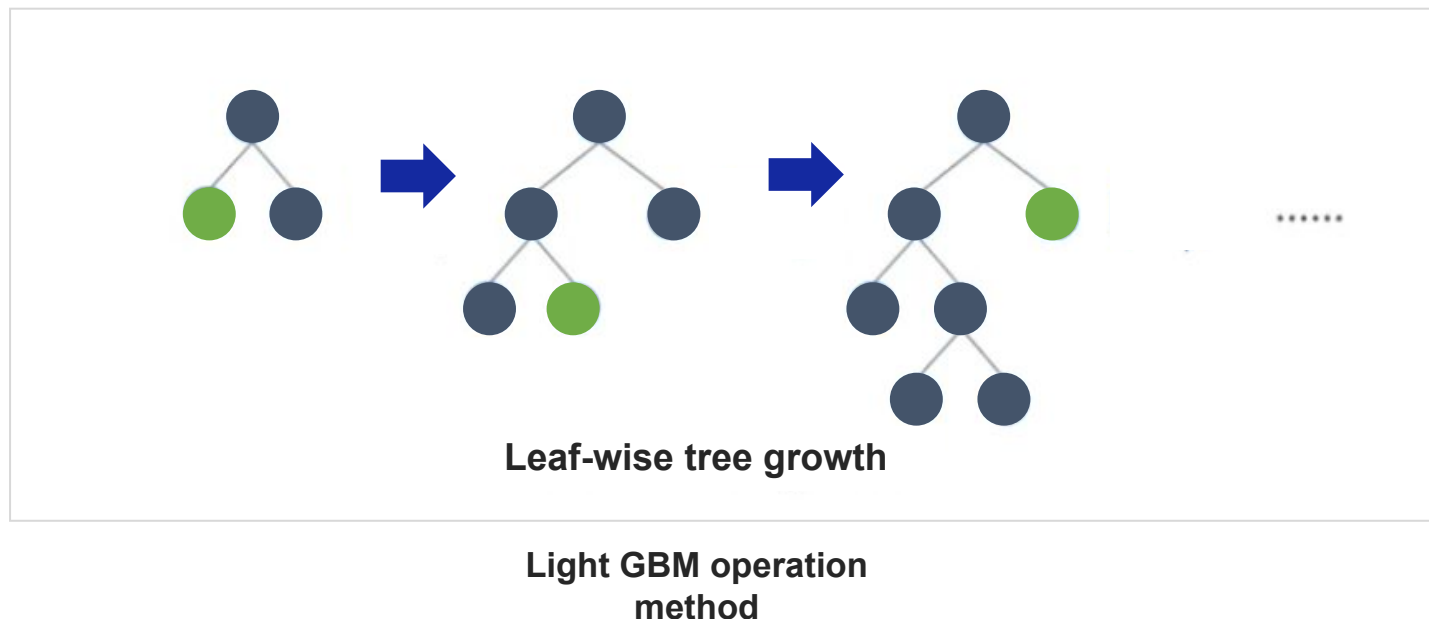
- Need to be tuned for optimized performance.
- More information can be found at: <https://xgboost.readthedocs.io/en/latest/python/index.htm>

- XGBoost

- XGBoost is a library that implements the gradient boosting algorithm to be used in the distributed system.
- It supports both regression and classification problems. This popularly used algorithm features good performance and resource efficiency.
- Gradient boost is a representative algorithm using the boosting method. The XGBoost is the library using this algorithm to support parallel training.
- It has been recently used a lot due to its high performance and computer resource application rate. It has become more popular as it was frequently used by top rankers of Kaggle.

- Light GBM

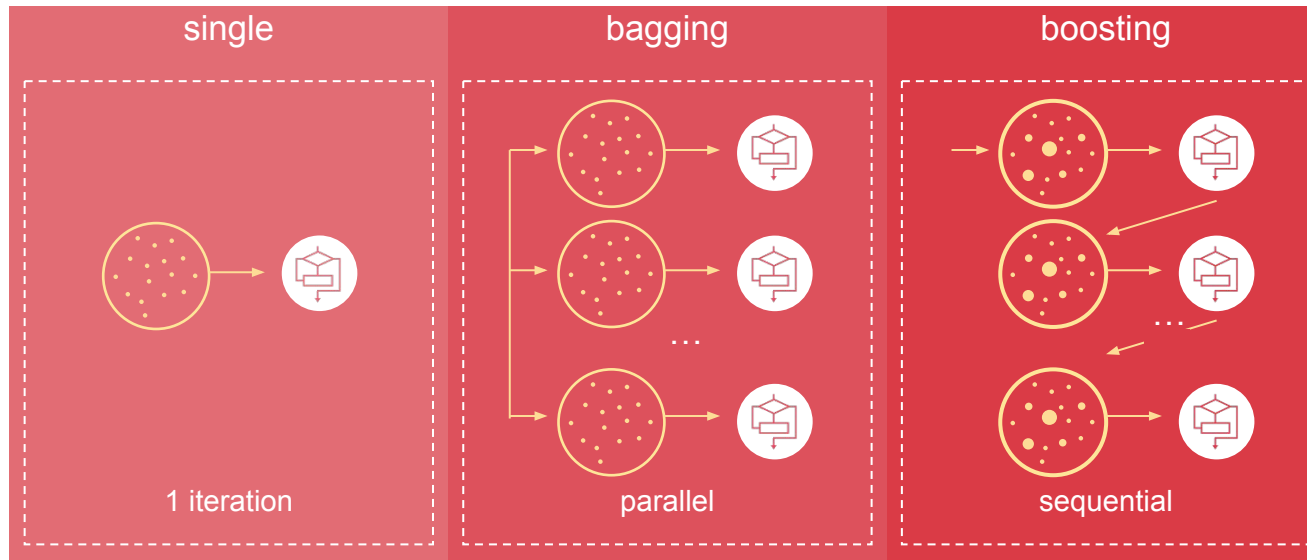
- ▶ While the tree is vertically expanded with Light GBM, other algorithms expand the tree horizontally. In other words, while the Light GBM is leaf-wise, other algorithms are level-wise.
- ▶ For expansion, the leaf with max delta loss is selected. When expanding the same leaf, the leaf-wise algorithm can reduce more loss than the level-wise algorithm.
- ▶ The following diagram shows how a boosting algorithm is implemented, which differs from the Light GBM.



- Boosting

- ▶ The boosting algorithm is also ensemble learning. After sequentially learning weak learning machines, it supplements errors by adding weight to inaccurately predicted data from the previous learning.
- ▶ The difference from other ensemble methods is that it performs sequential learning and supplements errors by adding weight. However, one of the disadvantages is that it is difficult to process parallel due to its sequential property, leading to a longer learning time compared to other ensembles.

Single Estimator, Bagging, Boosting

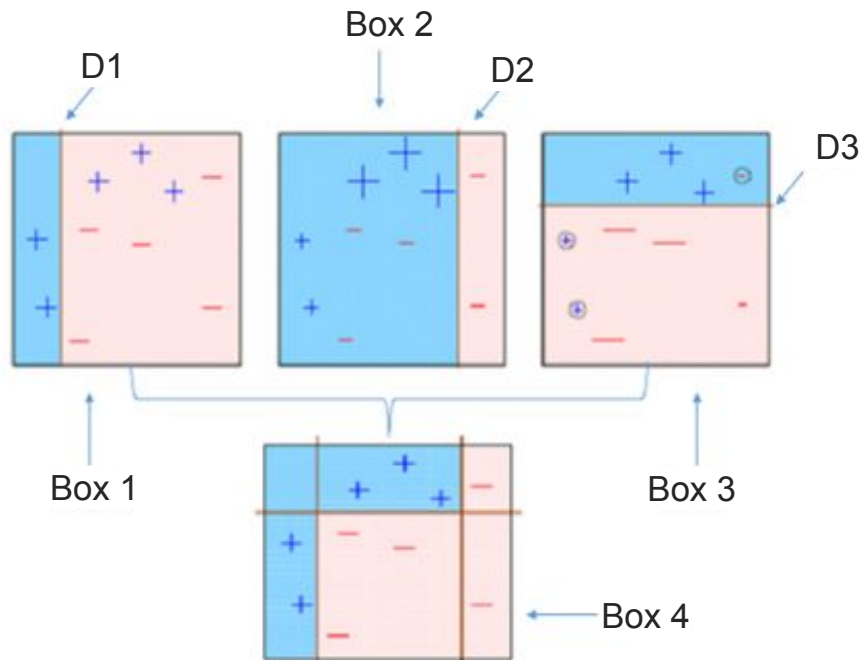


- Learning a series of predictors by supplementing previous models
 - AdaBoost
 - Train the first distributor (e.g., decision tree) in the training set and make a prediction.
 - Slightly increase the weight of the training sample with an inaccurately classified algorithm.
 - The second distributor uses the updated weight to be trained at the training set and makes a prediction again.
 - The weight is updated again, and the same process is repeated.
 - Gradient boosting
 - Similar to AdaBoost, gradient boosting sequentially adds predictors to the ensemble to correct the previous errors.
 - However, instead of modifying the sample weight repeatedly as AdaBoost, it trains a new predictor to residual error created by the previous predictor.
 - For regression problems – gradient boosted regression tree (GBRT)

- AdaBoost (Adaptive Boosting)

- AdaBoost (Adaptive Boosting) is an adaptive boosting method, one of the most representative boosting algorithms.
- As explained, the AdaBoost sequentially learns weak learning machines and supplements errors by applying weights to inaccurately predicted data.

• Principle of the AdaBoost



- Box 1 results from a weak learning machine classified as the D1 sector. However, the error rate is relatively high because some datasets are expressed in + distributed in the red sector.
- The D2 line in Box 2 moves to the right to supplement the error rate from Box 1. Here, the datasets expressed in – are distributed in the blue sector for better performance, but it is not satisfactory yet.
- The D3 line in Box 3 is horizontally drawn on the top. However, the dataset expressed in – is incorrectly classified.
- By training the previous Box 1, 2, and 3, Box 4 finds the most ideal combination. It shows much better performance compared to the previous three individual learning machines.

- Principle of the AdaBoost

- How is the weight applied for combination?

Ex Suppose that the following weight will be applied to the performance of Box 1~3.

- Performance of Box 1: weight = 0.2
- Performance of Box 2: weight = 0.5
- Performance of Box 3: weight = 0.6

It can be expressed as the following formula:

$$0.2 * \text{Box 1} + 0.5 * \text{Box 2} + 0.6 * \text{Box 3} = \text{Box 4}$$

- Gradient Descent

- The key to the boosting method is to supplement errors from the previous learning.
- The AdaBoosting and gradient descent methods are slightly different in how to supplement errors.
- Gradient descent uses differentiation to minimize the difference between the predicted value and actual data.
 - Weight
 - Input_data = feature data (input data)
 - Bias
 - Y_actual = actual data value
 - Y_predict = predicted value
 - Loss = error rate
- $Y_{\text{predict}} = \text{weight} * \text{input_data} + \text{bias}$
 - The predictive value can be obtained from the above formula.
Calculating the difference with actual data will result in the total error rate.
- $\text{Loss} = Y_{\text{predict}} - Y_{\text{actual}}$
 - (There are many different functional formulas to define error rate, including root means square error and means absolute error, but the above definition is provided for convenience.)
 - The purpose of gradient descent is to find the weight that makes the loss closest to 0.

- Gradient Boosting Machine (GBM)

- The boosting algorithm in gradient descent is called gradient boosting machine, abbreviated as GBM. It is provided in the sklearn package and applicable to both classification and regression problems.
 - GradientBoostingClassifier
 - GradientBoostingRegressor
- It is extremely easy to apply in actual problems.
- XGBoost and LightGBM are two major ML packages mostly used in Kaggle.
- XGBoost and LightGBM are not provided in the existing sklearn package. However, the sklearn wrapper class allows an easy application for fit & predict similar to the ML class of the existing sklearn package.

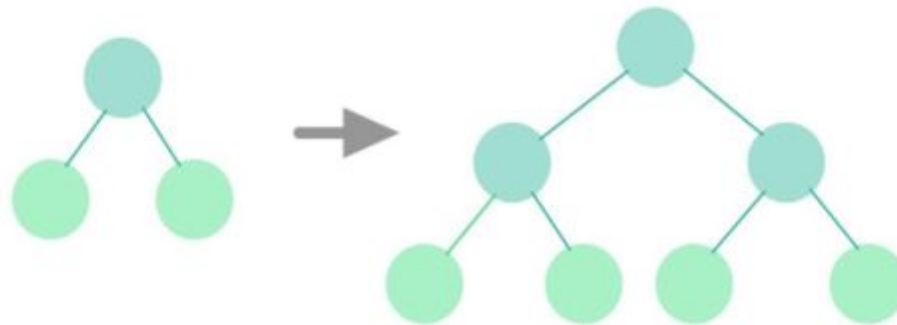
- Gradient Boosting Machine (GBM)

- Together with the XGBoost, LightGBM is the most highlighted boosting algorithm. Although XGBoost has an excellent performance, its learning time is too long.
- Advantages of the LightGBM
 - Short learning time
 - Relatively small memory use
 - Automatic conversion and optimal splitting of categorical features

- Characteristics of the LightGBM

- The existing tree-based algorithm uses a level-wise method. It splits but maintains a balanced tree as much as possible, so the tree depth becomes minimum. A disadvantage of the level-wise method is that it takes time to make a balanced tree.
- On the other hand, LightGBM uses a leaf-wise method. It does not balance the tree but continuously splits the leaf node with the max data loss. The tree depth becomes greater, and an asymmetrical tree is created. The repetition of the leaf node with the max data loss minimizes predicted error loss than splitting a balanced tree.

Level-wise



- Hyperparameter tuning methods

- The increased Num_leaves value raises accuracy. However, it also increases the tree depth and makes the model complex, thus increasing the probability of overfitting.
- Min_data_in_leaf is a significant parameter for overfitting. It depends on the Num_leaves and the size of training data, but in general, it prevents a deeper tree when setting it as a higher value.
- Max_depth constraints the depth size and improves overfitting when combined with the two parameters above.

•XGBoost

Parameter	Default Value	Description
num_iterations	100	Designates the number of trees for repetitive work. Overfitting occurs if the value is too high.
learning_rate	0.1	Updated when boosting steps are repeatedly conducted. The value is designated between 0~1.
max_depth	1	Identical to the max_depth of tree-based algorithms. No restriction is applied to tree depth when entering a value smaller than 0.
min_data_in_leaf	20	Identical to the min_samples_leaf of a decision tree. Used as a parameter to control overfitting.
num_leaves	3	Signifies the max. The number of leaves for one tree.
boosting	GBDT	
bagging_fraction	1.0	Designates the ratio for data sampling. Used to control overfitting.
feature_fraction	1.0	The ratio of the random feature for individual tree learning
Lambda_l1	0.0	The value for L1 regulation
Lambda_l2	0.0	The value for L2 regulation

Coding Exercise #0312



Follow practice steps on 'ex_0312.ipynb' file

Coding Exercise #0313



Follow practice steps on 'ex_0313.ipynb' file

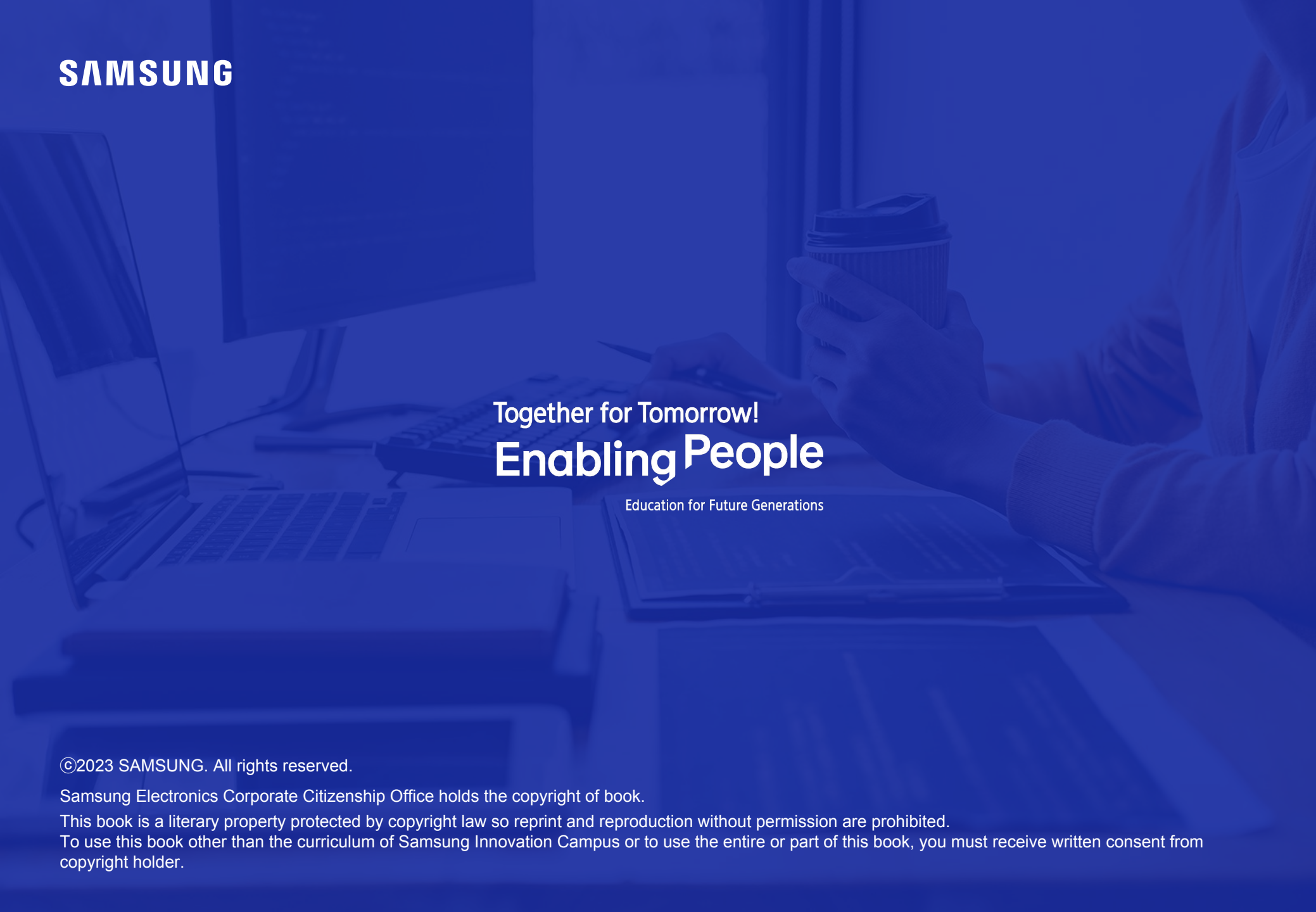
Coding Exercise #0314



Follow practice steps on 'ex_0314.ipynb' file

A person is sitting at a desk in a dimly lit office or home workspace. They are wearing an orange sweater and holding a brown paper coffee cup with a black lid in their left hand. Their right hand is holding a pen and is positioned over a black keyboard. On the desk, there is a laptop, a tablet, and some papers. The background is dark and out of focus.

End of Document



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